

The New Marketing Measurement Playbook

**How Attribution, Incrementality Testing, and MMMs Drive
Smarter Budgeting and Maximize Profits**

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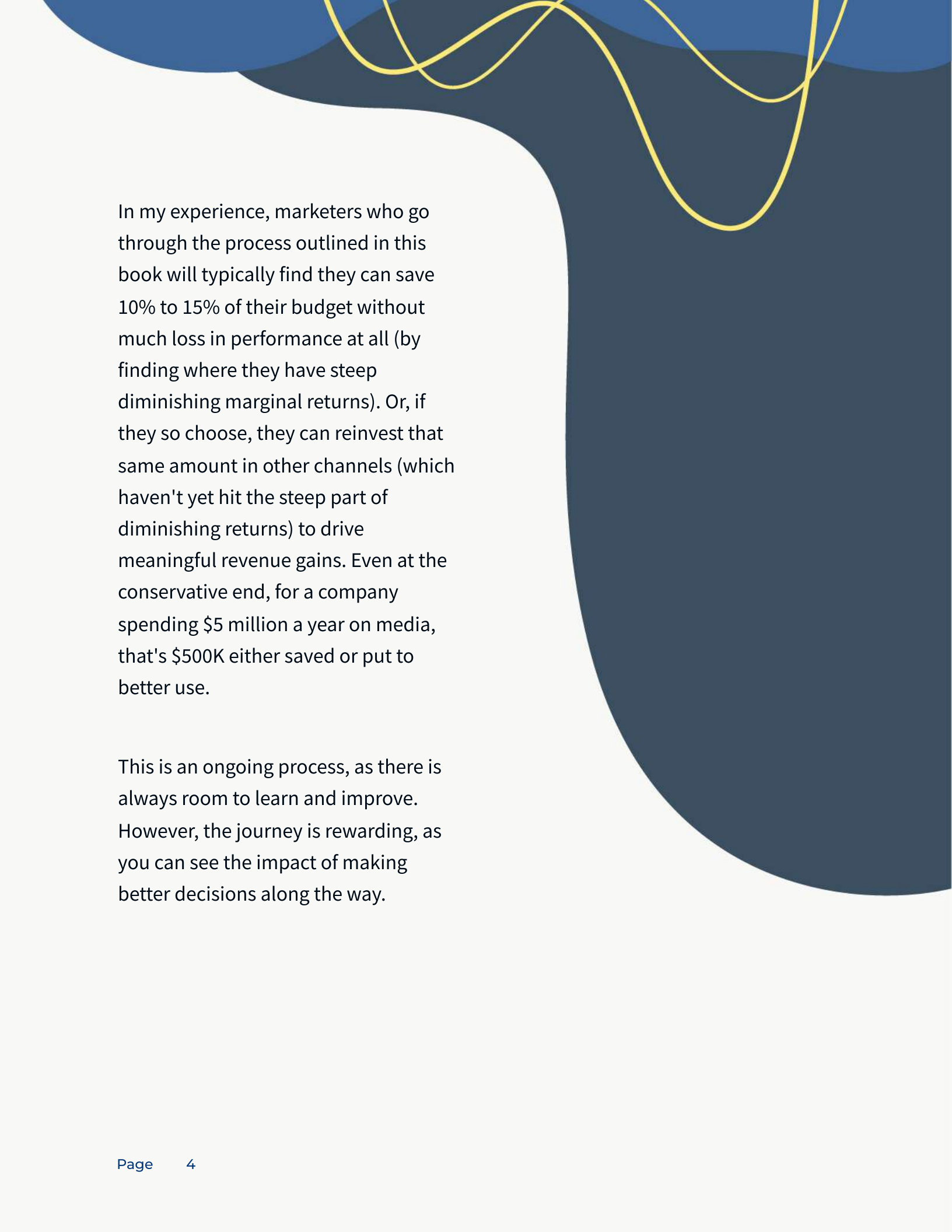
Introduction and Overview

Purpose

I hope this book will completely change the way you think about marketing and advertising budgets.

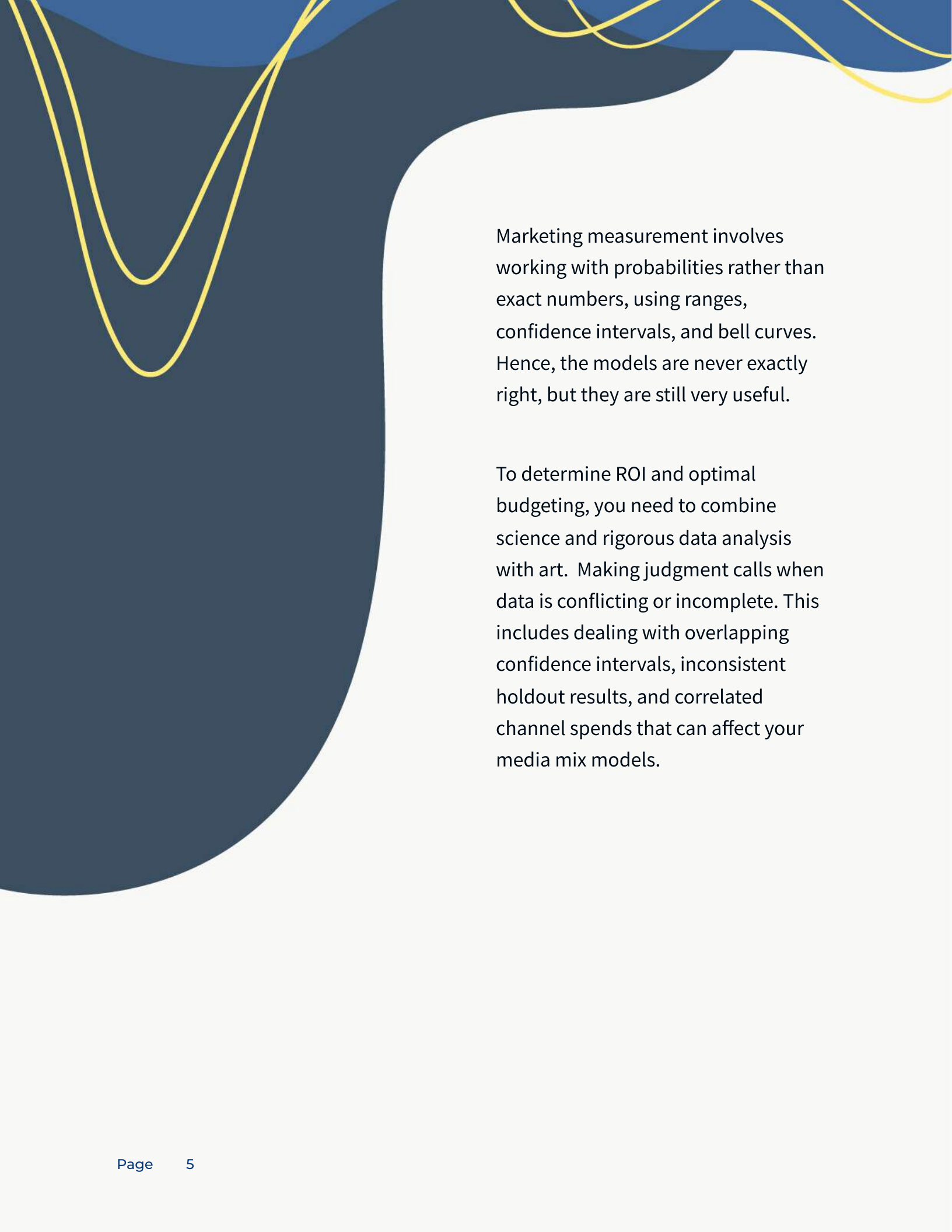
My goal is to provide a clear roadmap to help you make budgeting decisions, improve forecasting accuracy using the data you have, and identify any additional data or tools you may need. By the end of this book, you should be able to review your attribution reports, holdout tests, and media mix models and use them to create a coherent budget recommendation that drives performance and enables accurate forecasting.





In my experience, marketers who go through the process outlined in this book will typically find they can save 10% to 15% of their budget without much loss in performance at all (by finding where they have steep diminishing marginal returns). Or, if they so choose, they can reinvest that same amount in other channels (which haven't yet hit the steep part of diminishing returns) to drive meaningful revenue gains. Even at the conservative end, for a company spending \$5 million a year on media, that's \$500K either saved or put to better use.

This is an ongoing process, as there is always room to learn and improve. However, the journey is rewarding, as you can see the impact of making better decisions along the way.



Marketing measurement involves working with probabilities rather than exact numbers, using ranges, confidence intervals, and bell curves. Hence, the models are never exactly right, but they are still very useful.

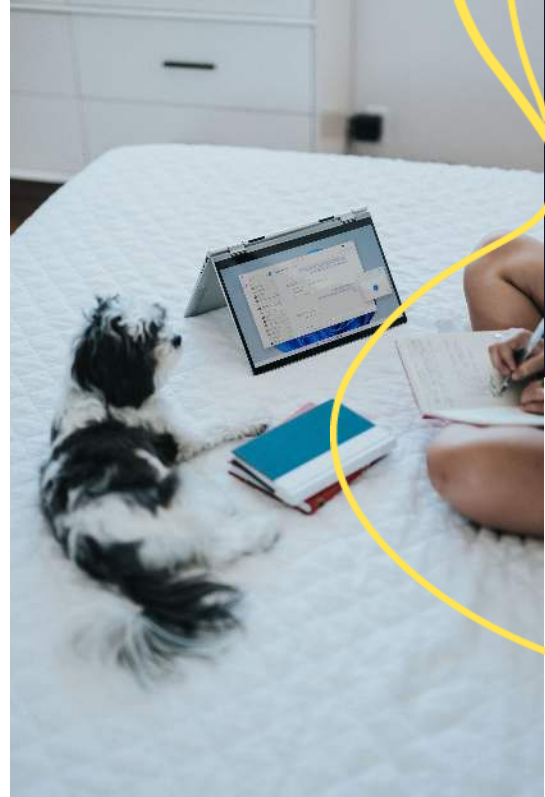
To determine ROI and optimal budgeting, you need to combine science and rigorous data analysis with art. Making judgment calls when data is conflicting or incomplete. This includes dealing with overlapping confidence intervals, inconsistent holdout results, and correlated channel spends that can affect your media mix models.

This book covers both the science and art of marketing measurement, starting with the science and then moving on to the art. Balancing science and art can be challenging, but it's a normal part of the process. As you become more successful at meeting this challenge, you'll feel a sense of real accomplishment. You will be at the cutting edge of Marketing Measurement!

Core Thesis

Marketing measurement is about assigning credit where it's due, with a clear goal: estimating the incremental, marginal, and often delayed impact. I'll break down what each of these terms means as we move forward. In any case, when you can't measure these credibly or accurately, you then can't justify to your executive team why you are even spending the money you are.

The ultimate goal then is to achieve top-notch performance marketing, figuring out how much to spend and which channels to use to meet your objectives - whether that's acquiring new customers, boosting monthly profit or revenue, or increasing lifetime value. Forecasting becomes a natural outcome.



One note: in this book, I specifically do not address brand marketing spending. While that is important too, its measurement follows a very different approach and cycle and is best left for another time (or book).

A key point I'll emphasize throughout is the importance of considering all available data. This approach combines multi-touch attribution, media mix models, and holdout tests (when available), with the art of using intuition and experience to bring everything together.

While this book focuses on marketing measurement, including attribution, incrementality testing, and media mix models, bid optimization and A/B testing are also crucial to performance marketing. That's why I've included them in the Appendix, hoping you'll find some valuable insights there as well.

By the time you finish reading, you'll be confident in your ability to analyze data, apply relevant scientific methods, and make informed judgment calls to drive the best decisions and predictions.

You can read this book in just 2 to 3 hours and gain a clear understanding of what needs to be done. Implementing everything may take a few months, but at least you'll have a clear path forward.

Intended Reader

In this book, I cover:



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1. Attribution Models, including a detailed look at a data-driven attribution algorithm.
 2. Incrementality Testing, including holdout groups, geo-based testing, on/off testing, and pulse tests.
 3. Marginal Analysis of budgets - a critical but often overlooked aspect of decision-making.
 4. Media Mix Models (defined as the media or advertising channel spend mix) bring everything together, including a step-by-step explanation of all the intricacies involved. So much so, that you can build one yourself with vibe coding.
 5. A 6-month roadmap (and beyond) for action.

The Appendices include:

1. Appendix 1: Key insights from bid optimization, which, although valuable, didn't fit neatly into the main book.
2. Appendix 2: Measurement through surveys, a related but distinct topic in the broader context of Marketing Measurement.
3. Appendix 3: A Statistics overview, including A/B testing of landing pages and important considerations, for those interested in the underlying statistics.

This book is designed for growth and performance marketers who oversee budgeting, media spend optimization, and forecasting across multiple channels.

I'll explain attribution approaches, holdout and geo-based testing, and the mechanics of media mix models, including the machine learning behind them. No programming knowledge is required. While the book does cover technical aspects of machine learning (enabling analysts can dive deeper into the details), CMOs and VPs can skim the technical sections and still gain a solid understanding of the frameworks.

The prerequisites are:

- Basic familiarity with marketing and advertising channels
- Comfort with Excel-level analysis, including data analysis. If the math gets too involved, you can still skim those sections and understand the overall concepts.
- A role in budget allocation, ROI optimization, or forecasting

Many growth and performance marketers struggle to make informed budgeting and forecasting decisions, despite having access to a wealth of data. I've been in the same situation, having spent over 2 decades navigating these challenges. When I started, I didn't know how to allocate budgets effectively, even in a single marketing channel.

In this book, I'm sharing my experience, lessons learned, and applicable theory, along with a Roadmap for Action, to help you make the most of your data and optimize your marketing strategy.

Note on AI Usage

I used AI tools (ChatGPT, Google Gemini, and Claude) to help write this book. Here's how I used them:

1. AI generated all the visuals and graphs based on my prompts
2. I used AI to simplify certain sections where I thought I was rambling
3. In some cases, I had AI write short sections based on my guidance, and then I revised them

Overall, the content is about 75% mine and 25% credited to AI.

Thank you, AI!

A Reasonable Decision, Revisited

I want to share a story from the fourth quarter of 2012. I had just joined Shutterfly, a highly seasonal ecommerce business where the holidays make or break the year. Since I wasn't there for the previous holiday season, my view of the business was based on Q4 2011 data. I was in charge of paid search, and as I looked at the prior year's numbers, something stood out. According to the data, we had overspent on Non-brand paid search during the 2011 holiday period and actually lost money on the channel.



From my perspective, the conclusion seemed clear. If a channel loses money during the most important quarter of the year, you should reduce spending. The numbers were significant - the Q4 paid search budget was in the millions for October, November, and December. I suggested cutting about \$1 million from the prior year's Non-brand paid search budget, and the recommendation was well received, as people rarely suggesting cutting channel budgets. At the time, it felt like a responsible decision based on the data.

Today, I'm not so sure. The problem wasn't that I was looking at Q4 2011 data, given the business's seasonality. The issue was how I interpreted that data. My conclusions were based on one assumption: that our measurement approach accurately captured the value of paid search. All of this analysis was based on last-click attribution.

To make things more confusing, Brand paid search was very profitable, but we couldn't increase spending on it. We were limited by the fact that only so many people search for "Shutterfly". What I didn't fully appreciate at the time was what happens in between. What if a customer starts with a Non-brand search, doesn't convert, and then comes back later to search for the brand and convert? Under last-click attribution, Brand search gets all the credit, but that conversion might not have happened if Non-brand search hadn't introduced the customer to Shutterfly first.

So, does last-click attribution give us enough information to make a multi-million-dollar budgeting decision? That's the question this book explores from every angle.





Before moving on, I should explain the term "convert" that I used earlier. It refers to the action you want the user to take, usually a sale. It can also be the user filling out a form, becoming a lead, and doing what you wanted them to do - converting. When users convert, we call it a conversion, and revenue, profit, or lifetime value can often be assigned to each conversion.



When Measurement Rewards the Wrong Channels

What made this situation uncomfortable wasn't anything unique to Shutterfly. The details may change from company to company, but the underlying problem remains the same. One channel introduces a customer, and another channel closes the deal. With last-touch attribution, the final touch gets all the credit, while earlier touches are blamed for high costs.

Paid search makes this issue more visible. Brand search usually looks good because it targets users who already know what they want. It converts efficiently and produces clean metrics. In contrast, non-brand search often looks messy, with lower conversion rates, higher costs, and questionable profitability on paper.

However, this doesn't mean non-brand search (or Meta ads, as the case may be) isn't important. In many cases, non-brand search or Meta is the first meaningful interaction a customer has with a brand. It creates awareness, frames the problem, and gets the brand into the consideration set. Even if the conversion happens later through a different channel, the journey often starts with a non-brand search or Meta.



Last-click attribution ignores this sequence. It treats each conversion as if it happened independently, without considering what came before. Channels that operate earlier in the funnel are penalized as a result. As customer journeys become more complex, this distortion grows. (Furthermore, with the increasing challenges of cookies, discussed a bit later, not all your marketing spend is even trackable.)

Retargeting, for instance, often looks very efficient because it targets users who have already shown intent. But with last-click attribution, we're using a framework that prioritizes convenience over causality. It provides easy-to-understand numbers, but it may not be answering the right question.

The Three-Legged Stool

In 2012, we were working to understand how to credit channels based on attribution when cookies were more reliable. Today, we're still trying to understand how credit channels, but we're now looking at attribution, incremental testing, and media mix models.

For about ten years, marketers had it easy with "deterministic" tracking using cookies. But with new privacy regulations like GDPR and CCPA, Apple's ATT update for iOS 14 and later, and the increasing complexity of the customer journey, there's no longer a single source of truth.



If you're relying on just one way to measure success now, you're not getting the full picture. To get a clear understanding today, we need to use a combination of methods, which I call the Three-Legged Stool. Each approach has its own flaws, but together they provide a solid foundation for budgets of all sizes.

- Leg 1: Attribution (the tactical view) answers which ads are driving results. It's fast, but it can be "greedy" and take credit for sales that would have happened anyway.
- Leg 2: Incrementality (the causal truth) through holdout groups (and A/B testing) answers whether a sale would have happened without the ad spend. It's the best way to prove value, but it's slow and hard to run constantly across every channel.
- Leg 3: Media Mix Modeling (the holistic strategy) assesses the total impact of all spending across all dimensions, including offline and external factors. It sees the big picture that attribution might miss, but it needs a lot of historical data and calibration.

This book aims to teach you how to build all three approaches and, more importantly, how to make decisions when they don't agree.

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Attribution Models



Attribution models are used to answer a simple question: how to assign credit for a conversion across multiple marketing touchpoints. Different models give different answers because they're based on different assumptions about customer behavior.

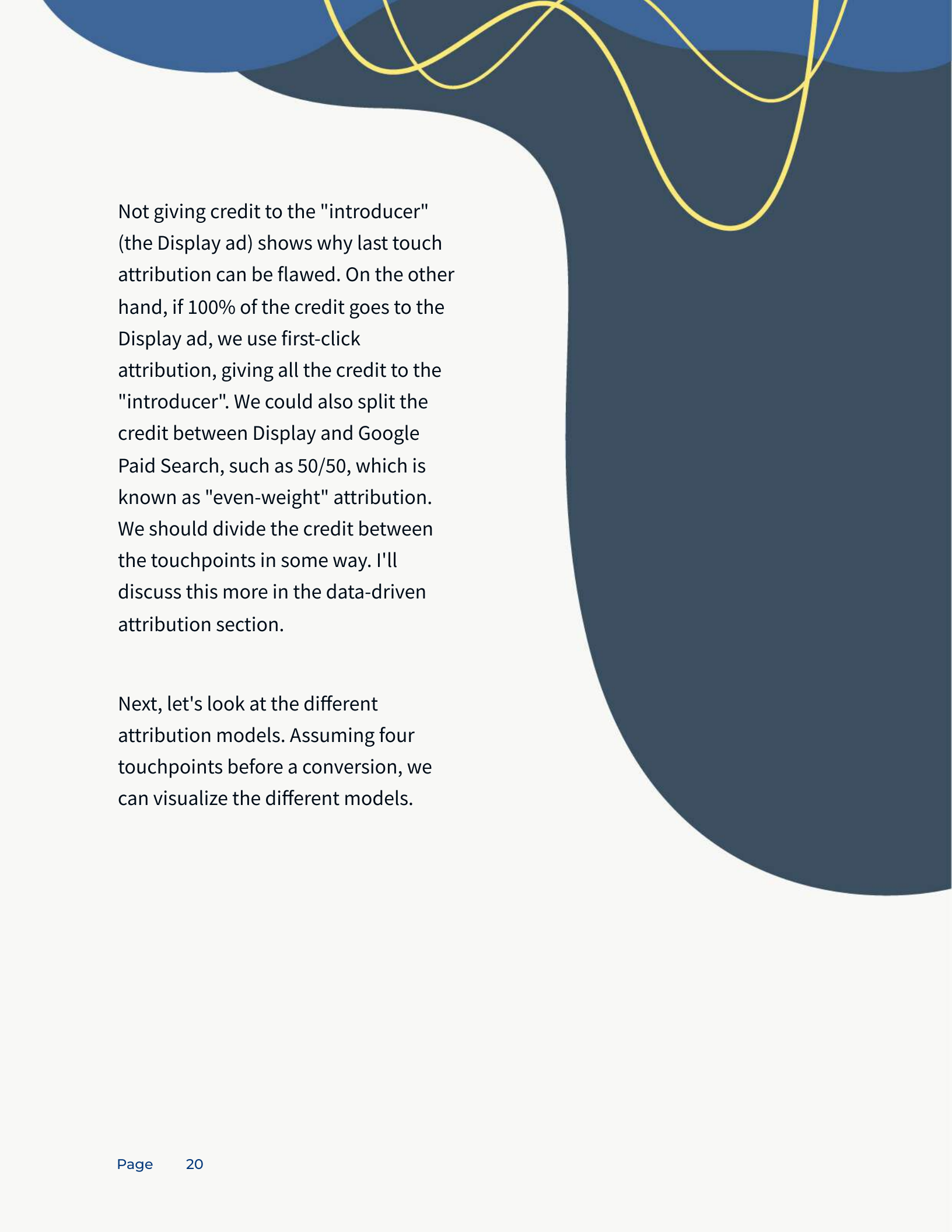
Let's consider an example. Imagine a user sees a Display Ad from Dell, clicks it, and visits the Dell website, but doesn't buy anything. The next day, the same user searches for "Dell" on Google, clicks the Dell ad, visits the website, and buys a computer. In this scenario, there are two marketing touchpoints: Display and Paid Search. Both could have contributed to the conversion. The key question is: how much credit should each touchpoint get for the conversion?

This is important because it affects budgeting and optimization decisions. If we give 100% of the credit to Paid Search and 0% to Display, Paid Search will seem to be driving most of our business, and Display will seem to be driving none. We'd likely reduce our Display budget and lower our Display ad bids. But if we split the credit 50 / 50 between Display and Paid Search, our decisions would be different.



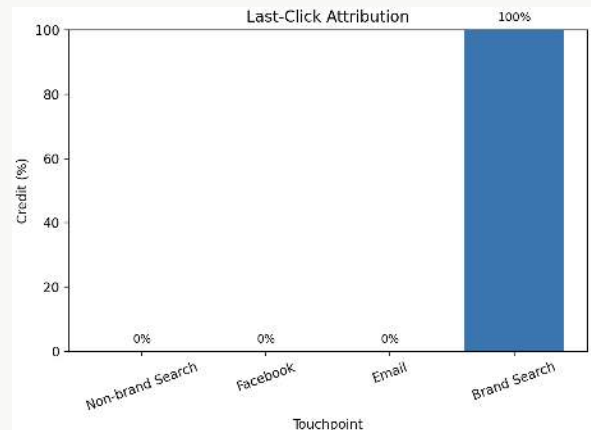
Using this example, if 100% of the credit goes to Google Paid Search, we're using last click attribution. This means we're giving all the credit to the channel that closed the sale. But shouldn't the first click from the Display ad be credited? It could be argued that the first click deserves most of the credit, since it led the user to interact with the website and eventually search for "Dell" on Google. The subsequent search was likely just for convenience, as the user already knew where they wanted to go. Last-touch attribution would give full credit to the last touch, which in this case is the second. Google Paid Search would get full credit for the conversion, even though it might have happened anyway.





Not giving credit to the "introducer" (the Display ad) shows why last touch attribution can be flawed. On the other hand, if 100% of the credit goes to the Display ad, we use first-click attribution, giving all the credit to the "introducer". We could also split the credit between Display and Google Paid Search, such as 50/50, which is known as "even-weight" attribution. We should divide the credit between the touchpoints in some way. I'll discuss this more in the data-driven attribution section.

Next, let's look at the different attribution models. Assuming four touchpoints before a conversion, we can visualize the different models.

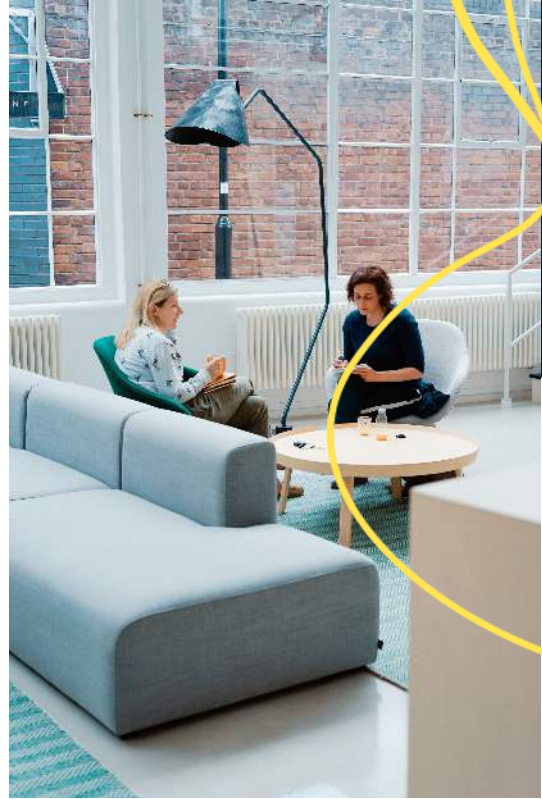
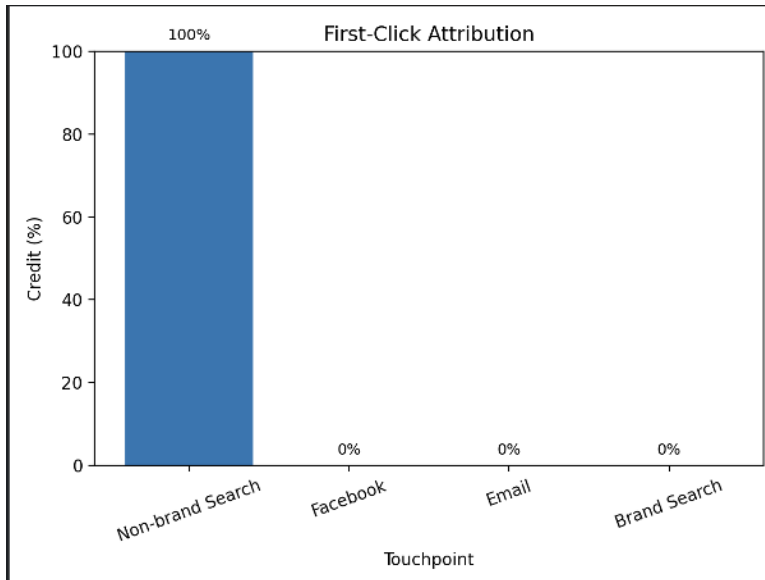


Last-click attribution assigns 100% of the credit to the final touchpoint before conversion.

This model is popular because it is simple, intuitive, and operationally convenient. If a user clicks an ad and then converts, that ad “worked.” No further interpretation is required.

Implicit assumption: the final touch alone caused the conversion.

First-Click Attribution



First-click attribution assigns all credit to the first touchpoint in the customer journey.

This model is often used to highlight discovery channels and upper-funnel activity. It emphasizes introduction over closure.

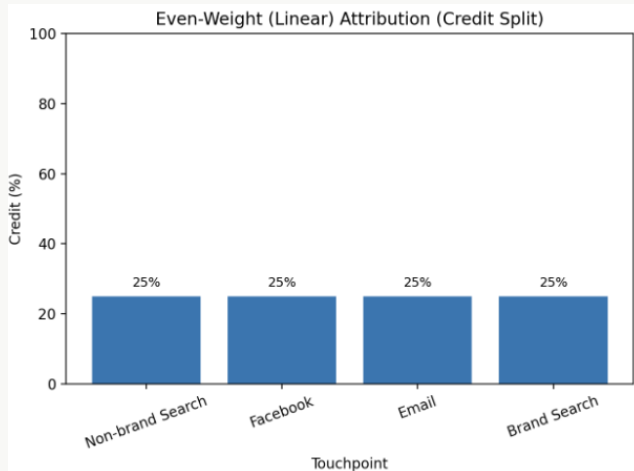
Implicit assumption: the first exposure alone created the customer, and all subsequent touches were irrelevant.

Even-weight attribution splits credit evenly across all touchpoints in the conversion path.

If four touches occurred, each receives 25% of the credit. The model attempts to be fair by construction.

Implicit assumption: all touches matter equally.

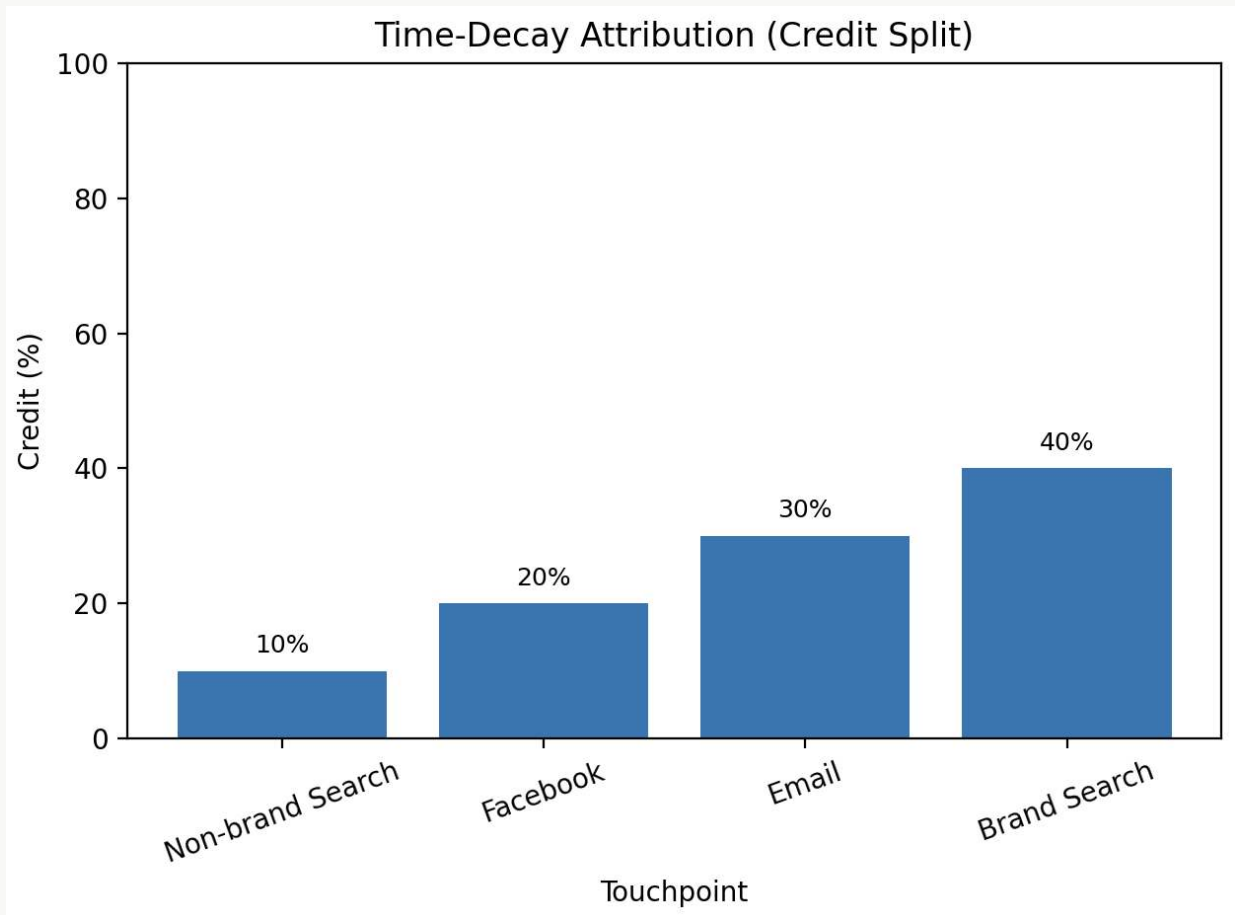
All touches mattering equally is rarely true, but in the absence of additional information, it is not unreasonable, which is why this is my second-favorite attribution approach.



Time-Decay Attribution

This model says that older touch points have less impact because people tend to forget them. For example, if someone searched and clicked a Google non brand ad a month ago and converted today, that ad probably didn't have much influence on the conversion, since it happened so long ago.



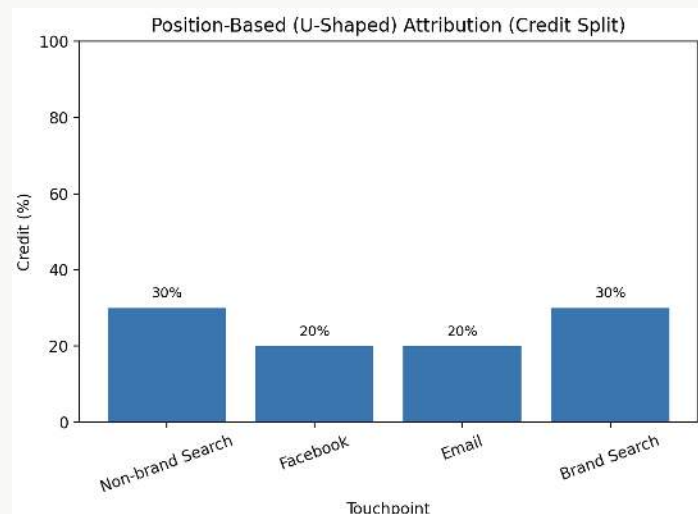


So time-decay attribution assigns more credit to touchpoints that occur closer to the conversion. The idea is simple: what happens recently matters more. What happened yesterday is more likely to influence something than what happened last month. The assumption is that influence grows stronger as the conversion approaches. For instance, if there were four touches before a conversion, the first touch might get 10% of the credit, the second 20%, the third 30%, and the fourth 40%. The credits add up to 100% and get bigger as the conversion approaches.

Time Decay with half-life

Time Decay attribution is often modeled using the concept of half-life. For example, a half-life of seven days means a click from seven days ago gets half the credit of a click from today, assuming both happen before the conversion. A click from 14 days ago would get 25% of the credit for a click from today, assuming both clicks occur before the conversion.

To illustrate, if a click was received 14 days ago, another 7 days ago, and a third today, and a conversion followed, the credits can be calculated. The first click would get $2/14$ of the credit, the second click $4/14$, and the click from today $8/14$. This sums to 1, which is correct since the total credit should be 100%, and it follows the Time Decay model with a half-life of 7 days.



Position-based models give more weight to the first and last touchpoints, with the remaining credit shared among the middle touchpoints.

The idea is that the first click introduces the user, and the last click seals the deal. The middle clicks have some influence on conversion, but not as much as the first and last clicks. For example, with four clicks before a conversion, the credit might be split like this: 30% for the first click, 20% for the second, 20% for the third, and 30% for the fourth. This model can be customized to fit different needs.

Position-Based (U-Shaped) Attribution

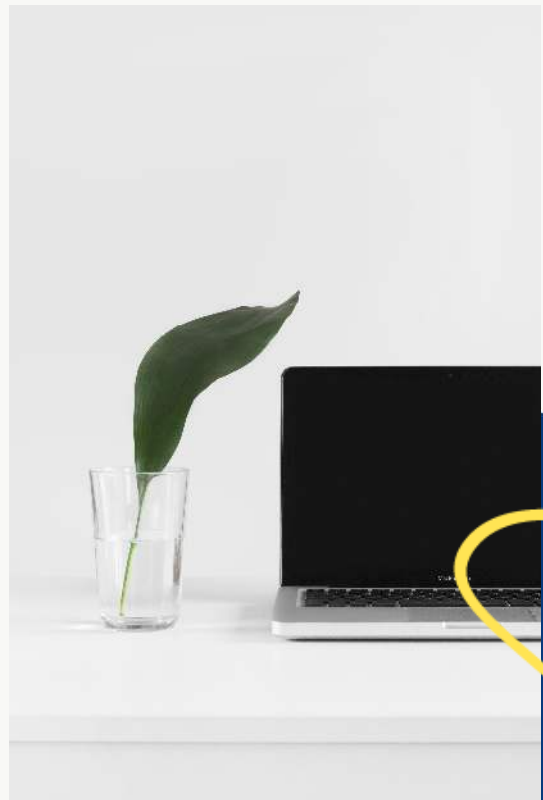
These models aim to balance the initial introduction and the final conversion, while accounting for interactions in between.

The basic assumption is that the introduction and closure are the most important parts.

Customized Attribution

You can also fully customize the attribution model you want to use. For example, you can decide that Paid Search is twice as important as other channels. This raises the question of what you would base this decision on. Any attribution model, such as last-click or first-click, poses the same question: "Why choose this model?" "What about your business makes it the right choice?"

From this perspective, the "even weight" attribution model seems fair. Since it's unclear which clicks deserve more or less credit, it simply splits the credit evenly.



Is there a better approach? This leads us to data-driven attribution, which we'll discuss after covering lookback windows.

The lookback window determines how many days to look back from the conversion date to find all the clicks that influenced the conversion. This window can usually be set to any period up to 90 days. Most businesses use a 30-day (or 28-day) window, assuming that touches more than 30 days ago are unlikely to influence conversions. However, the 30-day limit is arbitrary, and as long as we can measure the touch's influence, the window can be set as long as possible.

Before moving on to Data Driven Attribution, it's worth noting that there are two types of touches: Click and View. When we say "touch", we usually mean a marketing touch that brings a user to the website, such as clicking on a paid search ad. However, a "touch" can also be a user seeing an ad without clicking, like a Display ad or a Meta ad.

Can these ads influence a user to purchase later, even if they don't click on them? The answer is yes. However, the influence is generally less than if the user had clicked on the ad and visited the website.





Lookback Windows in Attribution Models

In Incrementality testing (discussed later), it's essential to measure whether ads seen but not clicked have an influence, and, if so, to what extent.

Data-Driven Attribution

Now that I've discussed various attribution models, which one should we use, and how do we decide? Is there a way to make this decision for us?

Choosing an attribution model can become a matter of personal preference, guided by some intuition. Trying different models can yield different insights into each channel's efficiency and its impact on your business. So, if you decide on a particular model, make sure to do your due diligence on why you're choosing it.

In some cases, a better approach called Data-Driven Attribution assigns a machine learning algorithm weights to each touchpoint before a conversion. This sounds great, but it has its limitations. For one, it might not be readily available on your platform, or it might not be free.

Let's take a closer look at how these machine learning algorithms work. They analyze all the touches that lead to a conversion and those that don't. Then they compare the two and assign weights to determine how much credit each touch should receive for a conversion.



For example, if a Paid Search click always precedes a conversion and never precedes a non-conversion, the algorithm will give Paid Search a lot of credit. On the other hand, if a Paid Search touch comes before both conversions and non-conversions, it might not be as valuable. After all, many people who saw or clicked the Paid Search ad didn't convert.

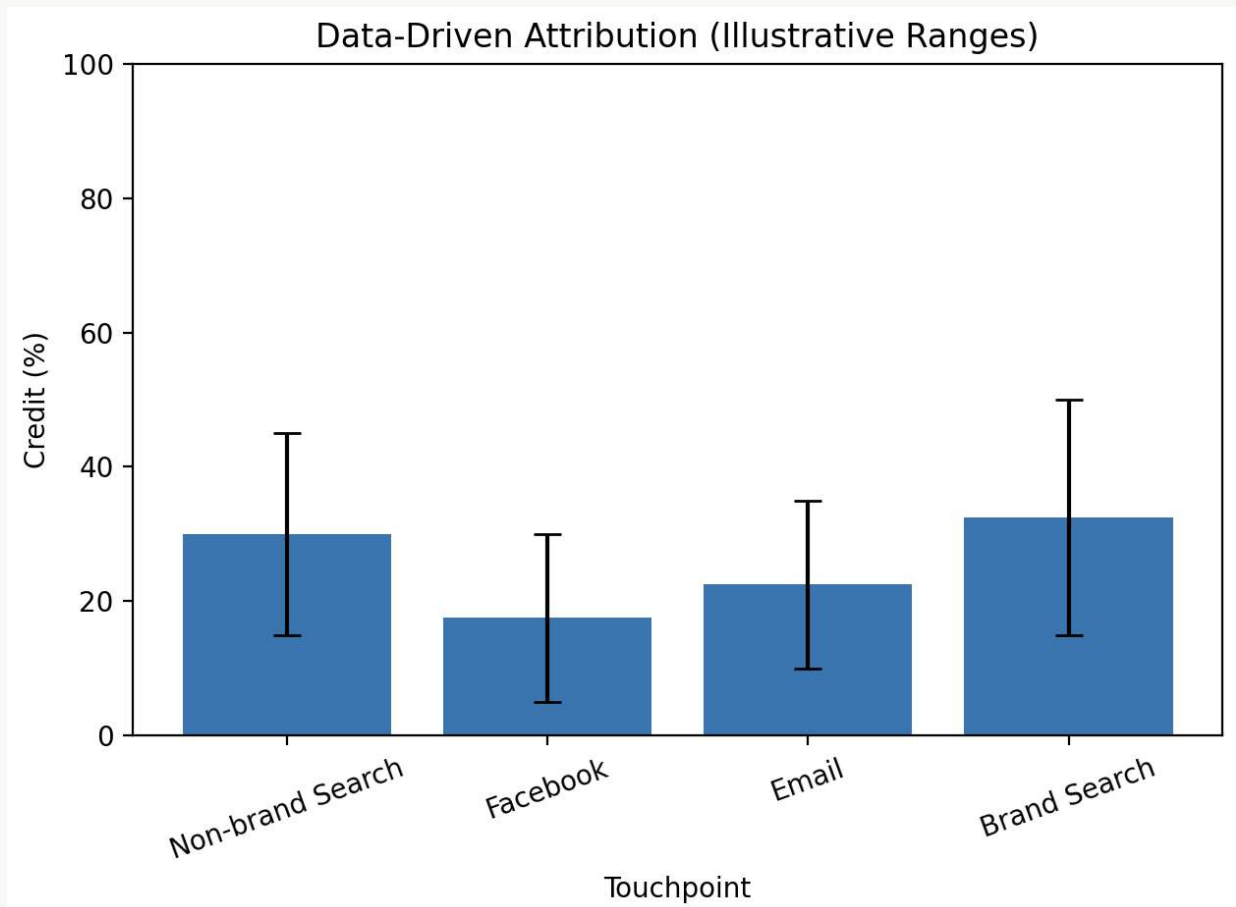
This approach can be improved by considering not just the touches, but also their order. For instance, the algorithm might find that a Display touch is very influential when it precedes a Paid Search touch, but not when it follows. Currently, data-driven attribution systems like this are the best, but they need data to work well.

One downside is that, while Data-Driven Attribution might be more accurate than other models, its results can be hard to explain. For instance, last-click attribution might not assign the right credit, but it's easy to understand. With Data-Driven Attribution, it can be tough to explain the numbers because it gives partial credit to touches based on a machine-learning algorithm.

The biggest limitation of Data Driven Attribution, and all attribution models, is that they can't always distinguish between causation and correlation. For example, the way we described Data-Driven Attribution makes it clear that we can identify channels that are highly correlated with conversions. We assign high value to channels that tend to precede conversions, not those that precede non-conversions. However, do these channels actually cause conversions, or do they just correlate with them? That's what we're trying to answer.

Consider the Affiliate channel. Affiliates are often paid on a revenue-share model, so they're motivated to show ads to users who are likely to convert. If they have their pixels on your website, they can get good at predicting who will convert and show ads to those users. As a result, their ads will be highly correlated with conversions, but they might not be causing them. Instead, they're just showing ads to users who were already likely to convert.

Despite this limitation, Data Driven Attribution is the most accurate attribution system when implemented correctly.





The data-driven attribution visual is shown above. This approach is the most sophisticated in principle, but in practice, it has its limitations. Although it is arguably better than other methods, it only shows part of the picture because it relies on observational data and lacks a true counterfactual. I will discuss these limitations in more detail later.

This is my preferred attribution model, not because it completely solves the problem, but because it is likely closer to the solution than other models.

Note that the range bars appear only in the data-driven attribution visual - we will revisit this point later.

Despite their differences, all attribution models share one key feature. They assign credit, rather than causality. In other words, attribution models describe how conversions are linked to touchpoints, but they do not tell us what would happen if a touchpoint were removed. This distinction is subtle yet a crucial flaw in attribution models.

Cookie deletion and cross-device

Attribution models also struggle with incomplete data. They rely heavily on cookies, which can be deleted or blocked. This means that user journeys across devices and browsers are often fragmented or invisible. As a result, credit is only given to touchpoints that appear on the same device and browser as the conversion.



You may remember that Apple's iOS 14 update, specifically version 14.5 and later, introduced App Tracking Transparency. This required apps to ask users for permission to track their data. Most users opted out, which reduced Meta's ability to track conversions. This meant that advertisers had to rely more on modeled data and Meta's Conversions API.

Cookie deletion and expiration, as well as the use of multiple devices and browsers, affect any kind of testing that uses cookies. This includes tests like Incrementality and Holdout, which will be described later. However, tests that use email to create control and treatment groups, like those in Meta, can still work because they rely less (or not at all) on cookies.

When users are on multiple devices, they may see an ad on their mobile device, click on it, but not make a purchase. Later, on their desktop device, they may visit the website directly and make a purchase. Because cookies are specific to each device and browser, it is hard to connect the desktop purchase to the mobile ad. This means that the value of mobile ads will be underrepresented in Attribution systems used by bid management tools.

In general, cookie deletion leads to under-representation of revenue or conversions from advertising. This can lead to underestimating the value of advertising, resulting in lower bids and budgets for ads if we rely solely on Attribution systems. However, Media Mix Models, which will be described later, do not rely on cookies and are not affected by this issue.



Using user IDs, such as email addresses, can help address cookie issues by enabling tracking across devices. Since users often use the same email address on all their devices, this can improve user detection and tracking. One reason why holdout testing on Meta can be accurate is that it uses email addresses and logins to track users across devices.

03

Deeper Dive into Data-Driven Attribution



I first learned about data-driven attribution in 2017. A data scientist from Shutterstock explained it to me like this: "It uses machine learning to calculate the chances of conversion after someone interacts with a channel, and the chances of not converting after the same interaction. Then, the machine learning looks at the differences in these chances to figure out how much credit each channel deserves."

This idea really interested me, especially the part about machine learning analyzing the likelihood of people not converting and comparing these across different channels to see what's working best.



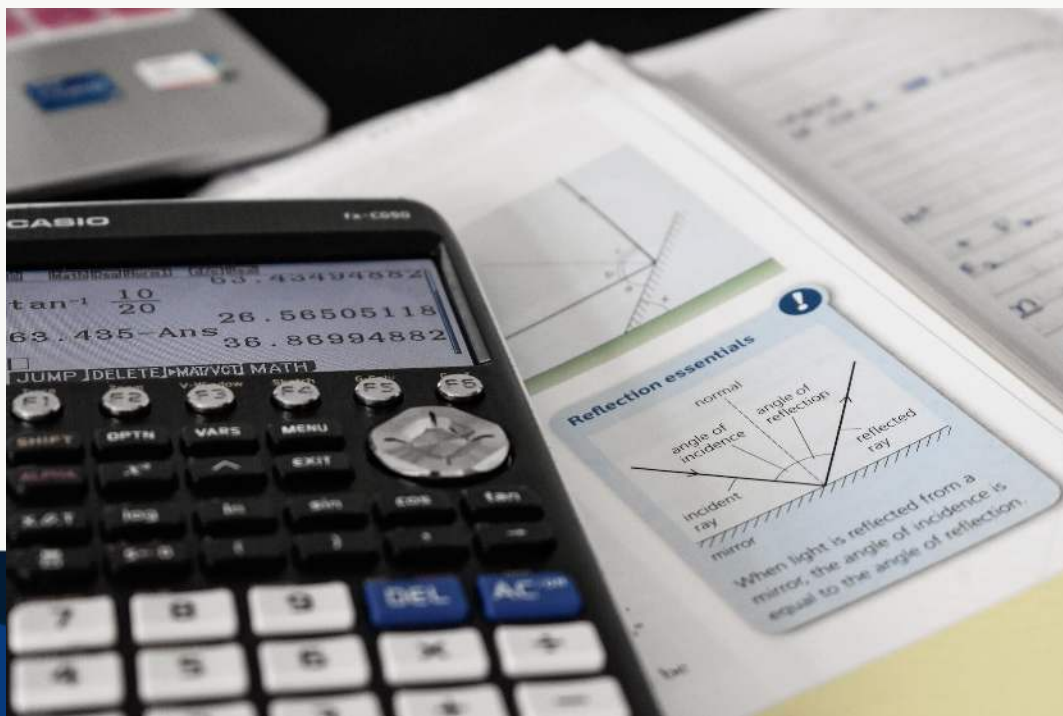
Let's take a closer look.

Markov Chain Attribution (Intuition, with a little Math)

There are several "data-driven" methods for assigning credit across touchpoints, including Shapley values, regression-based approaches, and Markov chains. These methods are "data-driven" because they don't rely on arbitrary rules, such as giving all the credit to the first or last touch. Instead, they use actual customer journey data to determine which channels drive conversions.

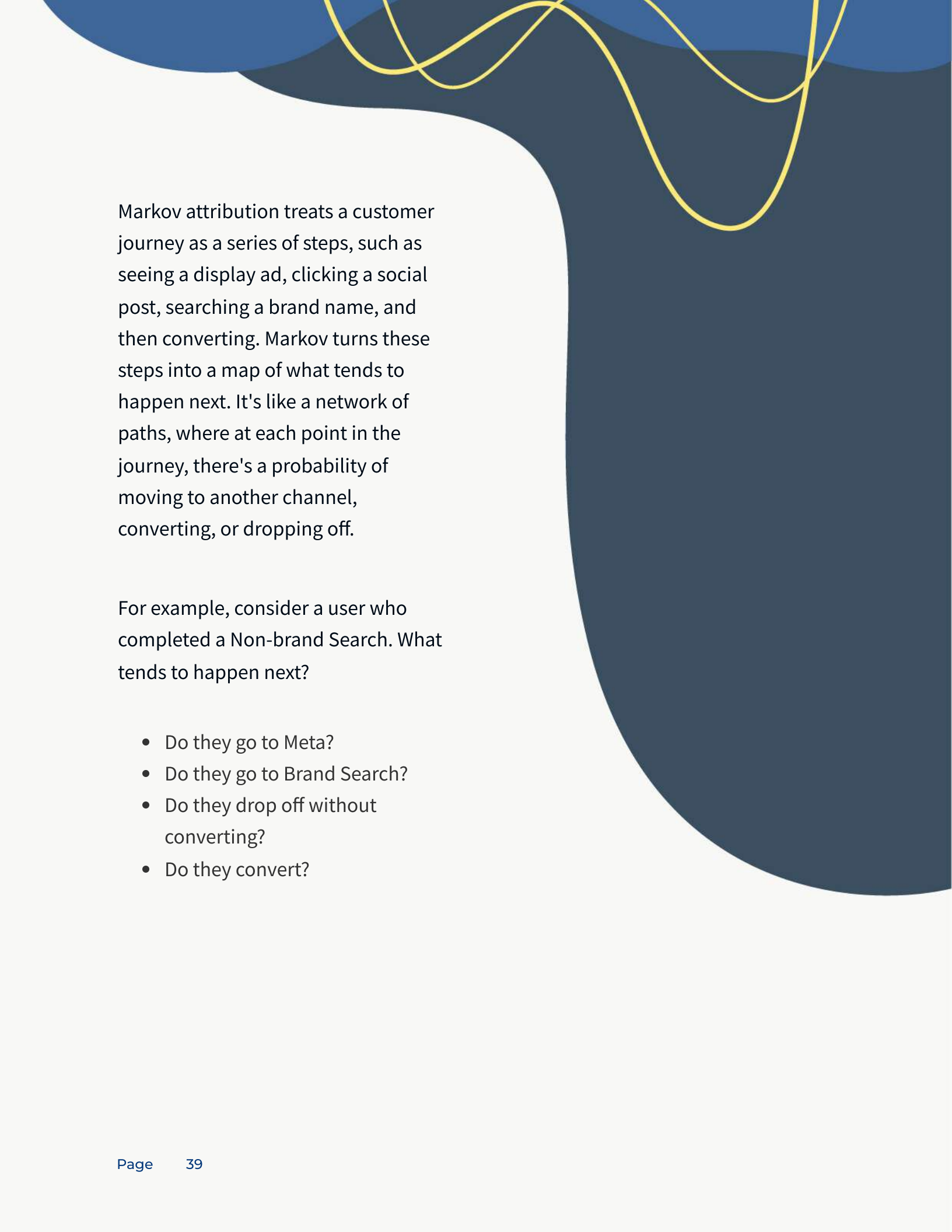
Each method has its own approach. The Shapley value, from game theory, considers all possible combinations of channels and measures each channel's contribution to the outcome. Regression-based approaches use statistical models to estimate the relationship between exposure to different channels and the likelihood of converting. Markov chains take a different approach, modeling the customer journey as a sequence of states and transitions that map how people move from one touchpoint to the next.

In this section, I'll focus on Markov chain attribution because it's a straightforward way to introduce two important ideas:



1. **Sequence matters** - the order in which someone encounters your channels can affect whether they convert. For example, seeing a social ad before a search ad might produce a different outcome than seeing them in reverse order.
2. **Credit can be defined through a thought experiment** - instead of asking "which channel was most important?", Markov attribution asks "what would happen if a channel disappeared completely?" The resulting drop in conversions shows how much that channel was contributing. This idea, known as the removal effect, provides a principled way to measure each channel's value.





Markov attribution treats a customer journey as a series of steps, such as seeing a display ad, clicking a social post, searching a brand name, and then converting. Markov turns these steps into a map of what tends to happen next. It's like a network of paths, where at each point in the journey, there's a probability of moving to another channel, converting, or dropping off.

For example, consider a user who completed a Non-brand Search. What tends to happen next?

- Do they go to Meta?
- Do they go to Brand Search?
- Do they drop off without converting?
- Do they convert?



Markov attribution learns these "next step" tendencies from observed journeys and updates the transition probabilities as more journeys become available.

How Markov Produces Credit ("Channel Removal Thought Experiment")

Once you have a "journey map," Markov attribution asks:

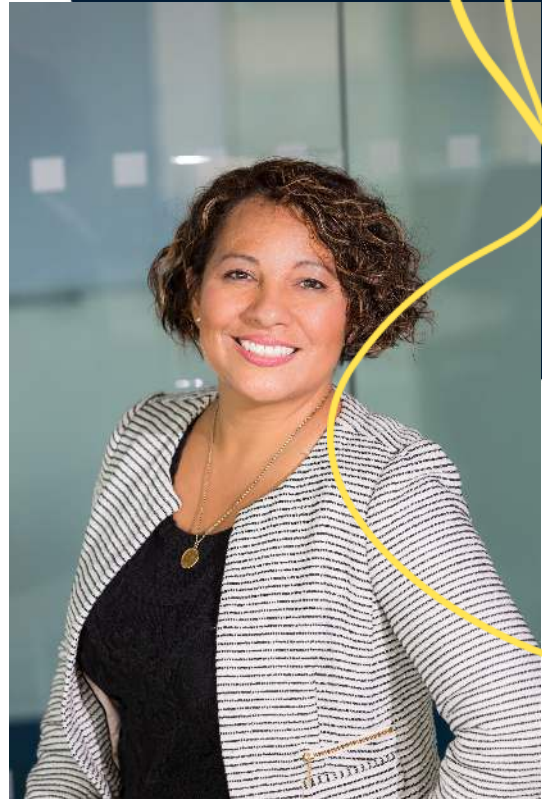
- If Brand Search were removed, what would happen to the conversion rate?
- If Non-brand Search were removed, what would happen to the conversion rate?
- If Meta were removed, what would happen to the conversion rate?
- If Email were removed, what would happen to the conversion rate?

You remove one channel from the journey system and recalculate the conversion probability. The "lost conversions" that result from this removal are the channel's credit.

So, Markov credit is determined by:

How much does the conversion probability drop if this channel is removed from the system of observed journeys?

To illustrate this, consider a "Markov transition matrix" for an e-commerce toy site. A "Conversion" means the user buys a toy.



Markov Transition Matrix (Toy Example)

From \ To	Start	SB	Meta	EM	BR	Conversion	Non-conversion
Start	0.000	0.500	0.250	0.125	0.125	0.000	0.000
SB	0.500	0.000	0.750	0.000	0.500	0.000	0.750
Meta	0.000	0.000	0.000	0.000	0.667	0.000	0.333
EM	0.000	0.000	0.000	0.000	1.000	0.000	0.000
BR	0.000	0.000	0.000	0.000	0.000	0.000	0.200
Conversion	0.000	0.000	0.000	0.000	0.000	1.000	0.000
Non-conversion	0.000	0.000	0.000	0.000	0.000	0.000	1.000

First, note that each row sums to 1. Once you're in a state, you must transition to another channel, conversion, or non-conversion.

From the Start row, the user's first touch is:

- Non-brand Search with a probability of 0.50
- Meta with a probability of 0.25
- Email with a probability of 0.125
- Brand Search with a probability of 0.125

These probabilities sum to 1.0.

Now look at the Meta row. After a Meta touch, the next step is:

- Brand Search with a probability of 0.667
- Non-conversion with a probability of 0.333 (this is important because it also shows the probability of a channel leading to a "failure path", which is necessary for the model to accurately calculate the "true value" of a touch point)

Again, this row sums to 1.0.





A key point to note is that sequencing is built into the model. "Meta followed by Non-brand Search" is a different transition from "Non-brand Search followed by Meta." In our matrix, Meta to Non-brand Search occurs with probability 0.000, while Non-brand Search to Meta occurs with probability 0.250. The model automatically distinguishes between the two.

If you solve the implied system of equations from the matrix, this matrix implies:

- The probability of Conversion from Start is 0.60
- The probability of Non-conversion from Start is 0.40

(Optional) Solving the System of Equations

Let $P(S)$ denote the probability of eventually reaching Conversion, starting from state S . By definition:

- $P(\text{Conversion}) = 1$
- $P(\text{Non-Conversion}) = 0$

Now we solve for the remaining states, working from the transition matrix:

$P(\text{Brand}) = 0.8$ (read directly from the matrix — 80% of Brand Search visits lead to Conversion).

$P(\text{Email}) = 1 \times P(\text{Brand}) = 0.8$, since every Email touchpoint transitions to Brand Search.

$$P(\text{Meta}) = 0.67 \times P(\text{Brand}) + 0.33 \times P(\text{Non-Conversion}) = 0.67 \times 0.8 + 0.33 \times 0 = 0.533$$

$$P(\text{Non-Brand}) = 0.25 \times P(\text{Meta}) + 0.5 \times P(\text{Brand}) + 0.25 \times P(\text{Non-Conversion}) = 0.25(0.533) + 0.5(0.8) + 0 = 0.533$$

$$P(\text{Start}) = 0.5 \times P(\text{Non-Brand}) + 0.25 \times P(\text{Meta}) + 0.125 \times P(\text{Email}) + 0.125 \times P(\text{Brand}) = 0.5(0.533) + 0.25(0.533) + 0.125(0.8) + 0.125(0.8) = 0.6$$

So the baseline probability of conversion from the Start state is 60%.

This is our baseline.

(Side Technical Note: The Standard Markov Chain used above is "memoryless," i.e., the model assumes the next step depends only on the current one. In reality, marketing often has "memory" (a user's conversion probability after the 5th touchpoint of Display is influenced by their specific 1st through 4th touchpoints). While we use First-Order chains here for simplicity, advanced DDA platforms often use Higher-Order Markov Chains. This involves redefining the "states" to account for the sequence of previous touches, providing a much more nuanced view of the customer journey.)

Now, let's run a thought experiment. Suppose we remove Non-brand Search as a possible state. We need to assume what happens to journeys that would have flowed into Non-brand Search.

A simple and easy-to-explain rule is:

Any transition that would have gone to the removed channel is treated as a Non-conversion.

This is not the only possible assumption, but it has two advantages:

- It avoids "free rerouting" into better channels.
- We apply the same rule to every channel removal to keep a level playing field.

Under this rule, the modified matrix becomes:

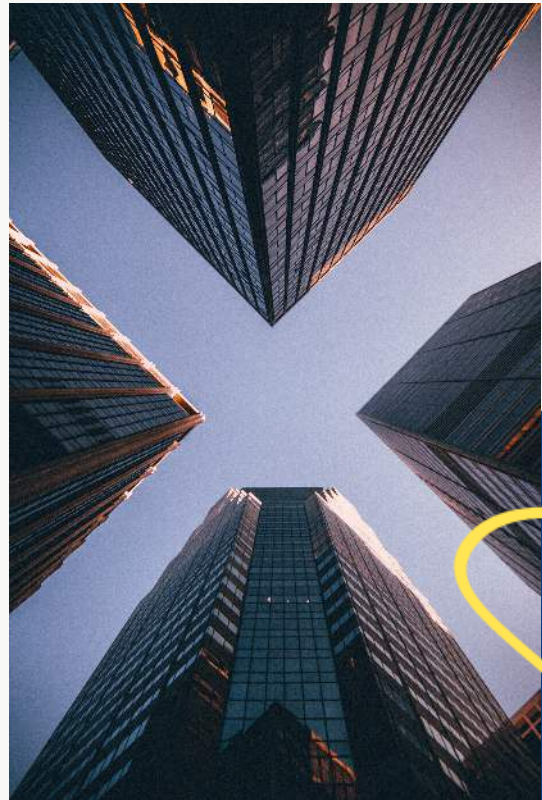
Markov Transition Matrix (NB Removed)						
From \ To	Start	FB	EM	BR	Conversion	Non-conversion
Start	0.000	0.250	0.125	0.125	0.000	0.500
FB	0.000	0.000	0.000	0.667	0.000	0.333
EM	0.000	0.000	0.000	1.000	0.000	0.000
BR	0.000	0.000	0.000	0.000	0.800	0.200
Conversion	0.000	0.000	0.000	0.000	1.000	0.000
Non-conversion	0.000	0.000	0.000	0.000	0.000	1.000

Solving the equations again, the probability of conversion drops to:

New probability of Conversion: 0.333

So, removing Non-brand Search reduced the conversion probability from 0.600 to 0.333, a drop of 0.267.

This drop is the "removal effect" for Non-brand Search.





Repeat the same exercise for each channel, one at a time. In this example:

- Removing Non-brand Search reduces conversion probability by 0.267
- Removing Meta reduces conversion probability by 0.200
- Removing Email reduces conversion probability by 0.100
- Removing Brand Search reduces conversion probability by 0.600

We convert these probability drop events into credit shares by normalizing them to sum to 1.

The ratios are:

0.267

0.200

0.100

0.600

Normalized (reducing in the same ratios so that they sum to 1), that becomes:

- NB gets 8/35 of the credit
- FB gets 6/35 of the credit.
- EM gets 3/35 of the credit.
- BR gets 18/35 of the credit.



Above sums to 1 (100%), as expected.

(Details skipped, but the logic is just “divide each drop by the total drop.”)

So if we were going to split credit for 100 conversions,

NB would get credit for $8/35 * 100 = 22.86$ conversions

FB would get credit for $6/35 * 100 = 17.14$ conversions

EM would get credit for $3/35 * 100 = 8.57$ conversions and

BR would get credit for $18/35 * 100 = 51.43$ conversions

A key reminder: this is system-level allocation, not a per-user explanation of any single conversion path.

How Markov Produces Ranges (95% Confidence Intervals)

Earlier, data-driven attribution was the only click-path approach where we showed ranges, not just point estimates.



The entries in the transition matrix are estimated from data. “NB→BR = 0.50” is what we inferred from the journeys we happened to observe. With more data, that estimate could shift, e.g., 0.45, 0.55, or 0.60.

Once you accept that transition probabilities are estimates, everything downstream becomes an estimate too:

- baseline conversion probability (0.60 here)
- removal effects (0.267, 0.200, ...)
- and therefore, channel credit too is an estimate

The practical way to quantify this is to repeatedly resample journeys (a bootstrap), rebuild the matrix, and recompute the credits. The spread of those credits becomes a 95% interval.

And you should expect:

- channels with lots of data → tighter ranges
- sparse channels → wider ranges

These intervals describe the model's uncertainty, given limited path data.



How Markov Can Use Priors (Holdouts)

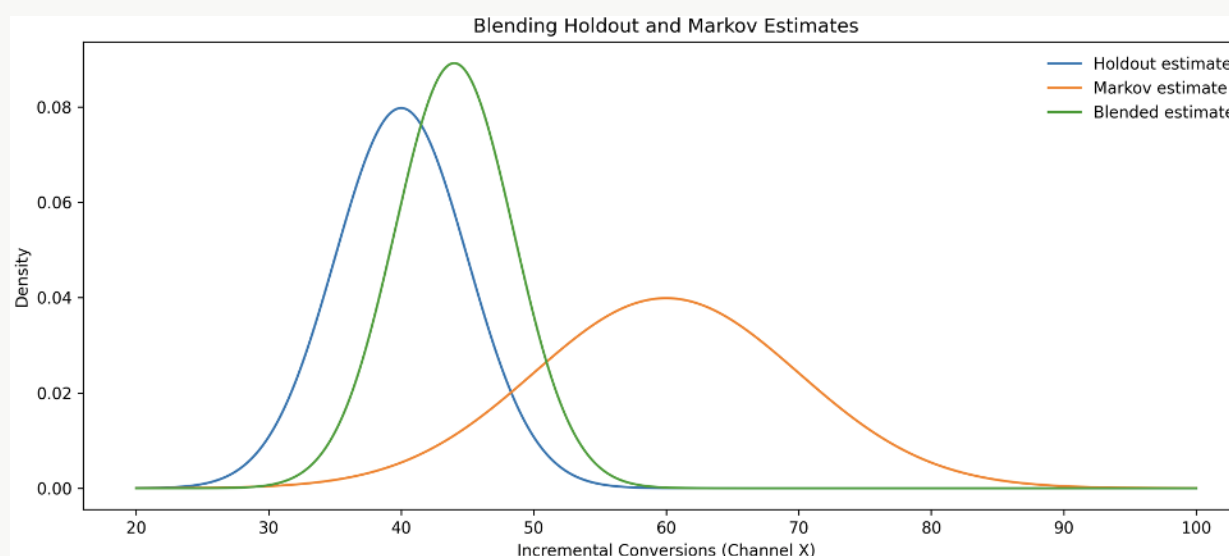
The beauty of data-driven attribution is that you can also include prior information, such as holdout-test results for a channel. This is something you can't do with any attribution model other than data-driven. To start, let's define the difference between Holdout and Markov:

- Holdout tests answer the question "What is the incremental lift of channel X?" which shows cause and effect.
- Markov answers the question "How does credit flow through observed paths?" which is more about the sequence of events.

In practice, the best approach is to use Markov to determine how credit is distributed across journeys, and use holdouts to determine the level of lift.

For example, suppose Markov suggests Channel X deserves credit for 60 conversions, but due to uncertainty, you might have it as " 60 ± 10 conversions". This uncertainty can be measured using a standard deviation calculated from the data. However, a holdout test over the same period might estimate the incremental lift of Channel X as 40 conversions, and including the standard deviations, this might be " 40 ± 5 conversions" due to uncertainty.

A reasonable step is to combine these two estimates, weighting them by your confidence in each. The holdout test is usually a stronger anchor because it's designed to estimate what would have happened if Channel X hadn't existed. So, the combined estimate might be around 48 conversions, with its own uncertainty. You would then reduce Channel X's credited conversions from 60 to 48 and redistribute the remaining credit to other channels proportionally, so the totals still add up. The resulting estimate is a range, not a single point; in this case, a bell-shaped curve combining the prior Holdout estimate and the Markov estimate. It's worth noting that all these estimates are bell-shaped curves, representing confidence intervals such as 95%.



One important note: many analytics platforms have "data-driven attribution" built in, but they usually don't let you add external information, like holdout lift distributions, directly into their attribution model. This blending often happens outside of the platform, which is where the art of using the data comes in, building on the science behind it.

04

What Attribution Still Can't Tell You: Incrementality

Attribution vs. Incrementality

I've already discussed incrementality a bit when explaining data-driven attribution. Now, let's take a closer look at what it is and what attribution can and can't tell us.

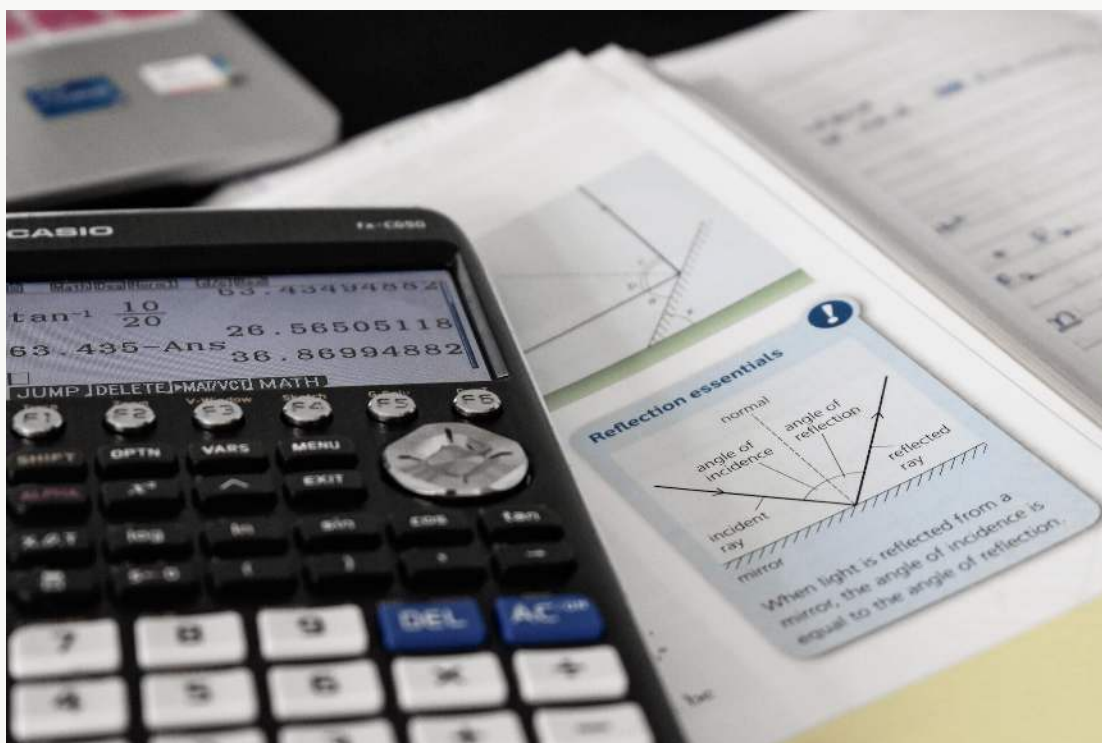
No matter what type of attribution model you use - last-click, first-click, time-decay, or data-driven Markov approaches - it can't give a precise answer to the question of incrementality.

It's helpful to step back and think about why attribution matters. Usually, it's to support a claim like this: "We spent \$100K on Meta ads and got 2,000 new customers. That's \$50 per customer, which is profitable for us. So, Meta ads are profitable." But there's an assumption behind that conclusion: without Meta ads, we wouldn't have gotten those 2,000 customers. That assumption is what we mean by incrementality. We're saying that Meta ads brought in customers who wouldn't have come otherwise. But the number "2,000" comes from an attribution model. We're using attribution results as a substitute for incrementality. Attribution analyzes conversions that have already occurred and determines how to credit each touchpoint. But the question we really care about is different: would this conversion have happened even if we didn't do anything? If the answer is no, then the conversion is incremental.





If the answer is yes, then getting attribution credit doesn't mean it added any real value. Incrementality is a question about what would have happened if things were different. It asks what would have happened if a certain channel hadn't been used. Attribution models try to approximate this by looking for patterns in the data we do have. This can be helpful ; data-driven attribution is the closest we can get, but it's still just an estimate.



A simple example

Consider Brand Search, which often seems extremely profitable in last-click attribution. Brand Search is usually the last step before a purchase, with high conversion rates and a low Cost per Conversion. But consider this: what if our Brand Search ad wasn't there? People searching for our brand are already looking for us, so if they don't see our ad, many will click on the organic listing or go straight to our site and convert anyway. This means some of those conversions would have happened regardless. Last-click attribution gives the Brand Search ad full credit for the conversion, simply because it was clicked right before.



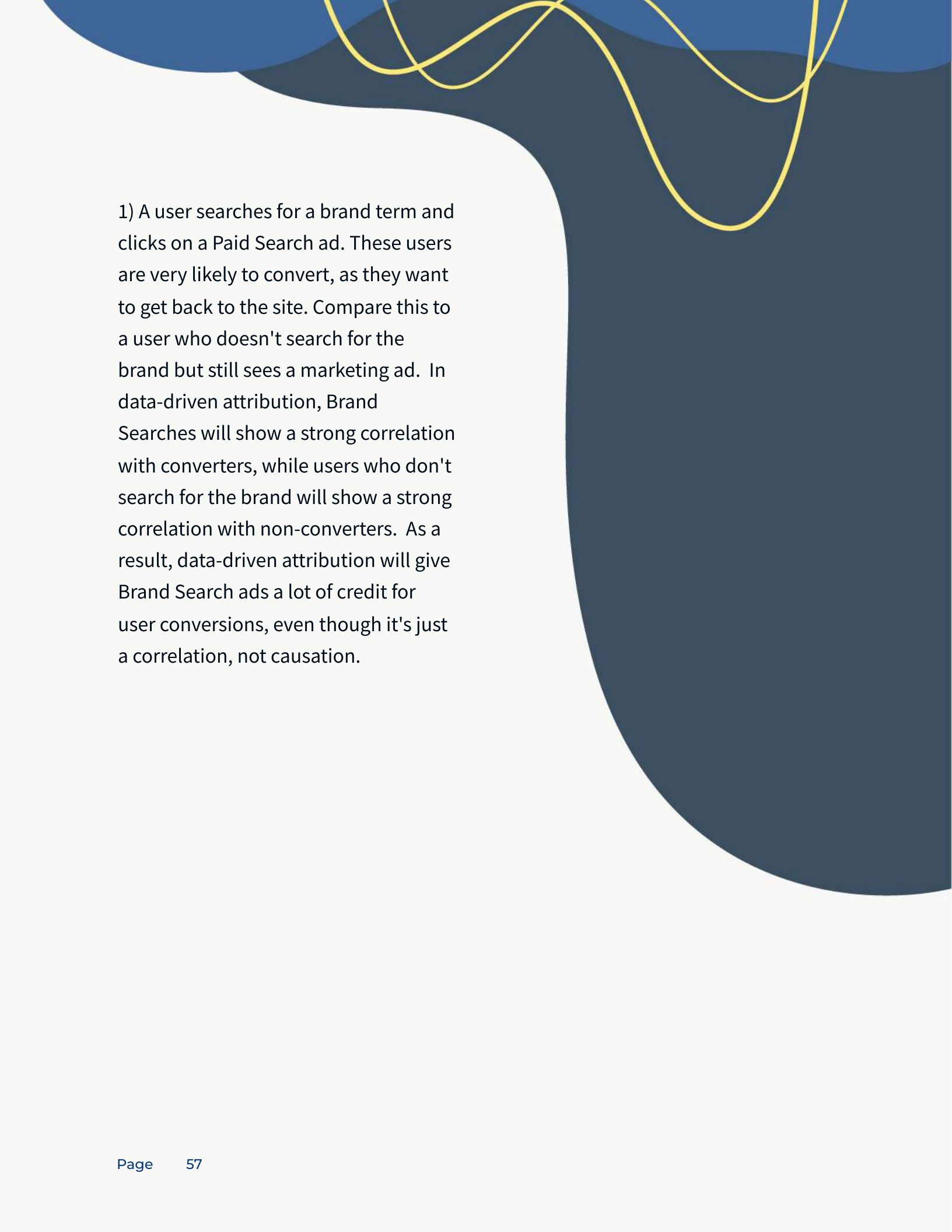
Attribution isn't "wrong" here - it's just answering the question it's designed to answer: which touchpoint is associated with the conversions we see?

Incrementality asks a different question: which touchpoint actually caused conversions that wouldn't have happened otherwise?

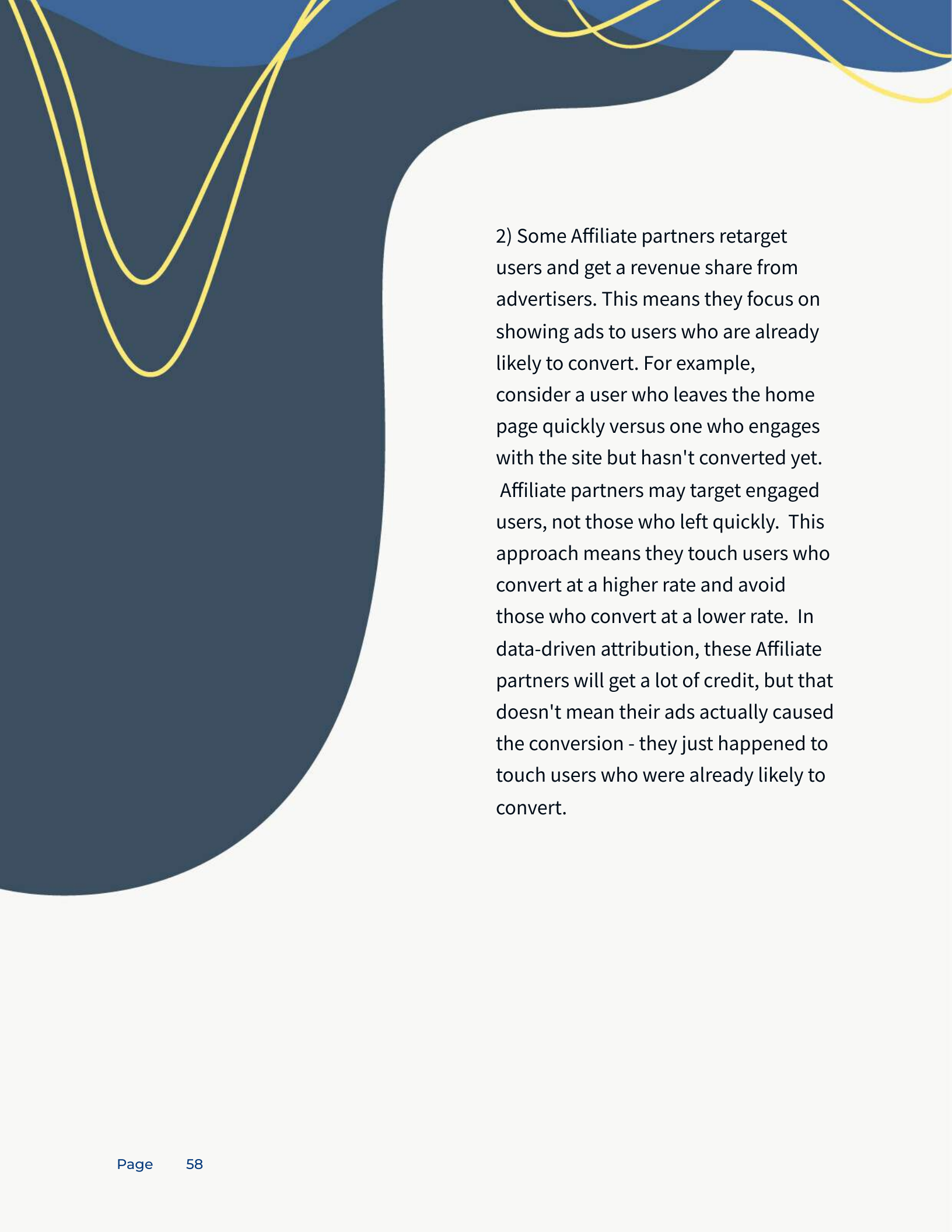
The same issue applies to retargeting. In most attribution models, retargeting appears right before conversion, but we need to ask: how many of those users would have converted anyway, just a bit later, without another reminder?

Let's make these examples more concrete:





1) A user searches for a brand term and clicks on a Paid Search ad. These users are very likely to convert, as they want to get back to the site. Compare this to a user who doesn't search for the brand but still sees a marketing ad. In data-driven attribution, Brand Searches will show a strong correlation with converters, while users who don't search for the brand will show a strong correlation with non-converters. As a result, data-driven attribution will give Brand Search ads a lot of credit for user conversions, even though it's just a correlation, not causation.



2) Some Affiliate partners retarget users and get a revenue share from advertisers. This means they focus on showing ads to users who are already likely to convert. For example, consider a user who leaves the home page quickly versus one who engages with the site but hasn't converted yet. Affiliate partners may target engaged users, not those who left quickly. This approach means they touch users who convert at a higher rate and avoid those who convert at a lower rate. In data-driven attribution, these Affiliate partners will get a lot of credit, but that doesn't mean their ads actually caused the conversion - they just happened to touch users who were already likely to convert.

Another challenge with attribution systems is that they're limited by the data they can use. For Paid Search, there's no view-through data in attribution systems. For Display ads, view-through data can be included, and when it is, attribution systems favor Display ads.

When Two Channels Are Each 100% Incremental and not

When channels work together, it's better to measure together, rather than one by one. Often, the best way to analyze is to look at the entire system of channels, since it's the combination that drives conversions.

A simple way to think about this is to consider what you actually use to make decisions: the cost per acquisition (CPA). Instead of looking at the CPA for each channel separately, consider the CPA across all channels.



Example 1: Non-Brand Paid Search + Email (an “AND-gate” conversion)

Consider a common pattern:

1. A user searches on Google and clicks on a Non-brand Paid Search ad.
2. That visit results in an email capture (registration).
3. Later, Email channel nudges the user who is “on the fence” and closes the sale (conversion).

In this example, both touches are required for at least some conversions:

- Without Paid Search, you never get the email address.
- Without Email, a meaningful portion of those registrants never convert.

So for those conversions, Paid Search and Email are jointly necessary. In a very real sense, both are “100% incremental” for that subset: remove either one, and those subset of conversions go to zero.

And that immediately exposes the limitation of attribution.





Attribution systems are built around a constraint: total credit per conversion must sum to 100%. But a joint-necessity conversion behaves like a 200% story:

- Paid Search was required.
- An email was required.
- Both statements can be true at the same time, i.e., without one of these channels, your conversions would go to 0

So if you insist on allocating partial credit to each channel and then judge each channel's CPA independently, you can make the wrong call.

Putting numbers on it

Assume:

- Paid Search spend: \$1,000
- Email marginal cost: \$0
- This system produces 100 conversions.

If you measure the system as a whole:

- Total cost = \$1,000
- Total sales = 100
- Blended CPA (Paid Search + Email) = \$10

And suppose you love a \$10 CPA. Great. Keep doing it.

Now look at what happens under attribution.



Last-click would give Email 100 conversions and Paid Search 0.

That would make Paid Search look like a money-losing channel—and you might cut it.

Which would lead to a very bad outcome: 0 registrations and 0 sales.

Even if you use split credit (including “data-driven attribution”), you can still get misled.

Suppose the model assigns:

- 50% credit to Paid Search
- 50% credit to Email

Now Paid Search appears to drive only 50 conversions:

Paid Search CPA = $\$1.000 / 50 = \20 .

If you don't like \$20 CPA, you might dial down Paid Search. For example, you lower bids or tighten your target CPA to “fix” the channel.

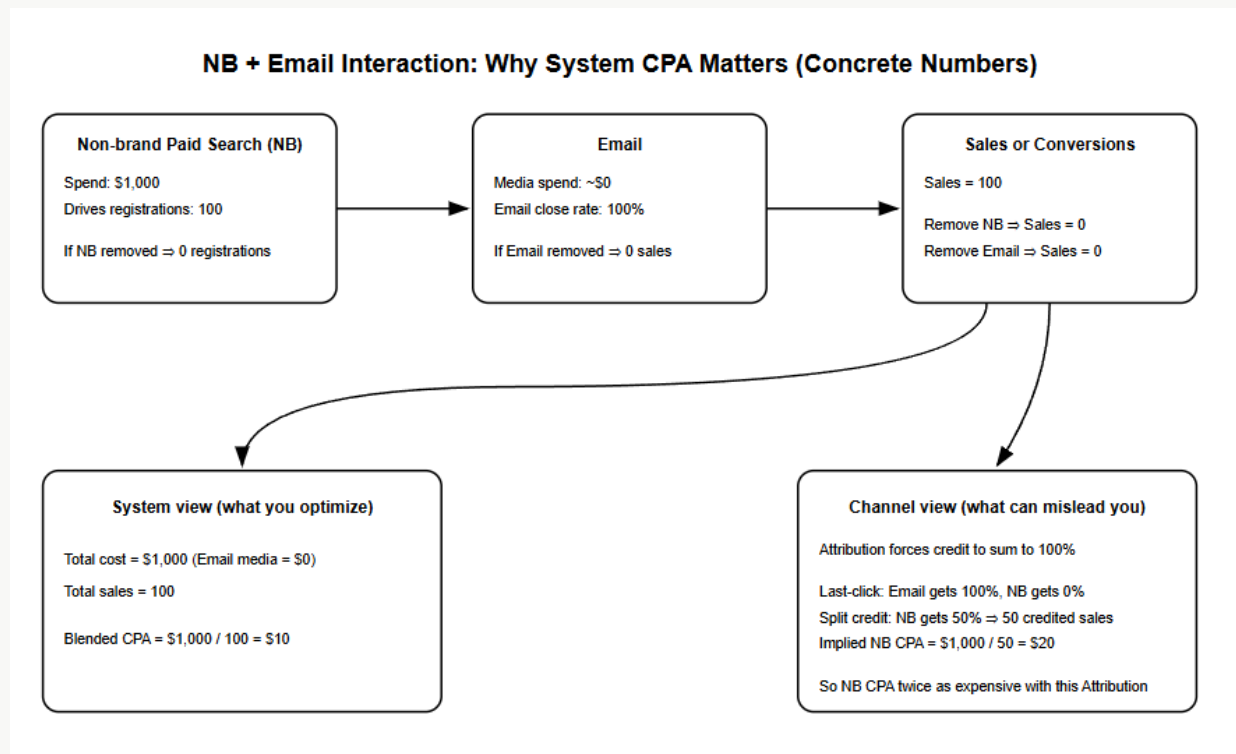
But that can reduce the top of the system.

Say cutting Paid Search reduces registrations from 100 to 50. If Email converts those 50 registrants, you now get:

50 sales instead of 100

You “improved” Paid Search efficiency on paper. Still, you shrank the system you actually cared about (i.e., the original blended CPA of \$10 and volume of 100 conversions across Paid Search + Email was very attractive).

This is the key point: when channels interact, evaluating them one at a time can starve the very mechanism that drives conversions.





It's especially dangerous when one of the channels has near-zero marginal cost (Email, Organic Search, Direct). Those channels can look like miraculous closers and cause upstream channels to look unprofitable, when in reality the upstream channel is doing the hard work of creating the opportunity.

So in these cases, a better decision framework is:

Add up the costs of the interacting channels, the conversions they jointly drive, and evaluate the blended CPA.

Example 2: Direct Mail + Display Retargeting (incremental CPA can increase)

Here's a related pattern that tells a very different story.

Suppose you run a Direct Mail holdout and learn that Direct Mail is incremental:

- Direct Mail spend = \$100K
- Incremental conversions attributed to Direct Mail = 10K
- Direct Mail incremental CPA = \$10

Fantastic. You want more Direct Mail.



But now introduce a common interaction: once those recipients visit your site, your systems start retargeting them in Display.

If Display spend increases as a result of Direct Mail traffic, then the true cost of generating those incremental conversions is not just Direct Mail. It's Direct Mail plus the downstream spend it triggers.

For example:

- Direct Mail spend = \$100K
- Display retargeting spend (triggered by those visits) = \$200K
- Total spend = \$300K
- Incremental conversions = 10K

Now the system-level CPA is: $\$300,000 / 10,000 = \30 .

Direct Mail can be genuinely incremental at \$10 CPA on its own, but when you factor in triggered retargeting, the business-level incremental CPA looks more like \$30. This difference can change the decision.

The takeaway

Attribution tools and path-to-conversion reports can hint at these relationships, but they do not automatically solve the problem. When channels interact, you have to ask actively:

- 
- Which channels create opportunities that other channels close?
 - Which channels are automatically triggered downstream?
 - What happens to costs across the rest of the system when I scale one channel?

And when those interactions are strong, the safest default is:

Measure conversions and costs in aggregate, at least for the interacting cluster of channels, and judge profitability at the system level, not channel by channel.

Section Summary: Attribution Models

Attribution models determine how credit for conversions gets divided across your marketing touchpoints. This credit allocation can sometimes directly control your budgets, bids, and channel investments. This section walked you through the full landscape, from simple rule-based models (last-click, first-click, even-weight, time-decay, position-based) to data-driven approaches like Markov Chain attribution, making clear that every model embeds assumptions about customer behavior that you should understand before trusting the numbers.

The section builds toward a critical insight: even the most sophisticated attribution model - data-driven attribution using Markov chains - assigns credit based on correlation, not causation. A channel can look like a star performer simply because it appears in front of users who were already going to convert. Brand Search is a classic example of this trap: it shows up right before the sale, attribution gives it heavy credit, but many of those conversions would have happened anyway.



The chapter also tackles a problem that most attribution discussions skip entirely: what happens when channels work together as a system. Through concrete examples (Non-Brand Search feeding Email registrations that later convert), it showed how evaluating channels independently can lead you to cut the very channel that makes the whole system work. When one channel creates the opportunity, and another closes it, both can be "100% incremental," but attribution's requirement that credit sums to 100% forces a false trade-off. The practical fix: treat interacting channels as a single cluster and evaluate blended CPA at the system level rather than channel-by-channel.

Why should you care?

Because attribution numbers are the foundation of most marketing budget decisions, yet they can systematically mislead you. Channels that merely intercept high-intent users get over-credited, channels that do the hard work of creating demand get under-credited, and channel interactions get ignored entirely. If you don't understand what attribution can and cannot tell you, you risk optimizing toward metrics that look great on a dashboard while destroying the engine that actually drives your business. Attribution is a useful starting point, but as this section makes clear, it's not the finish line.

05

**Incrementality
Testing - holdout tests,
geo tests, on/off tests**



If attribution can't reliably answer incrementality, what can?

Back in November 2014, we spent six figures a month on Meta (then known as Facebook) because its platform showed excellent CPAs for us. This was based on a one-day view-through, plus the standard 28-day click-through conversions. The one-day view-through measured conversions that happened within a day of viewing the ad, even if the person didn't click on it. Since there was no click, these conversions were credited only to Meta on its own platform, not to third-party analytics platforms like Adobe Analytics (or Google Analytics), which we were using.

We then worked with Meta's Engineering team to run a holdout test, which helped us measure the incrementality of Meta ads. This test was really eye-opening. Although the Meta team may have had a vested interest in showing that their ads were profitable for us, they were fully transparent and did a great job with the experiment (although we did have to go through a few iterations of the test at the time). The test results suggested that we should discount view-through conversions, but click-through conversions seemed to be fully incremental. This was a very valuable insight for us.

So let's take a closer look at holdout testing, or more generally, Incrementality Testing.

In Incrementality Testing, we typically have a Control group that doesn't see the ads we're measuring, and a Treatment group that's similar to the Control group but does see these ads. By comparing the revenues of both groups, we can measure the additional revenue generated by the Treatment group and conclude that it's the result of our advertising. This approach can accurately measure causation, not just correlation, making it a better way to assess marketing or advertising effectiveness than Data-Driven Attribution.

We can also use the results from Incrementality testing to improve the machine learning algorithm used in Data Driven Attribution. However, the accuracy still depends on how precise our Incrementality testing is, which is not perfect.

In holdout testing, as noted above, part of the population is randomly assigned to a control group that does not see the ad, while the rest is assigned to a treatment group that does. These two groups must be identical in every way except for ad exposure.

This mirrors how efficacy is tested in medicine. A control group receives a placebo, a treatment group receives the drug, and outcomes are compared. If the control group were elderly and the treatment group consisted of healthy 25-year-olds, any observed difference would be meaningless. The same logic applies here.

As indicated earlier, we have worked with Meta on such tests. We would target a specific audience, and Meta's data science team would randomly hold out, say, 10% of that audience as a control group that never saw our ads.

Holdout testing

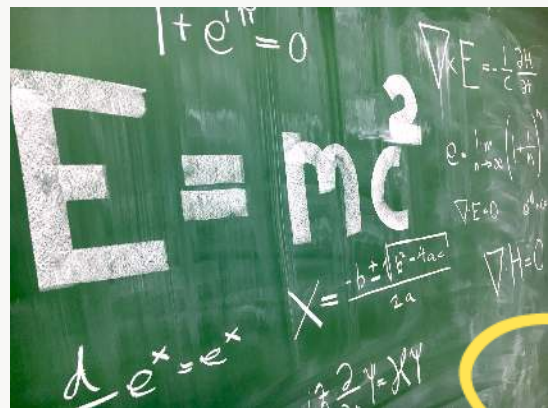
Suppose the holdout group generated 10 conversions and the treatment group generated 120 conversions. Because the treatment group was nine times larger, we would expect about 90 conversions even if the ads had no effect. The incremental impact of the ads is 30 conversions. (120 - 90)

Be Careful with Holdout Design

It is easy to get the holdout test design wrong.

One example from my experience in 2015 involved display advertising. A vendor proposed defining the holdout group as users who landed on a page but didn't scroll far enough to see the ad. The treatment group would be users who scrolled and saw the ad.

This is not a fair test. Users who scroll are systematically more engaged than those who don't. The two groups differ in ways unrelated to advertising exposure.



Correct randomization (or method of splitting between the Control and Treatment group) is extremely important. Coming back to the original question about splitting people who saw ads from those who did not, the randomization described above (Control group based on users who did not scroll down to see a “below the fold” ad) is not clean. For instance, this approach described will clearly lead to sampling bias (i.e., poor randomization), as users who don’t scroll down to see the ads are inherently different from those who do.

Furthermore, users on sites with ads below the fold are likely to differ from those on sites with ads above the fold in their browsing behavior. These seem like subtle points, but they can completely change the results of any experiment, so be careful.

A more subtle issue arises on platforms like Meta. If you split a 1,000-person audience into a 90 / 10 ratio, the comparison must be between the entire treatment group (900) and the entire control group (100). Even if only 300 of the 900 actually saw the ad, the baseline expectation remains $9\times$ the control conversions. Selecting only the subset that saw the ad introduces bias.

Why?

This is because we are using the entire Control group as the baseline, not just those who would have seen the ad. The reasoning here is that people who see the ad in the Treatment group are inherently different from those who don’t, and thus, only looking at people who see the ad would lead to sampling bias.





Why would people who actually see the ad be inherently different from those who don't in the Treatment group?

Well, because people who see the ad are perhaps more active on Meta and the Internet, and more open to buying online. Furthermore, Meta's own machine learning-based optimization may show ads to users (within the Treatment group) most likely to convert. So to reiterate, the Treatment group conversion rate must account for the entire Treatment group, not just those who saw the Ads.

At this point, Meta platform (within Experiments) handles this correctly, but some other Display Networks may not. So be aware.



Contamination of Cookie-Based Holdouts

Holdout tests based on user IDs or logged-in states can be fairly clean. Cookie-based holdouts often are not.

People delete cookies, block them, use multiple browsers, or switch devices. Cookies do not transfer across browsers or devices. As a result, someone in the holdout group may still see the ad elsewhere. If that person later converts on the device where they are classified as “holdout,” the test is contaminated.

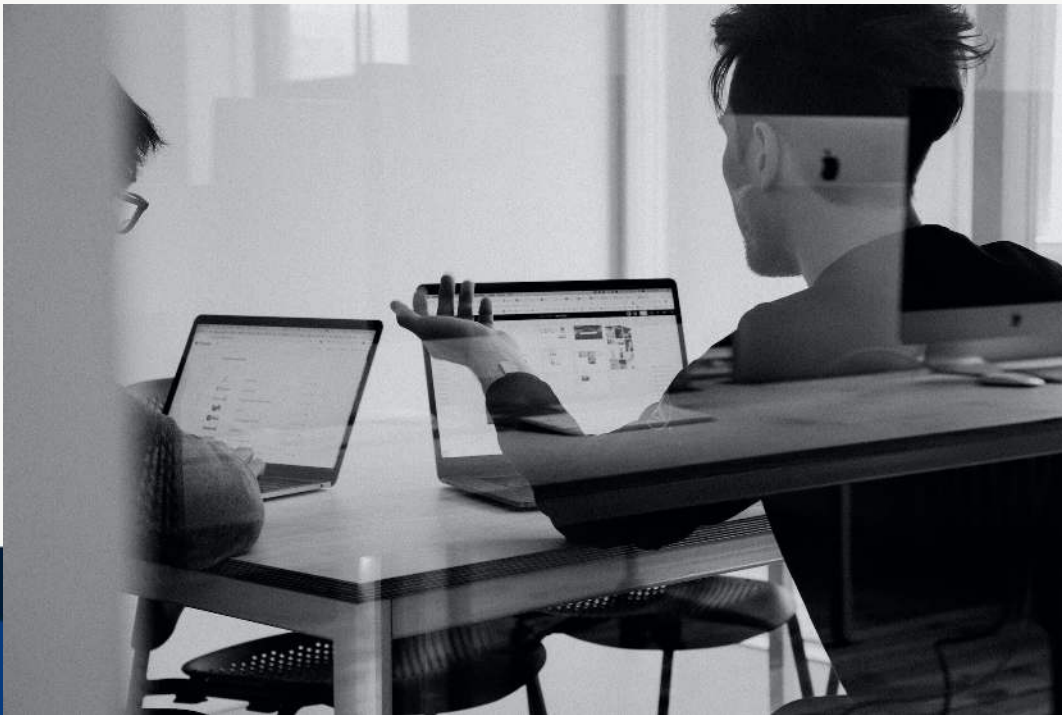
If enough of this happens (and you often won’t know when it does), the test becomes meaningless.

The Big Challenge of Holdout Tests

The most important thing to keep in mind about holdout tests - and why they are not, by themselves, a gold standard - is that they do not produce a single number. They produce a distribution.

Consider a test where both the control and treatment groups consist of 1,000 people. At the end of the test, say 30 people convert in the control group and 70 in the treatment group.

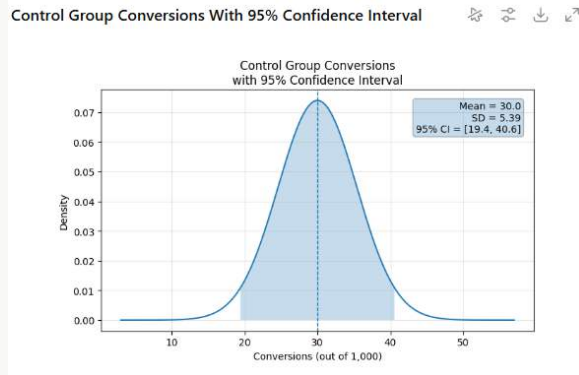
This is often summarized as:
Incremental lift equals 40 conversions.



This interpretation is incomplete.

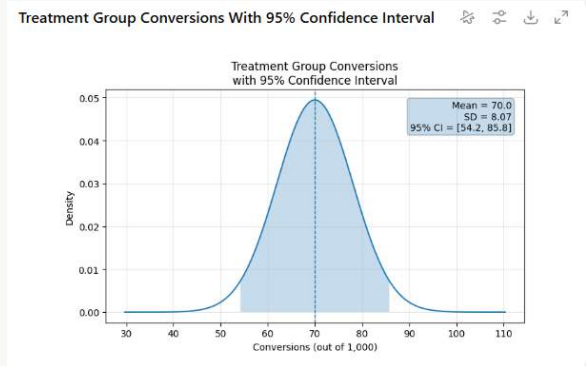
The number “30” (in the control group) is not a fact; it is an estimate. If we were to repeat the same test, we would almost certainly not get 30 again. As the visual below shows, the true control-group conversions plausibly lie between roughly 19 and 41 with 95% probability.



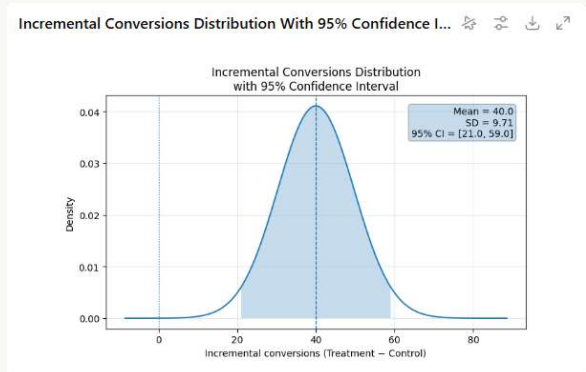


Think of it this way. Let's say we have no idea of the probability of getting a head in tossing a coin, and after tossing 100 times, we get 55 heads. If we toss the same coin 100 times again, we would very likely get a number different from 55 heads.

The same logic applies to the treatment group. The observed 70 conversions are our best estimate, but repeating the test would likely produce a different result. With 95% probability, that number lies between 54 and 86. See the visual below.



And below, the bell-shaped curve for the incremental conversion of 40.



Note a few things from this:

1. Incrementality inherits uncertainty from *both* groups. It compounds. (The standard deviation of the incremental distribution curve is more than either of the holdout or test groups.)
2. While 40 incremental conversions is our best guess, the true number plausibly lies between 21 and 59 with 95% probability. If we want to be even more confident, say 99%, the lower bound would be lower still. (And, by symmetry, there is also a chance it is higher than 59.)
3. Finally, a word on statistical significance.
When people say that the 40 incremental conversions are “statistically significant,” what they really mean is that the ads were effective at all. Important to stress that statistical significance does not mean that the ads produced 40 incremental conversions. Rather, it says the ads produced more than 0 incremental conversions, i.e., some positive lift. The question of *how much* remains uncertain, and that uncertainty is an important part of the result.



Practical Usage of Holdout Tests

Meta

As of this writing, Meta supports holdout tests through the Experiments section in Ads Manager. This hasn't always been the case. In the past, running true holdout tests on Meta often required a dedicated account representative who could connect you with their data science team.

If the option is available to you, I strongly encourage you to use it.

One immediate benefit is that you get a much clearer sense of how incremental Meta's view-through conversions actually are (conversions that occur after someone sees an ad but does not click it).

In my experience running holdout tests on Meta, the results tended to align reasonably well with click-through attribution reported in the Meta Ads Manager. For example, if a holdout test suggested that roughly 73 conversions were incremental during a given period (with a confidence interval around that estimate), that number was usually directionally close to Meta's reported click-through conversions for the same period, excluding all view-through credit.

That alignment is reassuring, but it shouldn't be taken for granted. Every brand is different, so you may get different results.





One important operational detail: you should include dark periods before and after the test. If you plan to run a four-week experiment, pause Meta ads for roughly two weeks beforehand. This allows any latent conversions (those that occur days after ad exposure) to play out before the test begins fully. Similarly, after the four-week period concludes, maintain another dark period of about two weeks to capture post-exposure conversions when measuring lift fully.

Direct Mail

Direct Mail is an offline channel, but in certain circumstances, its ROI can be measured quite cleanly with a holdout test. Let's say you have a list of 100K mailing addresses. Then, say, you randomly select 10K from them as a Control or Holdout group, and send your direct mail to the remaining 90K.

As long as you are collecting addresses of people who convert, you can then match back those addresses to your list of 100K for both the Control and Test groups. (This will require some sophisticated address matching, because people can vary addresses, eg, Ste vs Suite, but while this address matching is sophisticated, it is very doable.

Say, 100 users convert from the 10K holdout group without getting your direct mail. (Perhaps, because they were still exposed to your other advertising). This then creates a baseline of 900 expected conversions from the 90K test group. If, say, 1,700 converted from your Test group of 90K, then you know that Direct Mail really drove an estimated incremental 800 conversions, and you can measure your ROI based on that. (see table below).

As before, note that the 800 incremental conversion is really a point estimate in the middle of a bell shaped curve with a standard deviation.

Step 1: Test Setup

Group	Size	Treatment
Control (Holdout)	10,000	No direct mail sent
Test	90,000	Direct mail sent
Total	100,000	

Step 2: Observed Results

Group	Size	Conversions	Conversion Rate
Control (Holdout)	10,000	100	1.00%
Test	90,000	1,700	1.89%

Note: The 100 conversions in the Control group likely came from exposure to other advertising channels.

Step 3: Calculate Incremental Impact

Metric	Value	Calculation
Baseline conversion rate (from Control)	1.00%	$100 \div 10,000$
Expected conversions in Test group (without mail)	900	$90,000 \times 1.00\%$
Actual conversions in Test group	1,700	Observed
Incremental conversions from Direct Mail	800	$1,700 - 900$

Do keep in mind, though, that for those extra 800 who converted, you may incur additional Remarketing advertising costs, as these users may have visited the site and did not convert in the same session. With some cookie-to-address matching, you can estimate those additional costs as well to arrive at a better ROI estimate.

At the time of this writing, Google does not support true user-level holdout tests for paid search. If you bid on brand terms, for example, Google will not randomly withhold ads from a control group within its own systems.

This is unfortunate, particularly because many companies struggle with whether brand search is truly incremental. Other approaches, such as geo-based testing or on/off testing can still be informative, and we'll return to those later.

As a related aside, Paid Search (especially Non Brand) may have an incremental effect beyond what any click-based attribution might suggest, driven by the ads' view-through impact: people who view the ad but don't click still convert. Both Meta ads and Display Ads can measure view-through impact (Paid Search does not). In any case, any View Through credits are usually overstated and should be included only if supported by incrementality testing.

Display Advertising

Google

Google does not offer built-in holdout testing for display ads in the same way Meta does. However, there is a practical workaround.

Instead of holding out users from seeing any ad, you can hold them out from seeing your ad.

Here, you are doing Incrementality testing with PSA (Public Service Announcement) ads. These are “fake ads” (literally an ad for a non-profit, for example).

Consider randomly assigning every 2nd person who would have otherwise seen your real ad to a Control group, where the user sees the fake ad instead, and thereby becomes part of a Control group. By showing the user a fake ad, you can set a cookie to identify the list of cookies in the Control group.

Because every 2nd person also sees your real ad, the split is 50 / 50 between the Control and Treatment groups.

When doing Incrementality testing using PSA (fake ads in Display), Incrementality will likely be underestimated. Why?



Incrementality testing here uses cookies.

So we have a Control set of cookies, and a “Treatment” set of cookies. Cookies are set at a browser level. So if a user uses multiple devices (desktop, mobile) or even multiple browsers on the same machine, it is quite possible that they could end up in both the Control and Treatment sets of cookies. Similarly, if a user deletes cookies, they could end up in both the Control and Treatment groups. This means that users in the Control group will occasionally see real ads. This will result in a measured lift that is lower than its true value.

Note that even when we can measure the Incremental lift, keep in mind that the lift itself is statistical in nature. For instance, say that the Control and Treatment groups are split 50 / 50, and that the Control group generated 10K conversions during the experiment, while the Treatment group generated 15K conversions. Then the measured lift is 5K conversions from Display Advertising.

However, again, this 5K number is really the center of a Normal Distribution, and therefore itself has a variance and standard deviation. So if the standard deviation is measured at 1K, that means there is a 95% probability that the true lift is between 3K and 7K (approximately two standard deviations around the measured 5K cover a 95% interval).

PSA ads have the shortcoming that, in the real world, if your ad is not there, most likely there is a competitive ad (rather than a PSA). Can this affect lift results? Hard to say for sure, but these tests are still worth doing, as they yield insights that can inform budgeting decisions.





Geo-Based Testing: Incrementality Without Randomization

In Q4, 2013, we were spending a lot of money (again, 6 figures monthly) on traditional TV ads, with no way of knowing if we were getting our money's worth.

At the time, we were running these TV ads with an outside agency that handled all the detailed network negotiations and planning, including when and where our ads would air.



Technically, then, we couldn't run a regular holdout (with a completely randomized holdout group) test, as described in the earlier section.

We also tried to find a time-based correlation between TV ads and brand searches. After all, if TV ads were effective, they would cause more people to search for our brand terms. The correlation was clearly there. But we also found that we started the TV ads just when the holiday seasonality was kicking in for us. So we couldn't be certain whether the increase in brand searches was due to the TV ads or to seasonality, which would also drive an increase in brand searches.

We then compared brand searches to the prior year at the same time, and again, the data was inconclusive. Brand searches had increased at the same time in the prior year, but so had TV ad spend. So while we established a correlation between TV ads and brand searches, we couldn't establish causation.

We had to resort to a geo-based holdout group. While this wasn't perfect, we got some really good insights into how TV was working.

Let's jump in.

Regular holdout tests (with complete user-level randomization) are conceptually very clean when done correctly. Geo-based tests are different. They are often more practical, especially when user-level holdouts are not feasible, but they introduce an additional layer of uncertainty that is easy to underestimate.



Instead of randomly holding out individual users, geo-based tests hold out entire geographic regions. One set of geos receives advertising (the treatment geos), while another set does not (the control geos). Outcomes are then compared across the two.

This approach is often necessary when:

- User-level randomization is not available
- The channel operates at a geographic level (e.g., TV, radio)





It is important to note that the outcomes (e.g., conversions) compared between the 2 geos (control and treatment) are aggregated across the business. Whatever channel we are testing would need to generate enough incremental conversions to overcome the day-to-day variance in conversions in those geos.

How Geo-Based Tests Are Typically Constructed

Because users are not randomly assigned, geo-based tests rely on historical similarity to construct a credible control.

In practice, this means examining historical data and selecting geos that have shown:

- Similar conversion levels
- Similar trends over time
- Similar seasonality patterns



Often, geos are selected based on high historical correlation in conversions or revenue. If two cities, states, or DMAs have historically moved together, one may be designated as the treatment group and the other as the control group.

This matching can be done at different geographic levels:

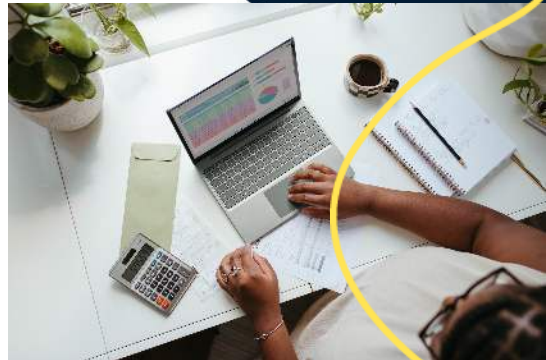
- City
- DMA
- State

The Granularity Tradeoff

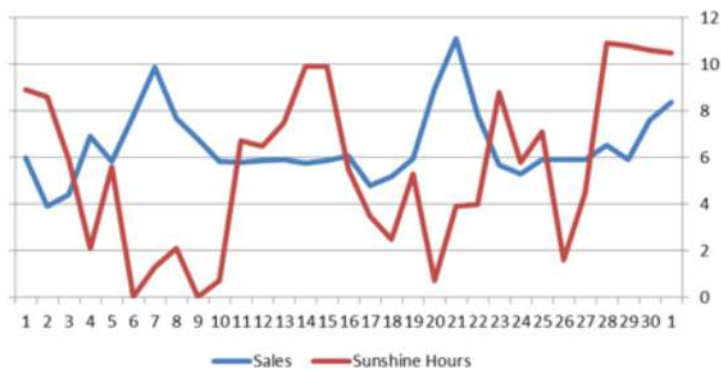
When setting up a geo-based test, you must decide how big a “geo” is. The finer the geographic unit, the more noise you introduce.

Smaller geos—such as cities—tend to be noisier because:

- Lower volumes
Fewer people mean fewer conversions. With small counts, random fluctuations matter more. A handful of extra conversions (or the absence of them) can materially swing results.
- Local shocks dominate
Smaller geos are more exposed to idiosyncratic events:
a local storm, too much sunshine (see image), a city-specific promotion, a store opening or closure. These affect conversions but have nothing to do with advertising.
- Less averaging out
In larger populations, unrelated effects tend to cancel out. In small geos, they don't.



Sales vs Sunshine Hours



As shown above, directionally, as sunshine hours decrease, sales tend to increase.

Such weather-related patterns can confound incrementality measurement from geo-based tests, especially at the city or DMA level.

On the other hand, larger geos (such as states or large DMAs) reduce noise, but introduce other problems:

- Fewer independent observations
If your test consists of one large state as the treatment and one as the control, you effectively have only one data point. That limits inference.
- Harder matching
Large geos are internally heterogeneous. Two states may look similar on average while hiding very different underlying dynamics.
- Less flexibility
With fewer geos, there are fewer credible test / control splits. One poor match can dominate the result.

The tradeoff only works in your favor if smaller geos are used in multiples, not as a single test / control pair.

Why Multiple Geos Matter

If you run one treatment geo against one control geo, you can get a result, but it is fragile.





You have no reliable way to distinguish whether the observed difference was driven by:

- the ads, or
- something else that happened in one geo (weather, competitor activity, local events)

With a single pair, you are implicitly assuming that nothing else important differed. Using multiple geos helps because each geo acts as its own data point.

If you have, for example, 10 treatment geos and 10 matched control geos:

- A local shock in one city becomes one noisy observation, not an entire noisy experiment.
- You can estimate not just average lift, but also variance across geos.
- You get a cleaner answer to the question: *Is this effect real, or are we seeing geo-specific drift?*

You do not need dozens of geos, but you usually want more than one pair.

Common patterns include:

- Matched pairs: selecting 10 to 30 similar DMAs and randomizing treatment within each pair.
- Grouped splits: assigning a group of matched geos to treatment and another group to control.

When selecting geos for a seasonal test (say, July), it is important to look not only at recent months (e.g., April to June) but also at the same period last year, especially if seasonality is present.

Historical Matching and Its Limits

Using historical correlation to select test and control geos is the best approach.

But historical similarity does not guarantee future similarity. Even the best-matched geos rarely exhibit correlations anywhere close to 1.0. This is why, when user-level randomization is available, user-level holdout tests should be preferred.

Because geo-based tests lack randomized sample groups, they almost always produce wider confidence intervals than holdout tests.

The uncertainty here is sampling noise plus environmental uncertainty - factors outside the experimenter's control. That makes geo-based results inherently more fragile.

Control and Treatment group pollution

Note that people may cross Geo lines and be in both Control and Treatment groups. For instance, they may work in one Zip Code that falls within a Treatment Geo and live in another Zip Code that falls within a Control Geo. Plus, IP Technology is not perfect, as it targets based on geos.

Being in both the Treatment and Control will dilute the measured lift. If, instead, you match/pair Geos between Control and Treatment at the State level, there is a lower likelihood of crossover. (After all, people don't usually cross State lines to go to work, but they still might travel).

Interpreting Geo-Based Results

When a geo-based test shows lift, the correct interpretation is narrow and specific.

It suggests that advertising likely had an incremental effect in that context, during that period, in those geos.

And, as with user-level holdouts, the result is not a single number, but rather a distribution, and often a wide one.

Despite these limitations, geo-based tests are often the best available option for certain channels, such as Brand Paid Search, where user-level holdouts are not possible.

Using Synthetic Controls to Improve Accuracy

When selecting control geos, you are not limited to a single city or region within a test/control pair. Because you are not running ads in the control, it can be constructed as a weighted combination of multiple geos.



Such a weighted blend (often referred to as a synthetic control) can track the treatment geo more closely in the pre-period than any single control geo.

If the synthetic control matches the treated geo well before the campaign, it serves as a more credible stand-in for what would have happened in the absence of advertising.

Constructing synthetic controls requires some math, as weights are chosen to best match pre-period behavior across correlation, trends, and levels.

But the intuition is simple. For example, if in the pre-period the treatment group consistently had 10% higher conversions than the synthetic control, and during the test period that gap widens to 15%, one can attribute the additional 5% difference to advertising.

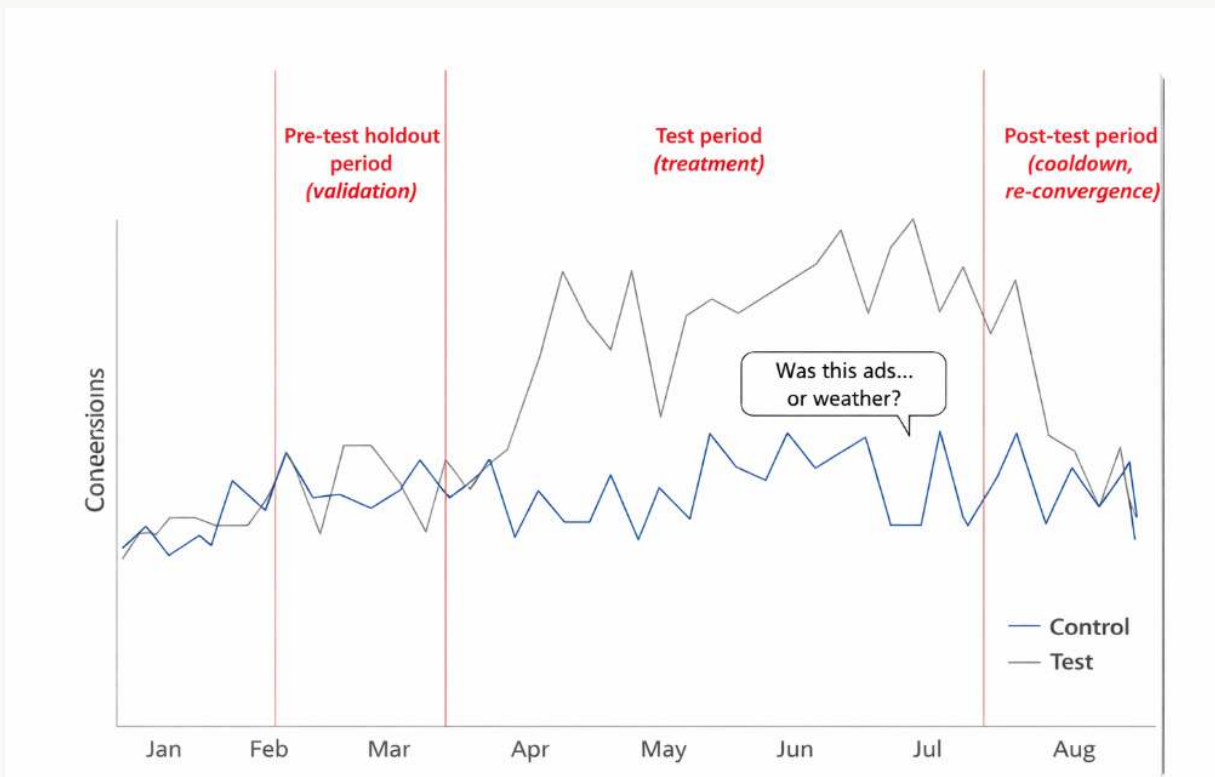
If, after the test ends and sufficient time is allowed for latent conversions, the gap returns to 10%, confidence in that estimate increases.

These results are still directional. Even the best synthetic control does not eliminate the underlying uncertainty. But they can provide valuable incremental insight when cleaner methods are unavailable.





In the visual below, you can directionally see that the ads helped. During the test period, there is a clear divergence in conversions between the Control and Test geos, suggesting that the ads are helping. But to what level? There is significant variance, and some of the conversion reductions on specific days could be due to weather. (We need to know how good the ads are, not just whether they helped at all, to make reasonable ROI calculations, as the ads cost money).



Multiple Treatment Groups for Geo-based testing

One additional thing to point out is that we don't have always just to do a two-way split between Control and Treatment Geos. We could, for instance, have Control, Treatment 1, and Treatment 2. There may be a couple of reasons for doing this:

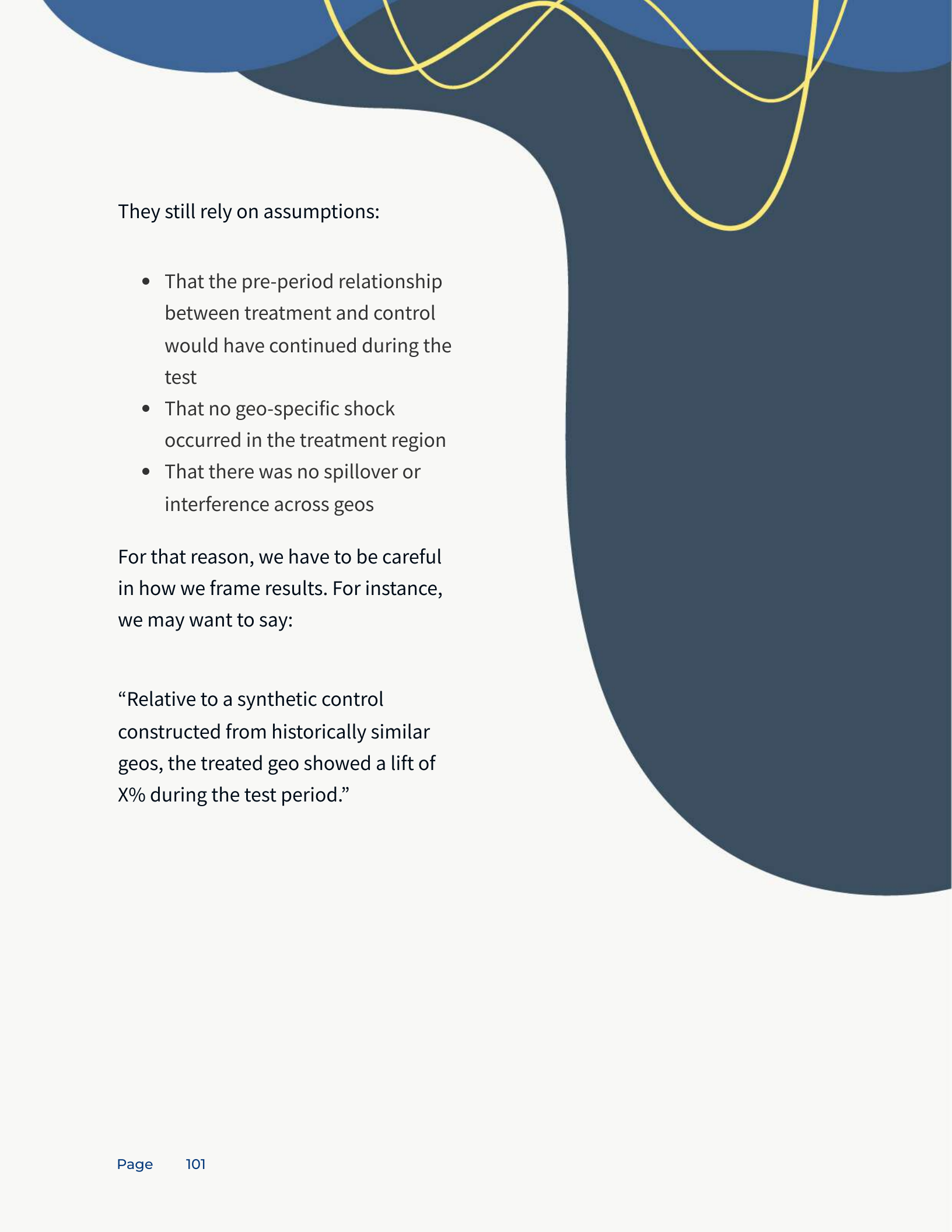


1. Control could have no TV ads; Treatment 1 could have low TV media spend; and Treatment 2 could have high TV media spend. This would allow us to measure TV effectiveness at various spend levels.
2. We could use Treatment 1 and Treatment 2 at similar levels of media spend and use more as an A/A test. For instance, if Treatment 1 shows a large lift and Treatment 2 shows a minor lift, even though modeling suggests they should show about the same lift as Control, then we know something is not quite right and requires further investigation.

Weaknesses of Geo-Based Tests

Even with synthetic controls, geo-based tests do not achieve true randomization.





They still rely on assumptions:

- That the pre-period relationship between treatment and control would have continued during the test
- That no geo-specific shock occurred in the treatment region
- That there was no spillover or interference across geos

For that reason, we have to be careful in how we frame results. For instance, we may want to say:

“Relative to a synthetic control constructed from historically similar geos, the treated geo showed a lift of X% during the test period.”



Measuring the Incrementality of Brand Paid Search Using Geo-Based Tests

One of the most common measurement questions in performance marketing is whether a company should buy its own brand terms on paid search.

On the surface, brand paid search often looks incredibly profitable. Conversion rates are high, CPAs are low, and attribution models, especially last-click, assign a large share of revenue to it. But this is exactly where attribution and incrementality can diverge most sharply.

The real question is: “how many of these conversions would have happened anyway if we had not run brand ads?”

For instance, if you don’t bid on your Brand Keywords on Google, will all those attributed conversions disappear from your business? Likely not, because even if your Brand Ad is not there in Google Ads, your SEO (Search Engine Optimization) listing will be present, and will pick up a lot of revenues (at 0 advertising cost) that would otherwise be attributed to your Paid Search Brand Keywords.

At the time of this writing, Google does not support true user-level holdout tests for paid search. You cannot randomly prevent some users from seeing your brand ad while allowing others to see it. That rules out the cleanest experimental design.

However, geo-based testing provides a practical alternative.



The Setup: Starting From Zero Brand Spend

Assume that a company is not currently buying its brand terms and is considering whether to start.

In this setup:

- The company is not bidding on brand terms anywhere.
- Brand traffic is flowing through organic search and direct navigation
- We introduce brand paid search only in selected geographies.

This allows us to ask a narrowly scoped counterfactual question:

“What happens in geos where we introduce brand paid search, relative to similar geos where we do not?”

Designing the Geo-Based Test

The basic design mirrors the geo-testing principles discussed earlier.



1. Select candidate geographies

These could be cities, DMAs, or States, depending on volume and business footprint.

2. Match test and control groups using historical data

Before the test, examine historical conversions, revenue, trends, and seasonality to identify geos that move similarly over time.

3. Assign treatment and control.

- In treatment geos: begin bidding on brand search terms.
- In control geos: continue not bidding on brand terms.

Keep everything else constant.

Budgets, creatives, landing pages, promotions, and non-search channels should remain unchanged during the test window.

What You Measure

The primary outcome is total conversions across the treatment and control geos.

If brand ads simply shift conversions from organic brand search to paid brand search, then paid conversions will go up—but total conversions will not. That shift is not incremental.

So the comparison should focus on:

Total conversions in treatment geos versus total conversions in control geos before, during, and after the test period.

A typical pattern might look like this:

- In the pre-period, the treatment and control groups track closely.
- During the test period, treatment geos show a modest uplift relative to control.
- After brand ads are turned off, the gap narrows again.

That pattern is suggestive evidence of incrementality from paid search brand ads.

Interpreting the Lift

Suppose that during the test period:

- Treatment geos show 5% higher total conversions than baseline would have, compared to control geos. For instance, if control geos' total conversions increase by 10% between the test and pre-test periods, then the baseline for treatment geos also increases by 10% during the same period. With paid search brand ads on in treatment geos, if total conversions increase by 15%, we can now say directionally that the paid search brand ads increased conversions by 5% relative to the baseline (and that 5% was incremental).
- Using this 5% number, say the brand paid search spend implies a CPA that looks attractive relative to that incremental lift.

Consider the following hypothetical example:

Geo Test Results: Paid Search Brand Ads

Metric	Control Geos	Treatment Geos
Pre-Test Conversions	10,000	11,000
Test Period Conversions	11,000	12,705
Growth Rate	+10%	+15.5%
Expected Baseline (using control growth)	—	12,100
Incremental Conversions	—	605
Incremental Lift vs. Baseline	—	+5%

CPA Calculation

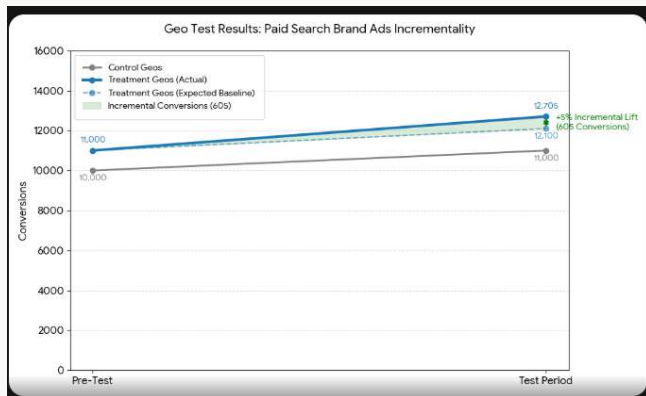
Metric	Value
Paid Search Brand Spend (Treatment Geos)	\$12,100
Incremental Conversions	605
Incremental CPA	\$20

The above is a quick directional way to come up with lifts and incremental CPAs on Paid Search Brand without getting into Statistics and confidence intervals (for now).



We need to determine next whether the \$20 Incremental CPA is profitable for us to buy our Brand terms. If it isn't, that doesn't mean we shouldn't buy our brand terms. We could still buy them, but at lower click costs until they become profitable.

Here is a graph showing the effects of the geo test:



It suggests:

- That brand's ads likely captured some incremental demand.
- That organic brand traffic did not fully substitute for paid brand traffic.

A 5% lift can be economically meaningful depending on how much our ad costs were. If the 5% lift suggests an increase in revenues that is less than our costs for buying these ads, then clearly, even though these ads generated incremental revenues, they weren't profitable.

Post Test Period

To be more confident in our measured lift, we could continue measuring in the Post Test period (after we stop buying our Brand terms in the Treatment Geos) to see if Conversions between the Control and Treatment Geos revert to the pre-test ratio. If yes, we can feel more confident about our results. If the ratio does not revert, more analysis will then need to be done. (Perhaps, we did not pick the right Geos to begin with, or perhaps, some seasonality or weather-related issue occurred, or perhaps, too many people crossed over between Treatment and Control Geos).





Sources of Uncertainty (Especially for Brand Search)

In the above example, we showed a point estimate of a 5% lift in conversions. Keep in mind that there will be plenty of uncertainty around this number; e.g., we may have a 95% confidence interval of 1% to 9% lift, suggesting that, with 95% confidence, the true lift is anywhere between 1% and 9%. While a 9% lift may certainly change our decisions about buying our brand ads, a 1% lift would yield negative ROI. What risk are we comfortable with?



Additionally, with Paid Search brand ads:

- Navigation behavior may vary by region
Some regions rely more heavily on bookmarks or direct navigation than others.
- Competitive brand bidding may differ across geos as well. A competitor bidding on your brand terms in one region but not another can distort results.
- Organic rankings are not uniform by geo. Small differences in SERP (Search Engine Result Page) layouts or sitelinks can affect click behavior.
- Seasonality and promotions interact with brand demand.
A regional promotion, PR event, or offline campaign can swamp the effect of brand ads. This was discussed earlier as one reason why geographically based tests produce more uncertainty than true randomization via holdout groups.

Because of these factors, confidence intervals around lift estimates are often wide.

How to Phrase the Result

A responsible conclusion might sound like:

“Relative to historically similar control geos, regions where brand paid search was introduced showed an estimated lift of X% in total conversions during the test period.”

This phrasing avoids implying certainty about:



- 
- the exact size of the effect
 - its permanence
 - or its universality across markets

Why This Is Still Worth Doing

Despite all the caveats, geo-based testing is often the best available method for evaluating brand search incrementality.

It can be superior to:

- relying on last-click attribution
- assuming brand ads are either fully incremental or fully wasteful
- making budget decisions based purely on reported ROAS

The test won't give you a clean yes-or-no answer. But it will usually give you a bounded, evidence-based range for how incremental brand paid search likely is.

And in practice, that is often enough to make a better decision.

An Aside on Paid Search Brand Ads

While we're on the topic of whether to buy your brand terms on Google (and Bing), it's worth stepping back and sharing a few insights.



In addition to the geo-based tests discussed above, brand search can also be evaluated using on/off testing, running brand ads one day, turning them off the next, and repeating this pattern. We'll cover the mechanics and limitations of on/off tests later, but for now, it's useful to keep them in mind as another imperfect, but sometimes informative tool.

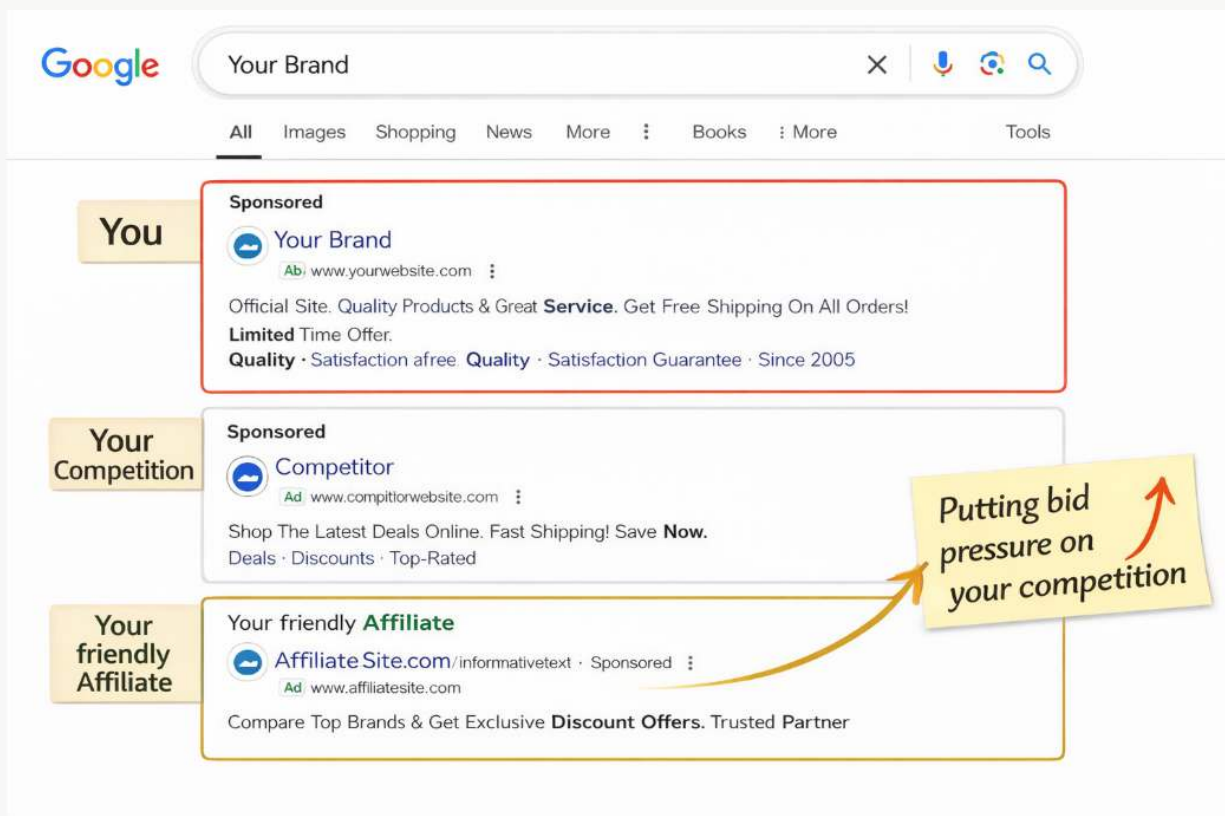
If competitors are bidding on your brand terms, you will often need to buy them yourself. Their ads are likely to appear above your organic listing, assuming you are ranked #1 organically for your own brand. (If you are not ranked #1 organically for your own brand terms, that is usually fixable and should be addressed regardless.)

1. Even in this competitive scenario, it is still worth testing, either via geo-based controls, on/off tests, or both. On the other hand, if no competitors are bidding on your brand terms and you rank #1 organically, it does not automatically follow that you should avoid buying your own brand terms.
2. There are cases where brand paid search is still incremental. One reason is message control. Paid Search ads allow you to control copy more precisely, whereas organic listings force the title to do double duty, both ranking and eliciting the click. In addition, Google often allocates more SERP (Search Engine Results Page) real estate to paid brand ads than to organic listings, which can materially change click behavior.
3. So again: test.



A related strategic point: if you are the larger brand, do not start a brand-bidding war by bidding on competitors' brand terms. They will almost certainly respond in kind. If there are more searches for your brand than theirs, you have more to lose. Of course, if they are already bidding on your brand terms, you may need to respond, but do so with discipline. Keep bids at levels that are economically justified, not emotionally driven.

Now for something slightly contrarian. If competitors are bidding on your brand terms, consider having one of your affiliates bid on those terms as well. This is “friendly competition.” While these affiliate-driven conversions may be more expensive than direct conversions, the revenue still accrues to you. More importantly, the presence of an affiliate can increase competitive pressure on rival bidders, sometimes enough to make brand bidding unprofitable for them.



And now for something even more contrarian. If there is only one competitor bidding on your brand terms, try lowering your bids so that they temporarily win the top position. This feels wrong, but here's why it can work. When your competitor sits in position two, they often enjoy very low CPCs (Cost Per Click). With no meaningful competition beneath them, the second-price auction keeps their click costs down. If you drop to position two, become the competition from below. That can push their CPCs higher; sometimes high enough that bidding on your brand terms no longer makes economic sense for them. This is not guaranteed to work. But it can work well, especially as a short-term experiment. Like most things in brand search, it's situational, and it's best approached as a test.



Measuring Incrementality with On / Off Tests

Are your App Install ads driving incremental installs, or would those installs have happened anyway? In cases where installs occur on the same day as ad exposure (which is common since an install is a relatively easy action compared to a purchase), an On / Off test can give you some really good insights into how those ads are performing.

In situations such as these, on/off tests can be better than geo-based tests.

At a high level, an On / Off test alternates advertising exposure over time rather than across users or geographies. It is not perfect, but when carefully designed, it can still yield highly actionable insights.

Test Design: Daily Rotation

The simplest and most robust version of an On / Off test is a daily rotation:

- One day, ads are on
- The next day, the ads are off
- Then On again, then Off again, and so on

This rotation should run for at least 2 weeks, with 4 weeks preferred.



With daily rotation and a two-week test, both on days and off days get exposure to every day of the week—Mondays, Tuesdays, weekends, etc. This automatically controls for day-of-week effects, which are often large and can easily swamp the signal you are trying to measure. A four-week test further reduces noise and improves confidence.

Control vs. Treatment

Conceptually, the setup is straightforward:

- Off days are the Control
- On days, the Treatment

You are measuring the incremental lift observed when advertising is turned on, relative to the baseline behavior when it is off.

You are asking whether the business performed differently on days when ads were present versus absent.

When On / Off Tests Work Well

On / Off tests work best when a large portion of conversions occurs on the same day as the ad spend.

This is often true for:

- Brand Paid Search
- App installs
- Certain impulse-driven ecommerce categories

If conversions tend to lag significantly, say clicks today convert three, five, or ten days later, then a simple On / Off design will break down. Conversions from “On days” will bleed into “Off days,” and vice versa, contaminating both Control and Treatment.

In those cases, On / Off tests will systematically underestimate incrementality, sometimes to the point of being misleading.

Using Proxies When Conversions Are Delayed

There is, however, a very useful workaround, particularly for Paid Search brand ads.

While conversions may be delayed, clicks almost always occur on the same day as spend. If that is true, you can:

1. Run the On / Off test as described
2. Measure the incremental lift in clicks between On and Off days.
3. Apply a stable, externally estimated conversion rate to those incremental clicks.



In other words, clicks become your proxy for intent, and conversion rate is layered on top.

This approach is often far more accurate than directly comparing daily conversions when conversion lag exists. It also aligns better with how Paid Search actually works: bidding decisions influence clicks today, and conversions several days from now.

A Caution

On / Off tests are simple, but they are not casual experiments. You still need:

- Clean execution (no partial days, no “almost off” days)
- Consistent budgets on the days
- No major changes to site, pricing, or promotions during the test window

Used correctly, on / off tests are a powerful addition to your incrementality toolkit, especially in environments like Search, where cleaner experimental designs are simply not available.

As with most measurement approaches discussed in this book, this is less about perfection, and more about getting closer to the truth, knowing exactly where the limitations are, and making better decisions than attribution alone would allow.



An example to measure App Installs with On / Off tests

On / off tests work particularly well for app install advertising because installs are usually observed very close in time to ad exposure. When most of the response happens on the same day as spend, rotating ads on and off by day can give a surprisingly clean signal.

Consider a simple setup.

We run app install ads every other day:

- One day On
- The next day off
- Repeated for four weeks (28 days)

As discussed earlier, alternating daily ensures that both On and Off days include Mondays, Tuesdays, weekends, and so on.

Here, Off days are the control, and on days are the treatment.

Example Data (28 Days)

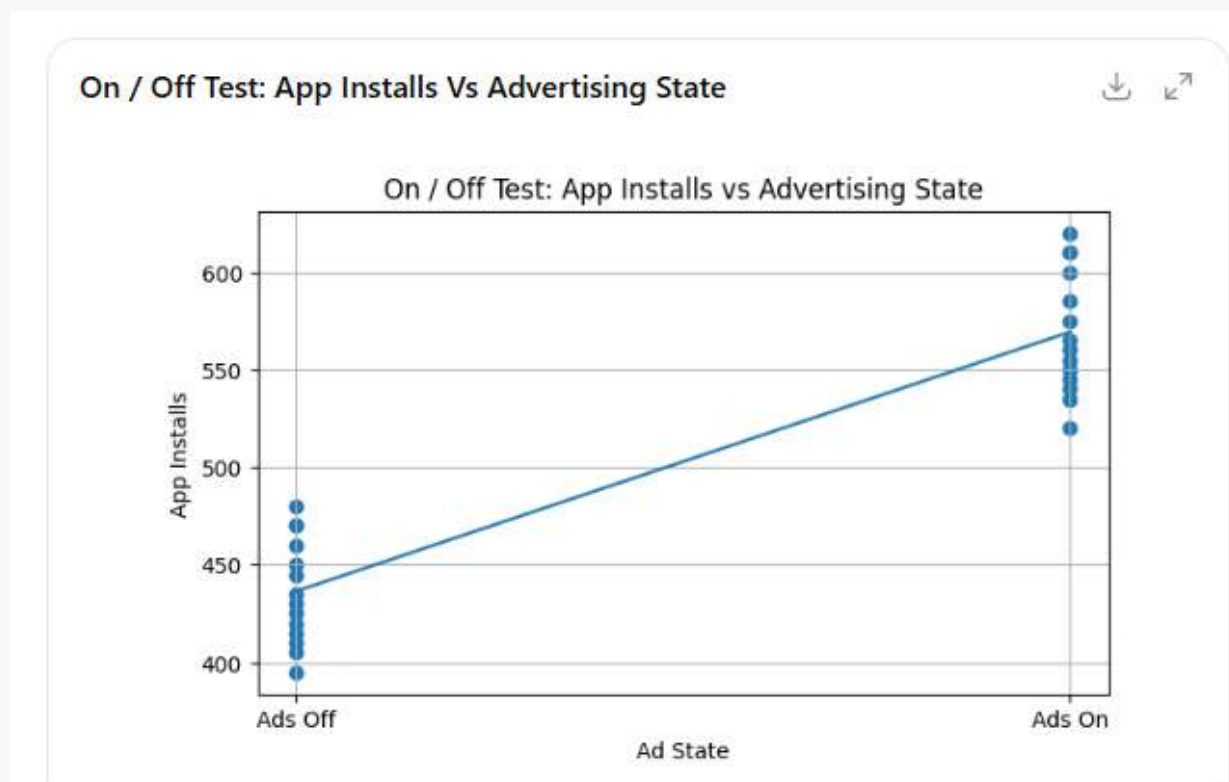
Below is a simple three-column structure you could have in Excel.



Day	Ads On (1 = On, 0 = Off)	App Installs
1	1	520
2	0	410
3	1	535
4	0	395
5	1	560
6	0	430
7	1	610
8	0	470
9	1	540
10	0	420
11	1	555
12	0	405
13	1	575
14	0	445
15	1	600
16	0	460
17	1	545
18	0	415
19	1	565
20	0	435
21	1	620
22	0	480
23	1	550
24	0	425
25	1	585
26	0	450
27	1	610
28	0	470

It may be hard to tell from just eyeballing the table above whether On days (2nd column, where value = 1) have an impact on App Installs (3rd column above) compared to Off days (2nd column, where value = 0).

So let's plot it.



In the above plot:

- Each dot is one day
- The X-axis is binary:
- 0 = Ads Off (control days)
- 1 = Ads On (treatment days)
- The Y-axis is App Installs
- The line is the linear regression fit. I'll say more about Linear Regression later in the book, but for now, it is simply the best line to pass through the dots. This can be plotted in Excel. (Mathematically, the best-fit line is found by looking at how far each dot is from the line vertically, squaring those distances, adding them up, and then choosing the line that makes that total as small as possible)

A few things to note:

- The Off days cluster lower and the On days cluster higher.
- The fitted line slopes upward, connecting those two clusters.

That upward slope is the incremental effect.

In our specific example, the equation of that Linear Regression line is:

$$\text{App Installs} = 436 + 133 \times \text{AdsOn}.$$

(In other words, the slope of the plotted line above is 133, and that is the incremental impact of having ads on). Another way of saying this is that without ads, our App Installs average about 436 per day, and with ads, our App Installs average 569 per day, or 133 more per day.



Note that, just like before, the incremental number 133 has a standard deviation. In this case, with a standard deviation of about 11, the 95% confidence interval (2 standard deviations) is 111 to 155 incremental app installs per day when ads are on.

If you want to lean toward the conservative side, assume 111 incremental app installs per day when ads are on. Based on ad costs, you can calculate the cost per (incremental) install and determine whether it is profitable. Ideally, you want to measure profitability, since those App Install Ads have a cost. If each App Download costs \$3 in advertising but is worth only \$2, even though the ads were effective, they weren't profitable.

Incrementality testing at the product level

Say you want to advertise a specific Product A. You could simply see the increase in Revenues resulting from the advertising. However, what if external factors in the environment (e.g., weather or competitive pricing) also impacted Sales? How can you tease apart the effect of advertising on Revenues from these external factors?

One way to do this is to also include another Product, B, in your analysis, which is as “close to Product A as possible, but which is not being advertised. Such a product is being used as a “pseudo control”.

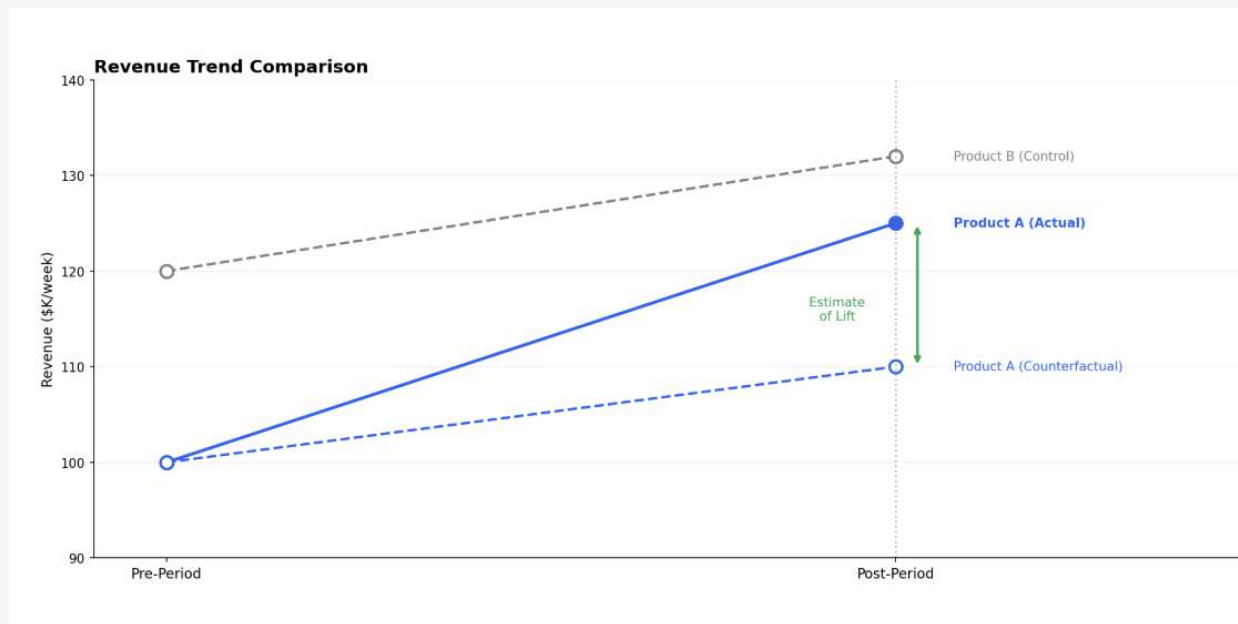
For example, in the Pre-period, before advertising Product A, revenues from Product A were \$100K per week, and revenues from Product B were \$120K per week. After advertising, sales of Product A rose to 125K/week (a 25% lift), while sales of Product B, which was not advertised, also increased by 10% to 132K/week. Then one can conclude that the real lift from advertising was really 15%, and that the other 10% increase that Product B also saw was due to external factors.

See table:

Revenue Data (\$K/week)		
Period	Product A (Advertised)	Product B (Control)
Pre-Period	\$100K	\$120K
Post-Period	\$125K	\$132K
% Change	+25%	+10%

Lift Attribution		
Component	Lift %	Explanation
Total Observed Lift (A)	25%	Product A's total revenue increase
External Factors (B)	-10%	Weather, competition, seasonality, etc.
True Advertising Effect	= 15%	Incremental lift from advertising

In the graph below, note the estimated lift from advertising Product A. (The word “Counterfactual” denotes the estimate of what Sales for Product A would have been without advertising.



Note that in this case, we need to be very careful about choosing Product B. If Product A and Product B are close substitutes, it is possible that sales of Product B decrease while sales of Product A increase, precisely because advertising of Product A causes more people to buy it, thereby decreasing sales of Product B. If this happens, sales of Product B are affected not by external factors but by advertising for Product A, leading to cannibalization. And then assuming that sales of Product B serve as a “pseudo control” would be flawed.

But in general, the above approach can work well when we use something else as a pseudo-control to tease apart any external factors that may be impacting our measurement from the effect of advertising or changes in advertising.



Based on the numbers above, it seems the lift from advertising for Product A is 15%. How confident are we, and what is the standard error/standard deviation around the 15% lift? This can be calculated from daily data and used to estimate confidence intervals.

As a slight aside, this Pre / Post approach can be improved with the Pre / During / Post approach. In this Pre / During / Post approach, the Pre time period remains the same, i.e., what were our metrics before we began the test, in this example, before we began advertising. Then the duration period is really what we saw during the advertising period (earlier defined as post). But then we add a third step: stop advertising to see if our metrics revert to what we saw in the Pre-period, with some lag. In this case, did the lift of Revenues for Product A over the pseudo-control (revenues from Product B) go back to 0%?

Please also avoid running multiple types of tests concurrently; e.g., don't run a User-Level Holdout on Meta while running a Geo-test across all Social channels. This will "pollute" your test. Your measurement roadmap must be on a calendar, ensuring that your tests have enough "breathing room" to provide a clean signal.



Section Summary: Incrementality Testing — Holdout Tests, Geo Tests, On/Off Tests

Attribution tells you which channels are associated with conversions.

Incrementality testing tells you which channels actually caused them. In this section, the book shifts from describing credit allocation to measuring real economic value; the difference between conversions you bought and conversions that would have happened anyway.

This section walks through three practical approaches to incrementality testing, each suited to different constraints. Holdout tests are the gold standard when available: randomly split your audience into a group that sees your ads and one that doesn't, then compare outcomes. The chapter draws on real experience running these on Meta, showing how view-through conversions often overstate value while click-through conversions tend to hold up. But holdout tests aren't always possible, which leads to geo-based tests; holding out entire regions instead of individual users. These are noisier, require careful geo-matching (ideally using synthetic controls), and produce wider confidence intervals. Still, they're often the only practical option for channels like TV or Brand Paid Search, where user-level randomization isn't available. Finally, on / off tests, i.e, alternating ads on and off daily, work surprisingly well for same-day response channels like app installs and brand search clicks, offering a simple but powerful design when other methods aren't feasible.



A theme that runs through every method is uncertainty. The section repeatedly emphasizes that incrementality results are distributions, not point estimates. Forty incremental conversions might actually range from 21 to 59. Statistical significance only tells you the effect is probably greater than zero; it doesn't pin down the magnitude. And every test design has its vulnerabilities: cookie deletion contaminates holdouts, local shocks distort geo tests, and conversion lag undermines on / off designs.

The section also includes a practical section on Brand Paid Search, walking through how to design a geo test to answer one of the most common and sometimes high-budget questions in performance marketing: are your brand ads actually incremental, or are they just intercepting traffic that would have converted organically? The chapter shows how even an imperfect test can replace guesswork with a bounded, evidence-based range.

Why should you care?

The gap between attributed value and incremental value can be substantial. Brand Search might appear to be your best channel in a last-click report, yet deliver only marginal lift. Display retargeting might appear to drive thousands of conversions that would have happened regardless. Without incrementality testing, you're flying blind on the question that actually matters: is this spend making me money, or am I paying for conversions I would have gotten for free? This chapter gives you the tools and the healthy skepticism to start finding it out.

06

Marginal Analysis: key to budgeting

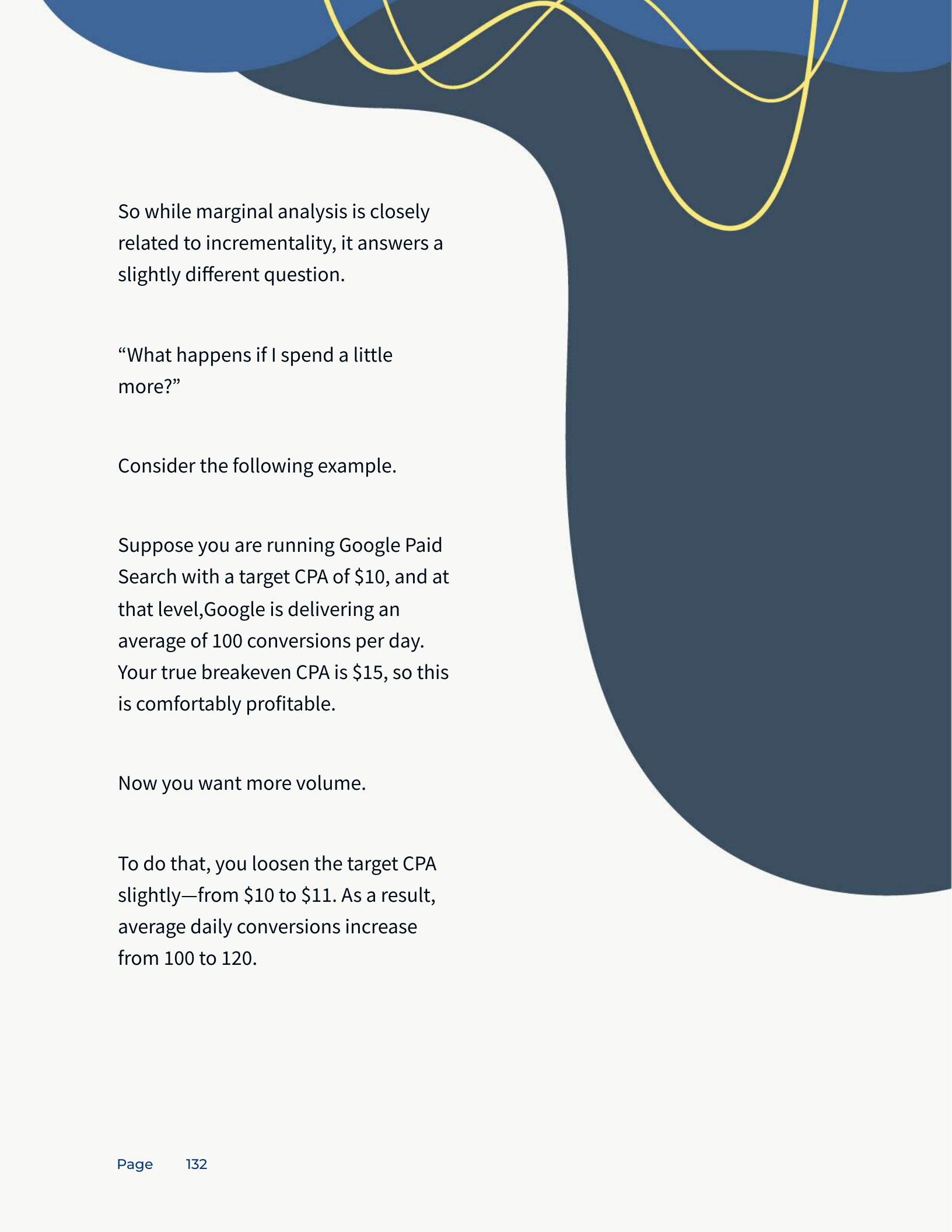
The Concept

Say, you are confident in your measurement that your YouTube ads are very profitable, and you are considering increasing your budget there. Should you?

At first glance, the answer seems simple: “yes”. But the real question you need to answer is: will these ads continue to be profitable, specifically given the additional spend vs. the current baseline?

Let’s see why this can be counterintuitive.





So while marginal analysis is closely related to incrementality, it answers a slightly different question.

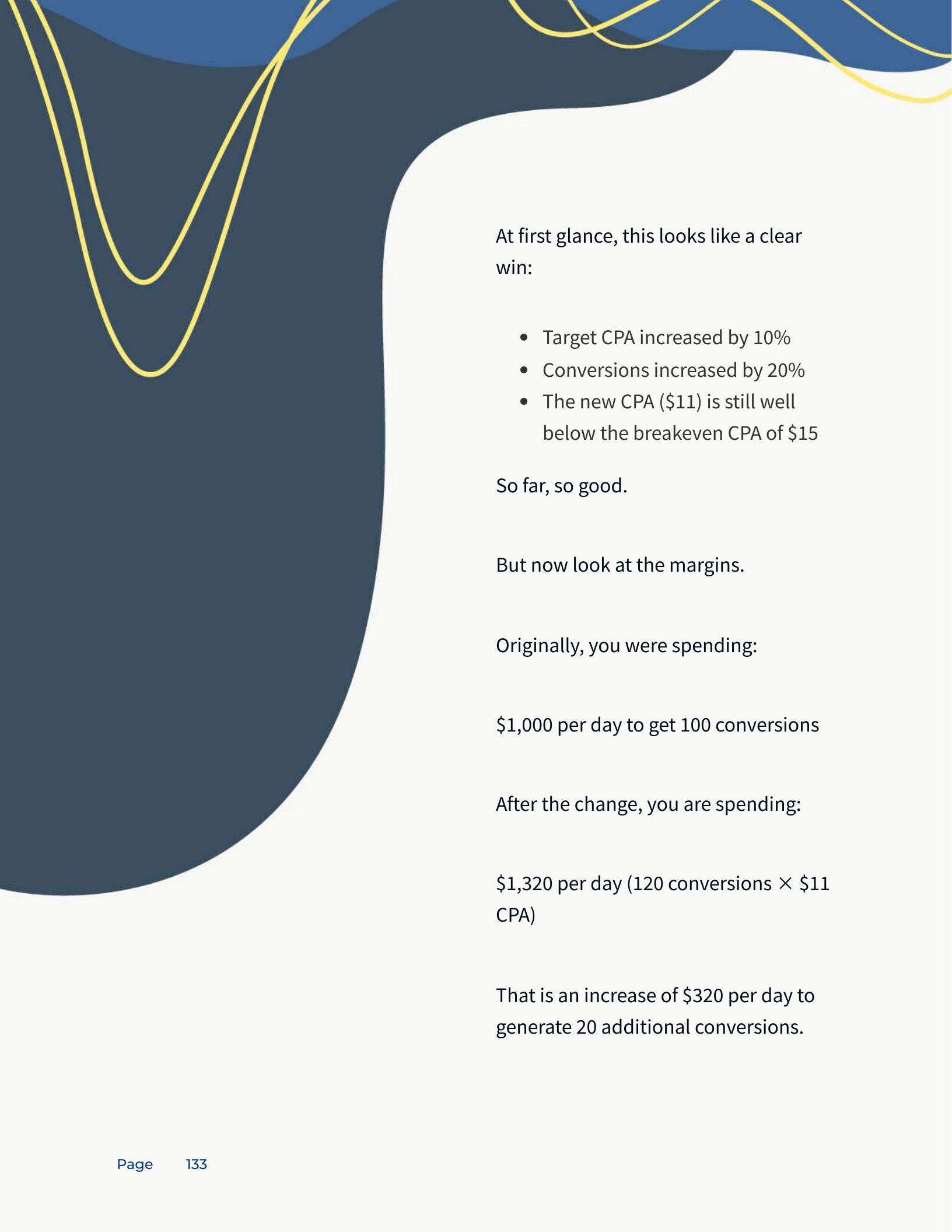
“What happens if I spend a little more?”

Consider the following example.

Suppose you are running Google Paid Search with a target CPA of \$10, and at that level, Google is delivering an average of 100 conversions per day. Your true breakeven CPA is \$15, so this is comfortably profitable.

Now you want more volume.

To do that, you loosen the target CPA slightly—from \$10 to \$11. As a result, average daily conversions increase from 100 to 120.



At first glance, this looks like a clear win:

- Target CPA increased by 10%
- Conversions increased by 20%
- The new CPA (\$11) is still well below the breakeven CPA of \$15

So far, so good.

But now look at the margins.

Originally, you were spending:

\$1,000 per day to get 100 conversions

After the change, you are spending:

\$1,320 per day (120 conversions $\times$ \$11 CPA)

That is an increase of \$320 per day to generate 20 additional conversions.

Which means the incremental CPA on those extra conversions is: $\$320 / 20$ or $\$16$

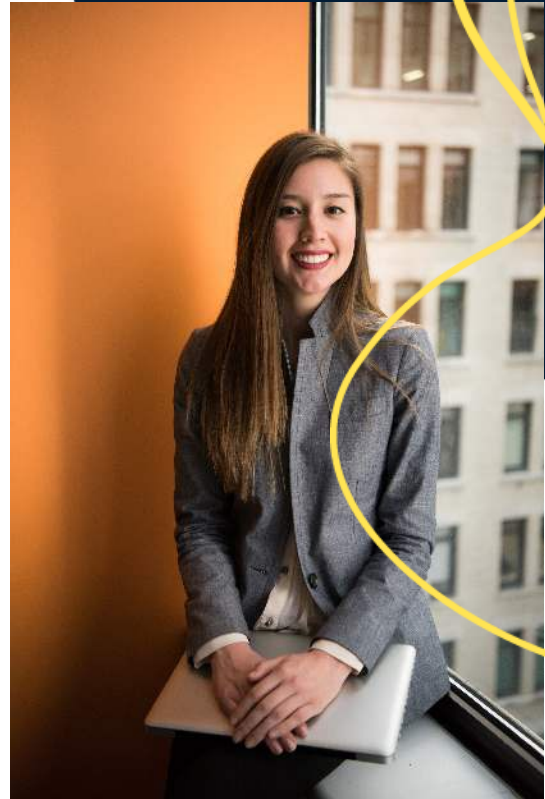
That is above your breakeven CPA of $\$15$.

In other words, while the average CPA still looks fine, the marginal (20 extra) conversions you just bought are unprofitable.

This is why marginal analysis can feel so counterintuitive. Everything about the aggregate numbers looks healthy, yet the incremental decision is destroying value at the margin.

Conceptually, this notion of marginal analysis aligns with diminishing marginal returns, but what is often surprising is how quickly it can set in.

(By the way, for those of you who are wondering why an increase from $\$10$ CPA to $\$11$ CPA resulted in $\$16$ CPA for the incremental conversions, here is what is going on. With the new CPA of $\$11$, it wasn't just the additional 20 conversions at $\$11$; even the original 100 conversions, which were at $\$10$, are now at $\$11$, so the incremental conversions end up being very expensive).



By the same token, you should also run the marginal analysis when reducing budgets or tightening efficiency targets. Take the same example. You are currently getting 100 conversions per day at a \$10 CPA. Your breakeven CPA is \$15, so this is comfortably profitable.

Now, suppose you tighten the target CPA from \$10 to \$9 (a 10% reduction). As a result, say, volume drops by 20%, from 100 conversions per day to 80. Your daily spend falls from \$1,000 ($\$10 \text{ CPA} \times 100 \text{ conversions}$) to \$720 ($\$9 \text{ CPA} \times 80 \text{ conversions}$).

What just happened on the margin?

You gave up 20 conversions, but you saved \$280 in spend. That means those last 20 conversions were effectively costing you \$14 each ($\$280 \div 20$). In other words, the marginal CPA of the conversions you cut was \$14.

Since \$14 is still below your \$15 breakeven CPA, those conversions were actually profitable. So even though the average CPA improved from \$10 to \$9, you are worse off from a profit standpoint by tightening the target.

So the bottom line is that it is not enough to ask: “Is this channel profitable on average?”





The more important question is: “Are the next dollars I spend still profitable?”

As you think about reducing or increasing budgets across channels, the central question is: what am I losing or gaining on the margins, and should I do it?

Why Marginal Analysis Is Hard in Practice

Doing this kind of analysis cleanly is not easy. Platforms do not naturally expose marginal CPAs, and running controlled experiments at many spend levels can be expensive and slow.

As a result, many teams implicitly assume that:

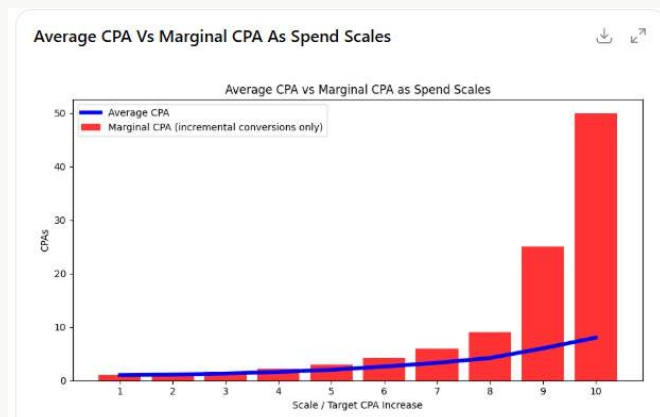
If the average CPA is below breakeven, then spending more must also be profitable.

That assumption is often wrong.

Note the visual below: As the target CPA increases, the average CPA rises slowly, while the marginal CPA can increase dramatically. As discussed, the reason is that higher bids reprice all conversions, not just the incremental ones.

Let's first make a few assumptions:

1. Our value or net revenue per conversion is \$15. In an e-commerce world, think of this as Revenues minus Cost of Goods Sold minus any other direct costs to support the e-commerce sale. (So would ignore overhead costs like rent, electricity, etc.). Let's call this "net revenue" and, as noted, set it to \$15 in this example. In other words, a CPA of \$15 is breakeven, and any CPA of over \$15 results in losses.
2. Profits are net revenues minus advertising costs.
3. When there is no advertising in any channel, there is no CPA and no revenue, so profits from advertising are 0.
4. We want to maximize Profits.



Later, when we discuss Media Mix Models, you'll see why marginal analysis is embedded directly into those frameworks. MMMs are trying to estimate what happens at the margin as spending increases or decreases.

For now, the key takeaway is simple:

Average performance tells you where you are.

Marginal performance tells you where you should go next.

How can we use the concept of Marginal Analysis to figure out where we really want to be on the CPA curve?

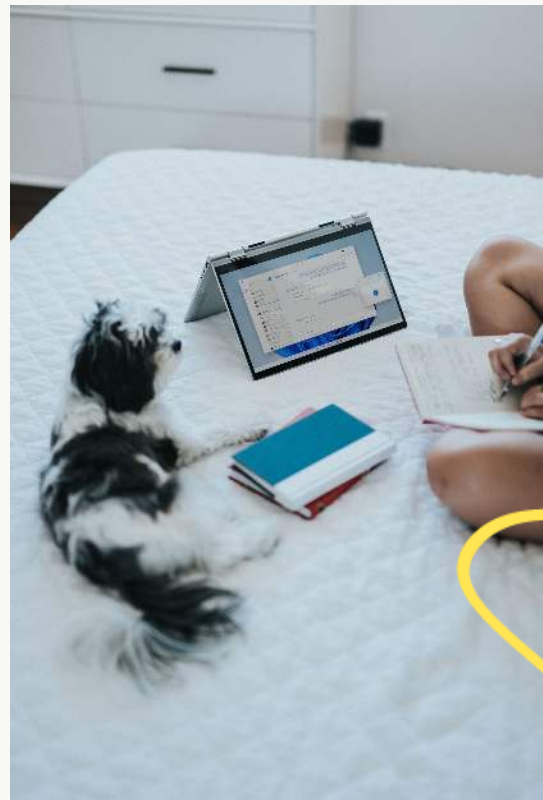
Profit Maximizing target CPA

As we start advertising with low budgets, our CPA can be really low, say even \$1, so our Profits per conversion would be \$14 in this example. (\$15 minus \$1). But because we have low bids (e.g, in Google Paid Search), we don't have much volume, perhaps only a few conversions per day.

As we increase our target CPAs and spend, say, to \$3 per conversion, our Profits per conversion reduce to \$12 (\$15 minus \$3). But quite often, the increase in conversions per day still increases our total profits.

But at some point, the decrease in Profit per conversion is no longer compensated by the increase in conversions.

Let's look at an example. Consider this data.



CPA	Net_Revenue_Per_Conversion	Profit_Per_Conversion	Conversions_Per_Day	Total_Profits_Per_Day
\$1	\$15	\$14	5	\$70
\$2	\$15	\$13	14	\$182
\$3	\$15	\$12	26	\$312
\$4	\$15	\$11	40	\$440
\$5	\$15	\$10	56	\$560
\$6	\$15	\$9	73	\$657
\$7	\$15	\$8	93	\$744
\$8	\$15	\$7	113	\$791
\$9	\$15	\$6	135	\$810
\$10	\$15	\$5	158	\$790
\$11	\$15	\$4	182	\$728
\$12	\$15	\$3	208	\$624
\$13	\$15	\$2	234	\$468
\$14	\$15	\$1	262	\$262
\$15	\$15	\$0	290	\$0

The data tells the story nicely:

- At \$1 CPA: Highest margin (\$14/conversion) but only 5 conversions → \$70/day profit.
- At \$9 CPA: The sweet spot where margin (\$6) × volume (135) is maximized → \$810/day
- At \$15 CPA: Maximum volume (290 conversions) but zero profit margin → \$0

The graph looks as follows:



While the above is theoretically valid, how do we actually figure out the profit-maximizing CPA? We need to constantly test and measure our Profits against different target CPAs. At the time of this writing, both Google and Meta platforms support CPA-based bidding. (Google calls it "Target CPA" while Meta calls it "Cost Cap."). The underlying economic concept from the profit maximization chart applies equally to both.

In Google, using experiments, you can even A/B test different target CPAs with their “built-in Experiments” feature to at least directionally arrive at your profit-maximizing target CPAs.





Meta doesn't have a direct built-in feature to A/B test different Cost Cap values against each other, but there is a workaround:

Run an A/B test with two ad sets: Set up both ad sets with the same creative, audience, and placements, but with different Cost Cap values.

Experiment with Cap Levels: Test different cap settings to strike the perfect balance between cost control and conversion volume.



The limitation is that Meta ensures the average CPA stays within the Cost Cap you set, so if you set a very low Cost Cap, the algorithm may simply not spend your budget rather than show you the volume you could get at that price point. This makes testing multiple points on the profit curve in Meta trickier than in Google.

And if nothing else, you can always do the following:

1. Run at one CPA target, observe volume
2. Adjust the target, wait for the learning phase, and observe the new volume
3. Repeat iteratively

The downside of this is that it is not a true A/B test, but rather a sequential design, which is susceptible to changes in the environment.

(Note that the above setups suggested for Google and Meta may not work by the time you read this as these can change often).

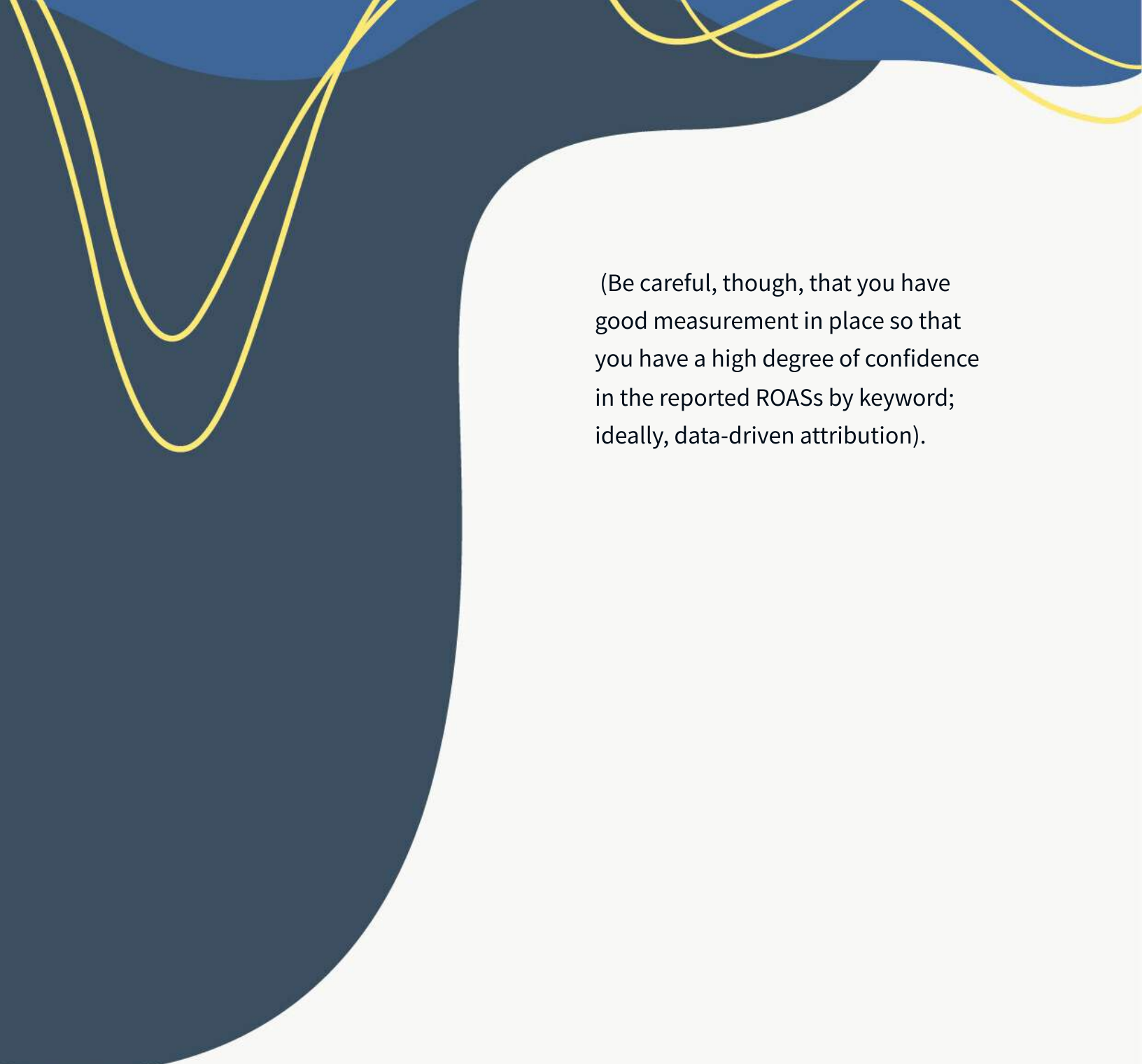
In e-commerce, another complicating factor in measurement is that revenue per transaction can vary widely. The question this raises is: as you increase your target CPAs, is revenue per conversion fairly stable, or is it affected as well? While it doesn't happen often, a higher target CPA can attract a very different audience (more likely on Meta), which then has a different revenue (or net revenue) per transaction. This, too, can be tested, but it does complicate the Math quite a bit.



Efficiency at a more granular level

Even though your overall ROAS might be at a 2.0, look for high variance or differences in ROAS at a more granular level, e.g. a keyword (in Google Ads, for instance), ad, or campaign level. For instance, if some keywords are coming in at a ROAS of 5.0 and others at 0.5, that should alert you that things could be better. In this example, keywords at a ROAS of 5.0 are subsidizing the keywords at a ROAS of 0.5 to still hit an overall ROAS of 2.0. Rather, you should expect the keywords with a 5.0 ROAS to be bid up to drive more volume. As they are bid up, their ROAS should fall (since cost per click would increase), but they should get more volume. And the opposite should happen with keywords at ROAS of 0.5, i.e., they should be bid down, so that their ROAS increases and volume reduces, after all, who wants volume at a ROAS of 0.5, which is clearly unprofitable.





(Be careful, though, that you have good measurement in place so that you have a high degree of confidence in the reported ROASs by keyword; ideally, data-driven attribution).

Marginal Analysis in Ad Copy Testing

In ad copy testing (A/B), the platform often declares a winner based on click-through rates. After all, if more people click Ad B than Ad A, Ad B is the winner. But this isn't the right way to look at it. The winning ad is the one that does better on your main metric of interest, e.g, CPA or Profits after subtracting ad costs. For instance, if you promise too much in Ad B compared to Ad A, more people will click Ad B. Then, if they are disappointed or even annoyed because you fail to meet their expectations on the landing page, then Ad B may not be the winner on the metrics that matter to you. And maybe even worse, if these very clicks cost you money (e.g, on Google or LinkedIn).

Consider the following metrics for Ad B vs Ad A.



Metric	Ad A	Ad B	Winner
Impressions	100,000	100,000	
CTR	2.0%	2.5%	Ad B → Platform says B wins
Clicks	2,000	2,500	Ad B
CPC	\$2.00	\$2.00	
Ad Spend	\$4,000	\$5,000	Ad A (lower)
Conversion Rate	4.0%	3.0%	Ad A
Conversions	80	75	Ad A
Revenue per Conversion	\$150	\$150	
Revenue	\$12,000	\$11,250	Ad A
CPA	\$50.00	\$66.67	Ad A (lower)
Profit	\$8,000	\$6,250	Ad A → A is true winner

Above:

Ad B would be the winner in Click Through Rates (CTR), and cost per click are the same for both ads. (It is possible that a higher CTR may result in a lower cost per click in some platforms; if so, use those numbers, but go through the above calculation).

As a result of the higher clicks, Ad B actually costs more in total. (\$5,000 vs \$4,000).

But perhaps Ad B promised too much, and the landing page couldn't live up to expectations. Whatever the reason, on conversion rates, Ad A outperformed Ad B (4% vs 3%). As a result, Ad B generated more revenue (assuming each Ad had the same revenue per conversion).

And on the bottom line of Profits, Ad A won by a huge margin. Its ad costs were lower, and revenues higher.





This example aims to perform the differential analysis shown in the table above (similar to the concept of marginal analysis) and then decide which ad is performing better based on your bottom-line goals, even if that may not align with the ad the platform determines as the winner.

Next, let's turn to "Pulse-ups," which are another way that tries to measure how expensive conversions may be "on the margin".

Pulse-Ups: Probing the Margins

Pulse-ups are another tool we often reach for when we want to learn more about incrementality, but cannot run clean holdout or geo-based tests.

Conceptually, a pulse-up is simple. You run a channel at some baseline level of spend, then temporarily increase spend for a short period of time, and then return to the original level. The increase is deliberate and time-bound. You are not permanently scaling the channel; you are "poking" it to see how it responds.

"What happens if I spend a little more, right here?"

In that sense, pulse-ups are fundamentally a marginal analysis tool.

What Pulse-Ups Can Tell You

A well-designed pulse-up can give you directional insight into the local slope of a response curve.

In other words, if you increase spend by 20% for a short window and see conversions increase by only 5%, that is an early signal that you may already be operating on a steeply diminishing part of the curve. If, on the other hand, conversions increase almost proportionally, that suggests there may still be room to scale a bit more.

This is valuable information. Many budget decisions are made at the margin. Pulse-ups are one way to gather evidence about those marginal dollars without committing to long, expensive experiments.

What Pulse-Ups Cannot Tell You

Unlike true and randomized holdout testing, pulse-ups do not create a clean counterfactual. There is no randomized control group. There is no guaranteed “what would have happened otherwise” baseline. Any lift you observe during the pulse could be influenced by:

- seasonality
- demand shocks
- platform learning effects
- regression to the mean
- coincident promotions or external events

As a result, pulse-ups cannot reliably estimate total incrementality. At best, they provide a directional signal about marginal response under current conditions.

If a pulse-up shows strong performance, then one could conclude:

“Under these conditions, increasing spending slightly appeared to generate additional response, but with uncertainty.”

Pulse-Ups and Marginal CPA

Pulse-ups are particularly useful when combined with marginal analysis thinking.

If you see that the average CPA only rises a little as you pulse spend upward, that does not automatically mean the incremental conversions are profitable. As we saw earlier, marginal CPA can rise sharply even when average CPA moves slowly.

The right way to read a pulse-up is to ask:

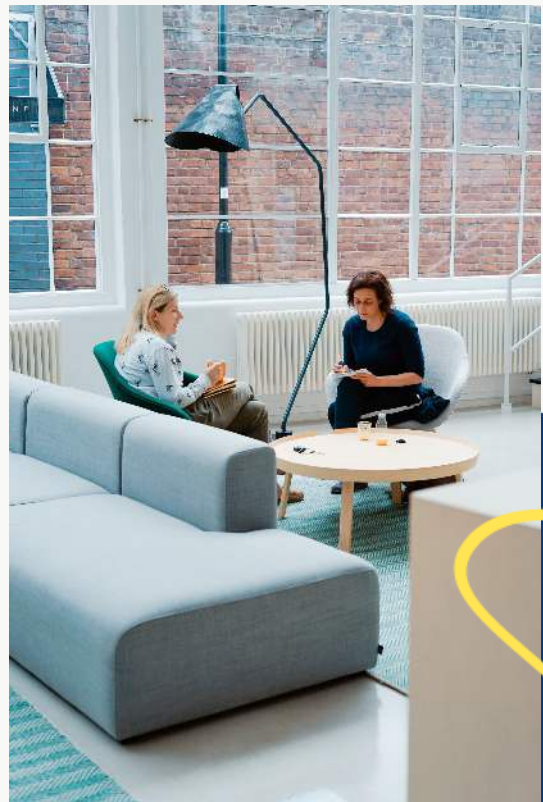
- How much additional spending did we add?
- How much additional outcome did we observe?
- What does that imply about marginal CPA or marginal ROAS?

Even then, the answer will be noisy, but it will often be more informative than relying solely on averages.

When Pulse-Ups Are Most Useful

Pulse-ups tend to work best when:

- conversions happen quickly after spend (or when good same-day proxies exist)
- Demand is relatively stable during the test window.
- pulses are repeated over time, rather than treated as one-off “proof points.”



Repeated pulse-ups at different spend levels can trace out a rough response curve. (A response curve describes how an outcome (conversions, revenue, installs, etc.) changes as advertising spend changes.)

Where Pulse-Ups Fit in the Measurement Toolkit

Pulse-ups sit somewhere between:

- On / off tests, which are cleaner but blunt, and
- media mix models, which are more comprehensive but slower to build.

They are lighter-weight, faster, and more fragile.

A good way to think about them is this: they help you decide whether the next few dollars are likely to be well spent.

Used carefully, they can improve marginal allocation decisions.

As with everything else in marketing measurement, pulse-ups are most powerful when interpreted alongside other evidence, not in isolation.

Geo-Based Pulse-Ups: Strengthening Pulse Tests With Controls





Pulse-ups on their own are lightweight and fast, but they rely heavily on time-based assumptions. One natural way to strengthen them is to combine them with geo-based controls.

A geo-based pulse-up follows the same basic idea as a standard pulse-up, with one important addition: not all geographies are pulsed.

Instead of increasing spending everywhere, you:

- keep spending unchanged in a set of control geos, and
- apply a temporary spend increase in a set of test (treatment) geos.

The pulse is still time-bound. After the pulse window ends, the spend returns to baseline in the test geos as well.

Conceptually, this creates a cleaner comparison:

“What happened in the geos where we spent, relative to similar geos where we didn’t?”

This still does not give you perfect randomization, but it is better than relying on time alone.



What Geo-Based Pulse-Ups Improve

By holding spending constant in control geos, you absorb many of the effects that plague pure pulse-ups:

- Seasonality at a country level
- macro demand shifts
- platform-wide changes
- unrelated promotions or news events

If both test and control geos experience the same external environment, then the difference between them during the pulse window is more plausibly attributable to the additional spend.

In that sense, geo-based pulse-ups are a “difference-in-differences” version of marginal probing.

What They Still Don’t Fix

Despite being stronger, geo-based pulse-ups still fall short of clean incrementality measurement.

They still assume:

- 
- The test and control geos would have continued to move together absent the pulse
 - No geo-specific shock hit only the test geos
 - limited spillover across regions (e.g., national brand effects, cross-geo behavior)

And as always, the result is still a distribution, not a point estimate.

Geo-Based Pulse-Ups and Marginal Decisions

Geo-based pulse-ups are particularly well-suited for answering questions like:

- “Should we push budgets another 10 to 20%?”
- “Are we already near saturation in certain regions?”
- “Do incremental dollars behave differently by geography?”
- “Should we actually test decreasing dollars?”

Geo-based pulse-ups help you decide whether to lean in or back off. (Important to stress that similar to pulse-ups, we can also test “pulse-downs”, with and without geo-based controls).

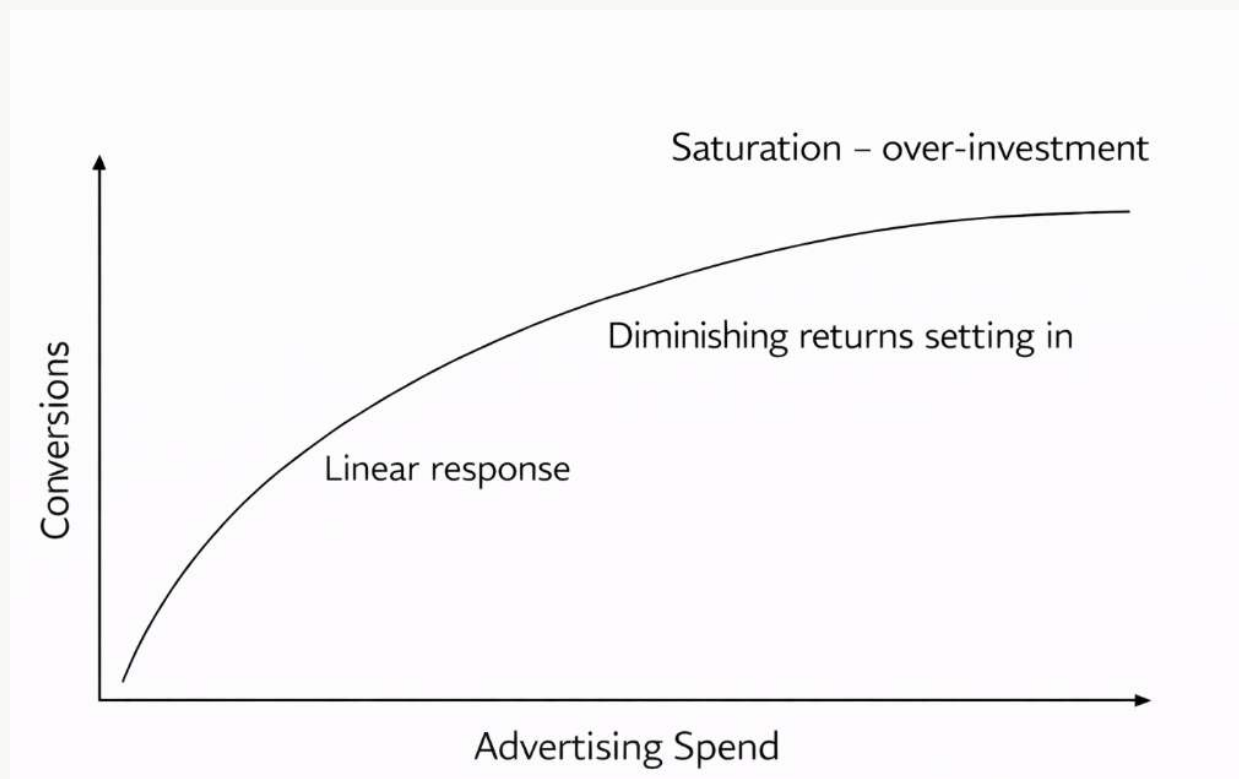
Where Geo-Based Pulse-Ups Fit

If we line up the tools discussed so far:

- User-level holdouts are the cleanest
- Geo tests are next-best when randomization isn't possible.
- On / off tests trade precision for simplicity
- Pulse-ups probe the margin quickly.

Used thoughtfully and repeated over time, geo-based pulse ups can meaningfully improve allocation decisions.

In a really good case scenario, with repeated testing on a channel, you can carve out the response curve below, but no matter what, there will be some level of uncertainty. Nevertheless, once you have something like the below, it would be easy to figure out mathematically where you want to be on this curve, i.e., how much you want to spend on this channel.





But remember, these are imprecise tools when used in isolation.

Examples Where Diminishing Returns Don't Apply

Throughout this chapter, we've emphasized diminishing marginal returns: as you spend more, each incremental conversion usually costs more than the last. This pattern holds remarkably well across most channels and spend levels.

But there are important exceptions. The examples below are taken from Google Paid Search (non-brand), but exist in other channels.

Exception #1: Campaigns Limited by Budget

Suppose you're running Google Search with a target CPA of \$20, and the platform is consistently hitting that target. You also notice the campaign is flagged as "Limited by budget."

(By the way, this should excite you as it gives you 2 different opportunities to improve performance - see below).

In this situation, increasing the budget does not necessarily introduce diminishing returns.



Why?

Because Google is throttling traffic randomly across the same underlying auctions. Roughly speaking, your ads might be showing 3 out of every 4 times they otherwise would.

That missing 4th impression isn't inherently worse traffic. It's drawn from the same keyword mix, same intent, same audiences, and expected CPA. So when you lift the budget constraint, Google simply allows more of the same opportunities to clear.

Result:

You get more volume at roughly the same CPA of \$20.

In other words, you move horizontally along the response curve, not up the marginal CPA curve. Diminishing returns haven't kicked in yet because you were never fully participating in the auction to begin with.

Exception #2: Lowering Target CPA When Budget-Constrained

There's a second, slightly counterintuitive case that also avoids diminishing returns.

Assume: (similar to the above example).

- 
- Your campaign is limited by budget.
 - You're using Google's automated bidding.
 - Your daily budget is fixed at \$180
 - Your current target CPA is \$20

At \$20 CPA, you're getting about 9 conversions per day ($\$180 \div \20).

Now, suppose you lower your target CPA from \$20 to \$18.

What happens?

Google responds by lowering bids. Click costs drop. And because the same clicks are now cheaper, your fixed \$180 budget lasts longer throughout the day. Instead of running out of budget early, you buy more total clicks.

At \$18 CPA, that same \$180 now yields:

10 conversions per day instead of 9

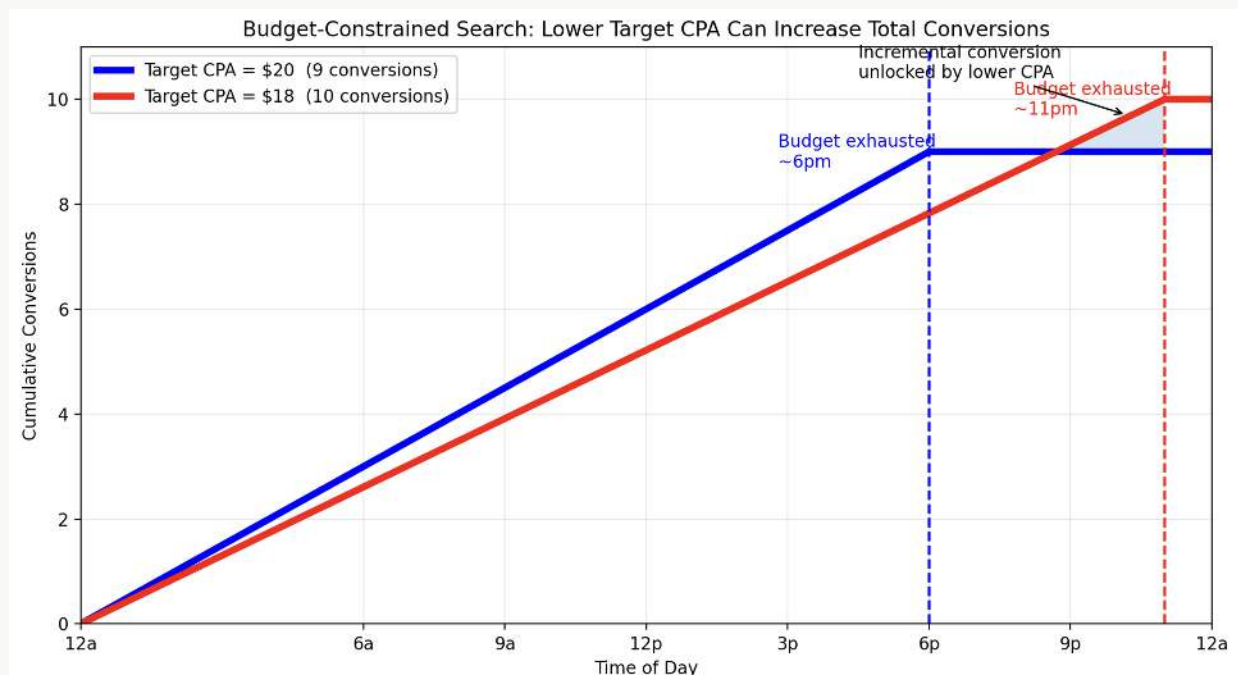
You just gained incremental volume without increasing spend. Wonderful!!

This works only because:

- You were budget-constrained to begin with
- Lower click costs allowed the same budget to stretch further.

See the visual below, which assumes Google uses your budget early in your day (versus spreading the lost impressions throughout the day, but that doesn't change the conclusions). At \$20 CPA, you run out of budget by 6 p.m. (see below), and at \$18 CPA, your budget lasts until 11 p.m., allowing you to get that 1 extra conversion from 9 to 10.

In this example, even at the reduced target CPA, your budget still ran out by the end of the day. It ran out at 11 p.m. (instead of 6 p.m.). There may still be room for improvement, but you get the idea.



Once the campaign is no longer limited by budget, this effect disappears. At that point, lowering the target CPA simply means fewer auctions won, less traffic, fewer conversions (and less spend per day), leaving more unused budget.



The Big Aha!

Both of these cases share the same underlying theme:
You weren't operating on the true margin yet.

Diminishing marginal returns kick in after:

- You're fully participating in the relevant auctions without being budget-constrained (in Google).
- Incremental volume requires reaching into less efficient demand and/or higher click costs.

This is why understanding where you are on the response curve matters just as much as understanding the curve itself.

Section Summary: Marginal Analysis — Key to Budgeting

Knowing that a channel is profitable isn't enough. The question that actually drives smart budget decisions is whether the next dollar you spend there is still profitable. This section showed why the answer is often surprising.



The core insight is relatively simple but is routinely ignored: average CPA and marginal CPA differ significantly. A channel running at a comfortable \$10 average CPA can simultaneously be generating incremental conversions at \$16 each; i.e., above your breakeven. This happens because raising bids or loosening targets reprices all conversions, not just the new ones. The math is straightforward, but it catches even experienced marketers off guard because the aggregate dashboard still looks healthy while the margin turns negative. The same logic works in reverse: tightening a CPA target might improve your average but cut conversions that were actually profitable at the margin.

From this foundation, the chapter builds outward. It introduces a profit-maximizing CPA, which is the specific target where margin times volume peaks, and shows how to approximate it through iterative testing on Google and Meta.

This section offers a practical diagnostic: look at ROAS variance at the keyword or campaign level. If some keywords are running at 5.0 ROAS while others sit at 0.5, and the tool is content hitting 2.0 on average, it's not doing its job. The high-ROAS keywords should be getting bid up for more volume, and the unprofitable ones should be bid down.



This section reframes ad copy testing as a marginal analysis problem, demonstrating that the ad with the higher click-through rate can easily be the loser in terms of profit once you account for cost differences and conversion rates. It then covers pulse-ups (and pulse-downs) as lightweight probes of the response curve: temporarily bump spend, measure what you get for the extra dollars, and use that signal to decide whether to scale. Adding geo-based controls to pulse-ups further strengthens the signal.

The chapter also highlights important exceptions to diminishing returns. When a Google campaign is flagged "limited by budget," increasing spend doesn't necessarily push you up the marginal CPA curve; you're simply buying more of the same traffic the platform was throttling. Even more counterintuitively, lowering your target CPA while budget-constrained can actually increase volume, because cheaper clicks let a "limited by budget" stretch further through the day.

Why should you care?

Because nearly every budget decision you make - scaling up, pulling back, reallocating across channels - is a marginal decision, yet most reporting only shows you averages. If you optimize for averages, you might overspend on channels that don't deliver profitability. Marginal analysis is the bridge between knowing a channel works and knowing how much to spend on it. Without it, even perfect incrementality measurement won't save you from misallocating your budget. This chapter gives you the framework and the practical testing approaches to stop guessing where you are on the response curve and start measuring it.

07

Media Mix Models



What is needed for good MMMs

My first exposure to Media Mix Models (MMM) was back in 2017, but I was merely reading about them then. I was hooked. The Math was elegant, and the Machine Learning built on top of Linear Regression was very appealing to me. I hadn't developed software since the 90's, so I couldn't build a real MMM, but I would frequently experiment with building pseudo ones in Excel. I use the word "pseudo" because while I was able to build the Math in Excel, I couldn't perform cross-validation testing in Excel (see later in this section).



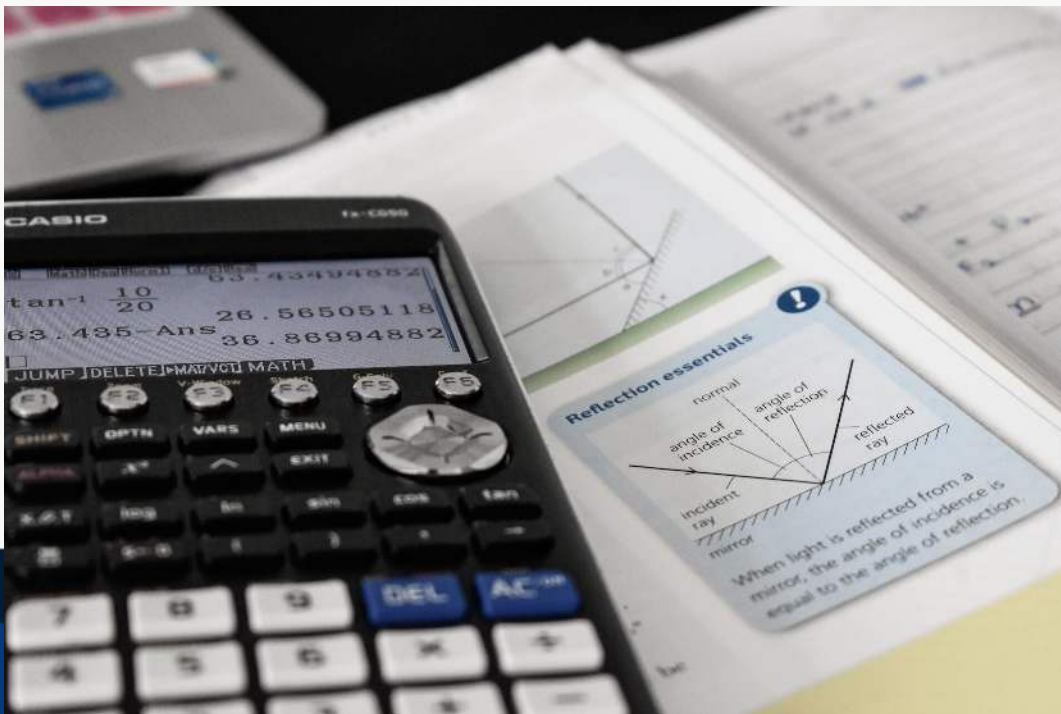
Then, in 2023, at Green Dot, we had to build one to measure TV impact and get a clearer picture of how our channels were performing. Our Data Science team developed the software to build one.

Now with vibe coding, it is much easier to build MMMs. Let me say a word about vibe coding here, and then I will go into more detail on how to implement with a Product Requirements Doc (PRD) in the next chapter.

Vibe Coding puts you at the helm. You are the product manager describing what you want built, and the AI builds it. You still need to have domain knowledge. You use your domain expertise to provide "priors" or constraints to a machine-learning model. You are in full control, using your knowledge and iterating to build the MMM that makes the most sense for you.

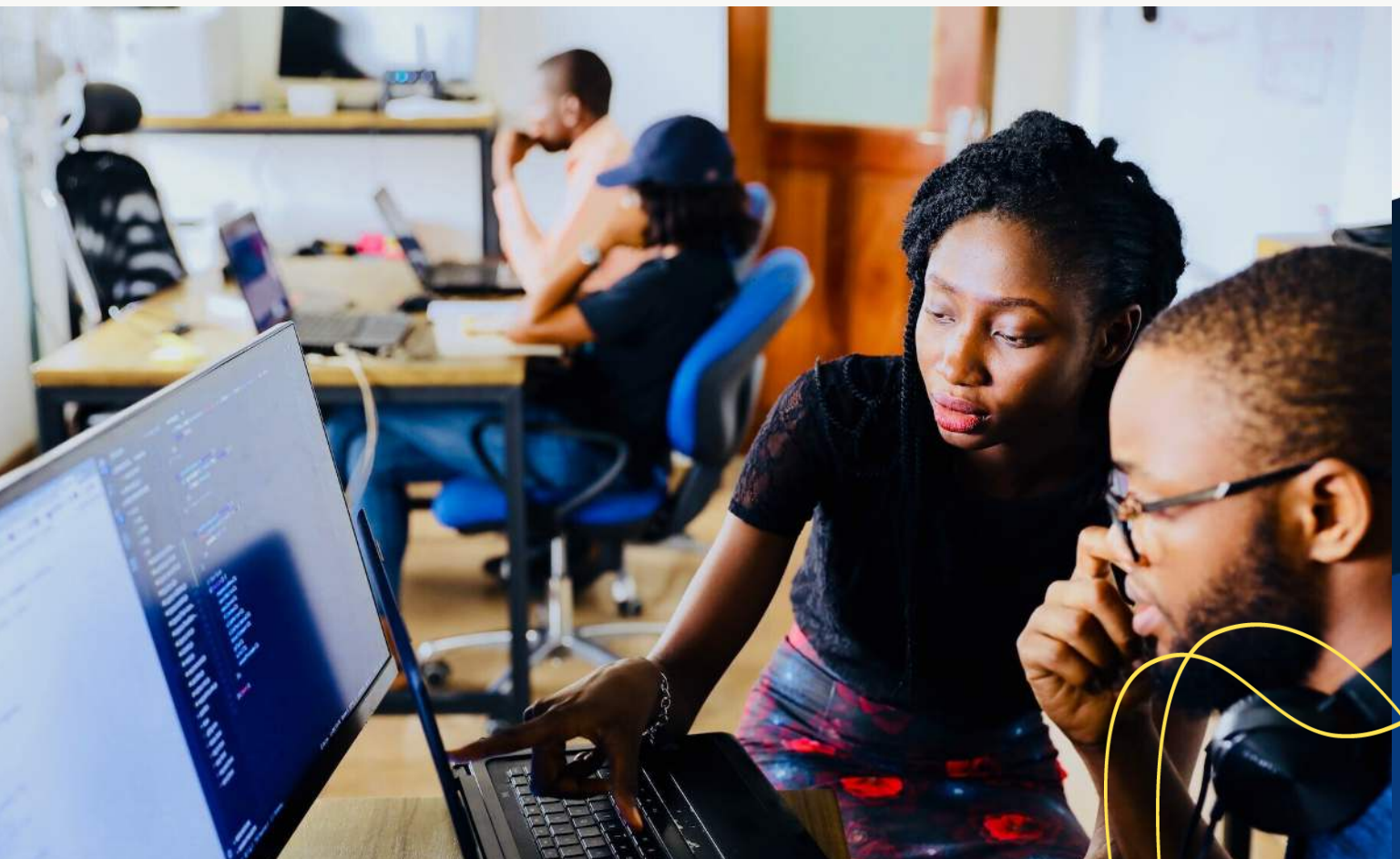
MMMs are very powerful, because they have the potential to bring it all together - incrementality, marginal analysis, exact spend for each channel, etc.

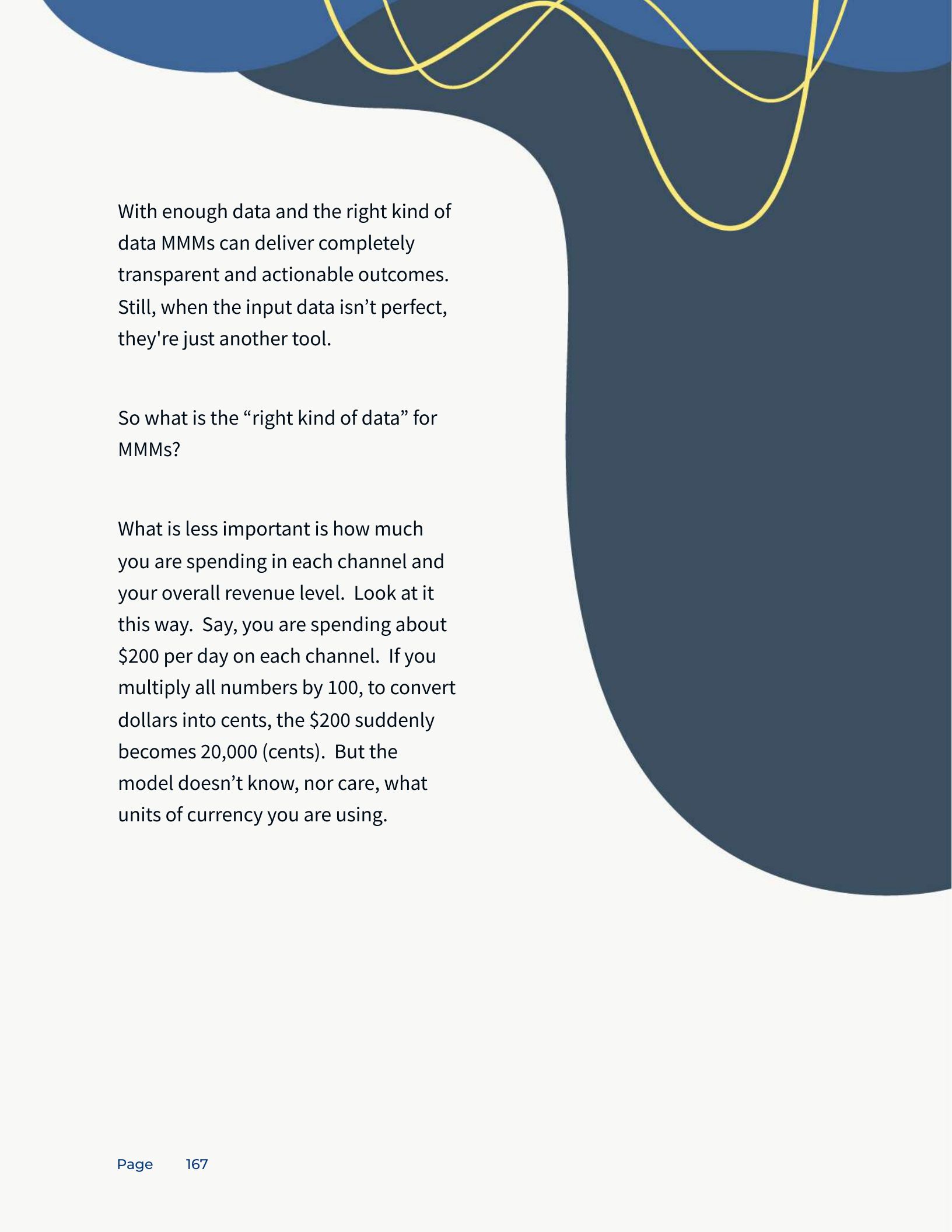
You can even feed your MMMs with the prior data you may have, e.g, data-driven attribution, holdout tests, etc. Now you might ask: “If holdout tests are the most accurate (even if with ranges), why not just use them, rather than first feed them into an MMM. There are two reasons for this:



1. Given the ranges for Incrementality that are inherent in Holdout tests, quite often, you can get better results if you use some kind of averaging with what the MMM would say for the channel or campaign you did a holdout on. This is conceptually similar to the wisdom of crowds, where averaging various estimates yields a more accurate assessment or prediction than using a single estimate.
2. By entering holdout test results for a specific channel or campaign and allowing the MMM to use them, the MMM can improve its results for that channel or campaign without holdout test results.

Before I get into the details of building an MMM from the ground up, let's first discuss why I say that MMMs have "the potential" to bring it all together, rather than actually bringing it all together.

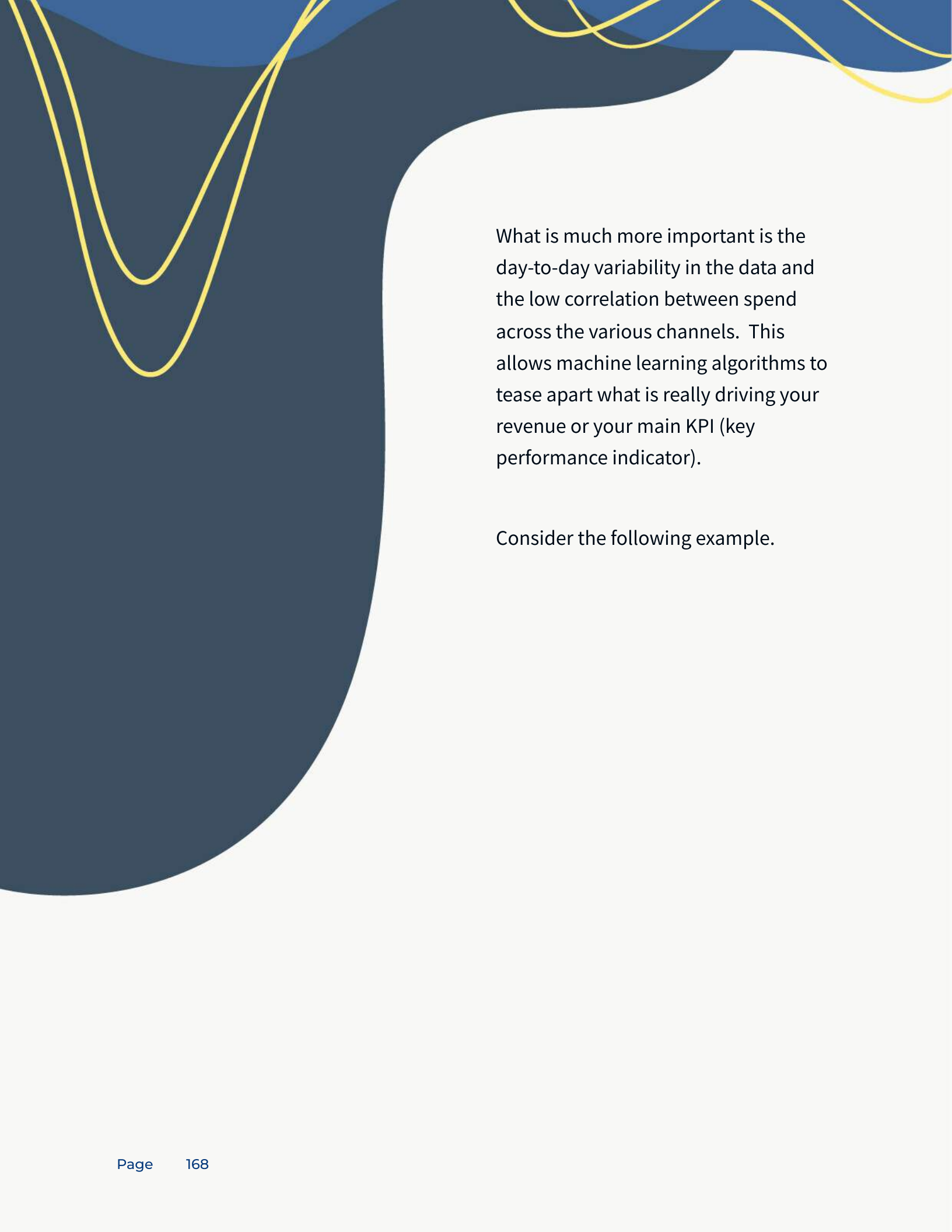




With enough data and the right kind of data MMMs can deliver completely transparent and actionable outcomes. Still, when the input data isn't perfect, they're just another tool.

So what is the “right kind of data” for MMMs?

What is less important is how much you are spending in each channel and your overall revenue level. Look at it this way. Say, you are spending about \$200 per day on each channel. If you multiply all numbers by 100, to convert dollars into cents, the \$200 suddenly becomes 20,000 (cents). But the model doesn't know, nor care, what units of currency you are using.



What is much more important is the day-to-day variability in the data and the low correlation between spend across the various channels. This allows machine learning algorithms to tease apart what is really driving your revenue or your main KPI (key performance indicator).

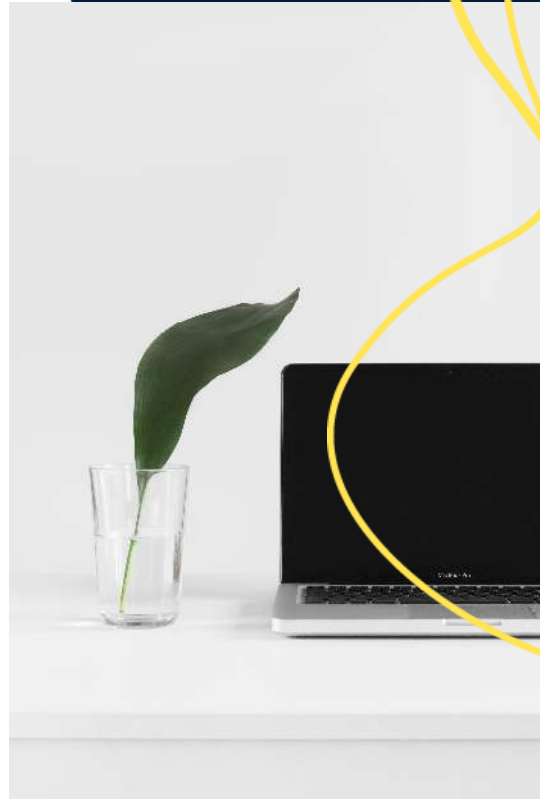
Consider the following example.

Let's say you have no variability in your data and are spending \$100 on TV and \$200 on Podcast ads every day. And your revenue is \$1K per day. You can well imagine, as you think through this, that there is no way any machine learning model can tease apart which channel is actually driving revenues and to what extent. For instance, TV could be driving all the revenue at a 10x ROAS ($1,000 / 100$), while podcast ads are driving nothing. Or perhaps TV is driving no revenue, and Podcast ads are driving all the revenue at a 5x ROAS ($1,000 / 200$). Or something in between.

As soon as TV spend varies, and with it Revenues, then the machine learning model can start learning what TV is actually driving.

So we need the Revenues to vary by day, the channel spends to vary by day, and ideally, the channel spends to have low correlations with each other.

Let's build out more examples for more solid insights here.

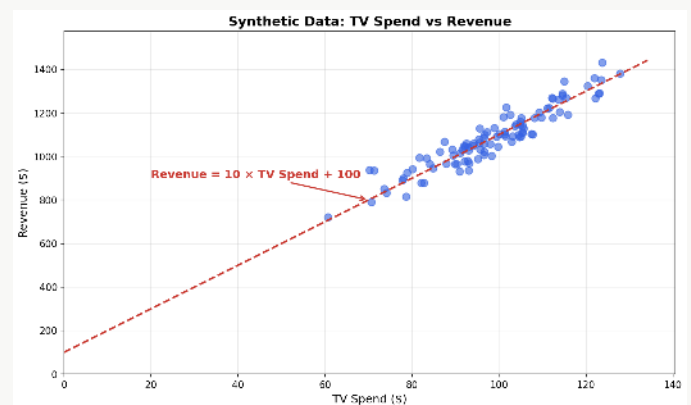


Linear Regression is conceptually simple to understand, yet is frequently known as the “workhorse of Statistics,” and together they make it so very powerful.

Let's start with just one channel and build from there.

Say, you look at your TV spend and Revenue by day, and then plot them as a scatter plot. You then decide to pass a trend line through the data (you can do all of this in Excel).

And you see the below.



The above is a Linear Regression, since you fit the best straight line through a bunch of dots. The equation of this line ($\text{Revenue} = 10 \times \text{TV Spend} + 100$) suggests that for every dollar you spend on TV, you get \$10 additional revenue.





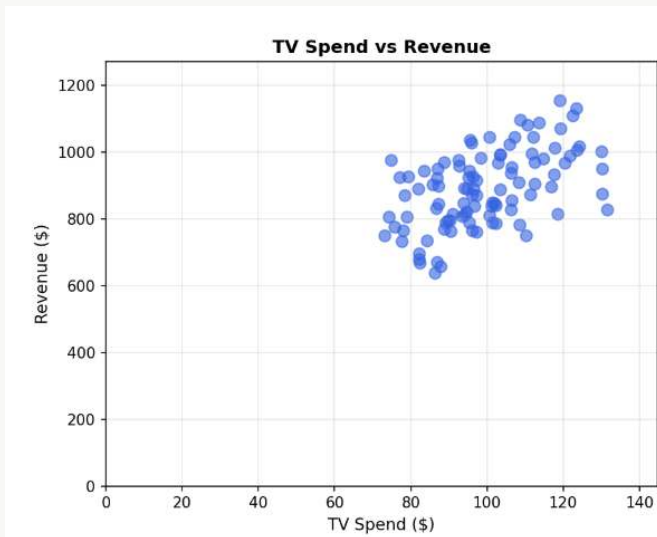
As an aside: "the 'best' straight line is the one that minimizes the sum of squared distances between each point and the line. (We square the distances rather than simply sum them for mathematical reasons we won't cover here.)"

So we now have the first MMM, which tells us that for every dollar spent on TV, we get \$10 in revenue. Yes, it does not allow for diminishing returns, ad stock (tomorrow's impact from today's dollars spent), and is only one channel, but this simple linear regression is an MMM, nevertheless.

Let's build further.

Let's now assume you have 2 channels, TV and Podcast.

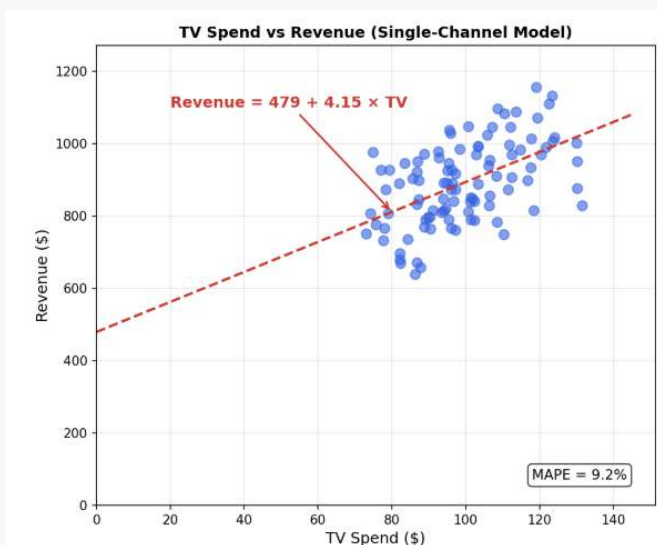
You plot revenue vs. daily TV spend, and get the scatter plot below.



The equation suggests that for every dollar spent on TV ads, you get \$4.15 in Revenues, but the fit isn't that great. (Dots aren't that close to the line, although you can still clearly see the linear trend).

The MAPE at the bottom right needs to be explained. It is the “Mean Absolute Percentage Error”. It tells you, on average, how far off your line is from the dots, expressed as a percentage—so a MAPE of 9.2% means the line here is typically about 9% above or below actual revenue.

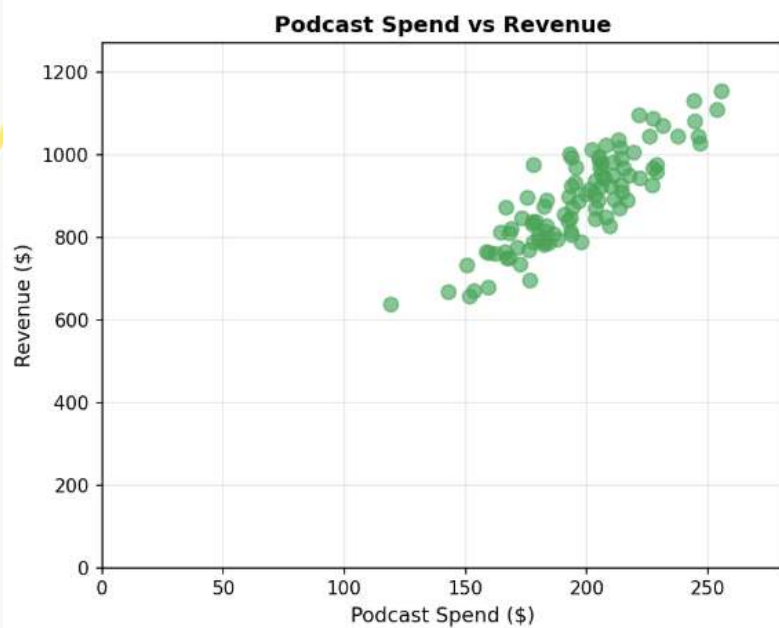
You then find the best-fitting line (trend line in Excel) and get the following:



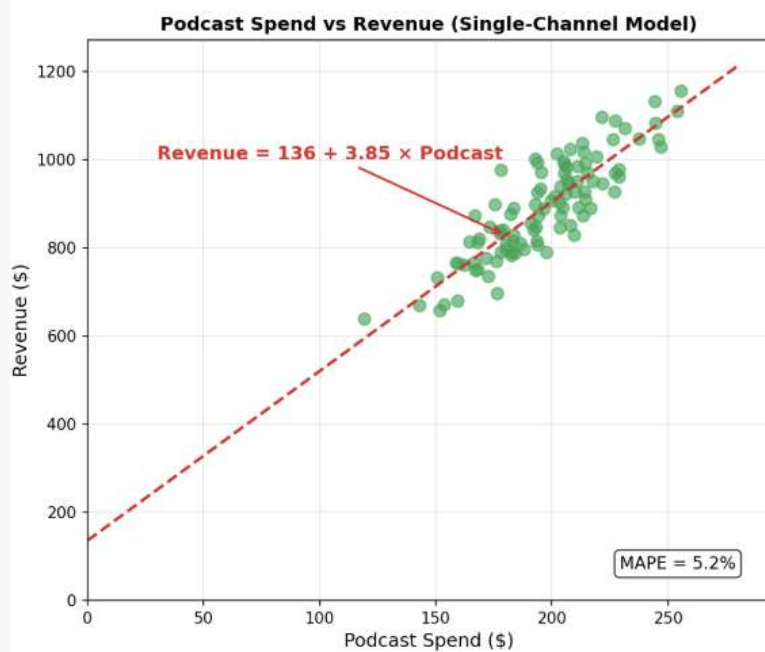
Mathematically, it is calculated as: average of $(|Actual - Predicted| / Actual) \times 100$.

In words: for each data point, calculate how far off you were as a percentage of the actual value, then average those percentages across all points.

Now, let's repeat for Podcast ads.



The best-fitting line for the above gets us this:



This suggests that for each dollar spent on podcast ads, we get back \$3.85 in revenue.

You can see that the line seems to be a better fit than what we got for TV, and the resulting MAPE is also lower (i.e, better) than what we had for TV.

But now, what if we combined both channels (TV and Podcast ads)?

While we can't plot a 3d scatter plot here, we can still get the linear regression equation, and here it is.

Now we get:

$$\text{Revenue} = 63.04 + (1.43 \times \text{TV}) + (3.49 \times \text{Podcast})$$

Since we have both channels here, these numbers are more accurate than when we only look at them channel by channel. This is because when we look at it channel by channel, we miss the impact that the other channel(s) might be having.

The single-channel models gave us TV's impact as 4.15 dollars per dollar spent, and Podcast's impact as 3.85 dollars per dollar spent.

The full model gives TV impact as 1.43 dollars per dollar spent and Podcast's impact as 3.49 dollars per dollar spent.

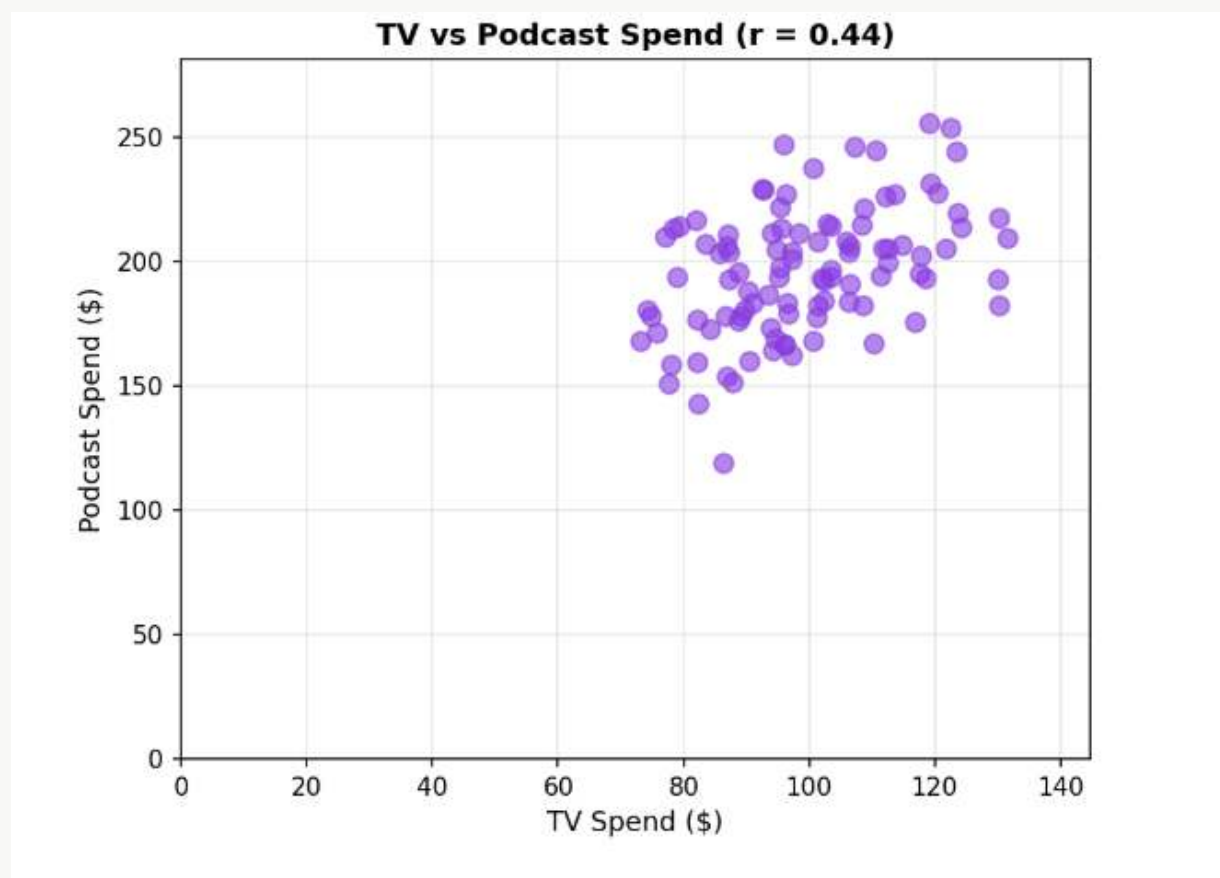
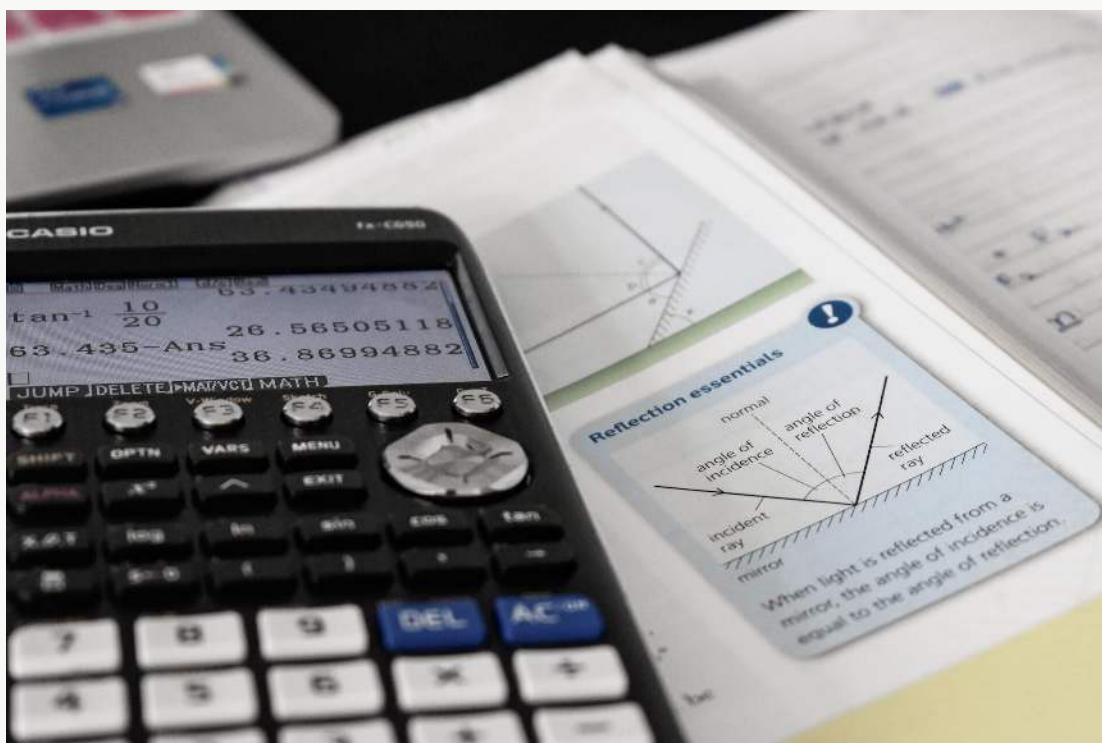
This suggests that, for every dollar spent on TV, we get \$1.43 in Revenues, and for every dollar spent on a podcast, we get \$3.49 in Revenues. Podcast now seems better than TV at generating revenue.





And the MAPE also reduced, suggesting that our 2-channel model is better than taking it one channel at a time. (The MAPE is not guaranteed to improve as we add more channels, but usually does, except for some edge cases).

How good is our model now? Again, it doesn't allow for diminishing returns or ad stock, which we build later. Furthermore, spending across the channels (TV, Podcast) is correlated. (See below).



As you can see from the scatter plot above, the Podcast spend and TV spend move together. When one increases, the other tends to increase as well.

(The exact correlation here is 0.44, and a correlation of 0 means no Correlation; and a correlation of 1 means a perfect Correlation, i.e, a straight line. Note that if the two were to move in opposite directions, then Correlation would be negative).

Now, a high correlation is not good. This is because when channels spend tends to move together, any machine learning MMM will have a harder time figuring out exactly which channel(s) are driving incremental revenues and which channel(s) are not, as the channels tend to move together.



Here, the correlation of 0.44 is okay, but would be much better if we could spend in a way that makes the channel spend less correlated. We can't change history, but we can plan for the future to get more accurate MMM results.

I want to stress one thing in this equation before we move on:

$$\text{Revenue} = 63.04 + 1.43 \times \text{TV} + 3.49 \times \text{Podcast}$$

We are getting a precise mathematical representation of how Revenues change based on spend by each channel (by day). This is very transparent, and as we build the model, it can tell us everything we need to know, including how much to spend in each channel and what the forecasted revenue is.

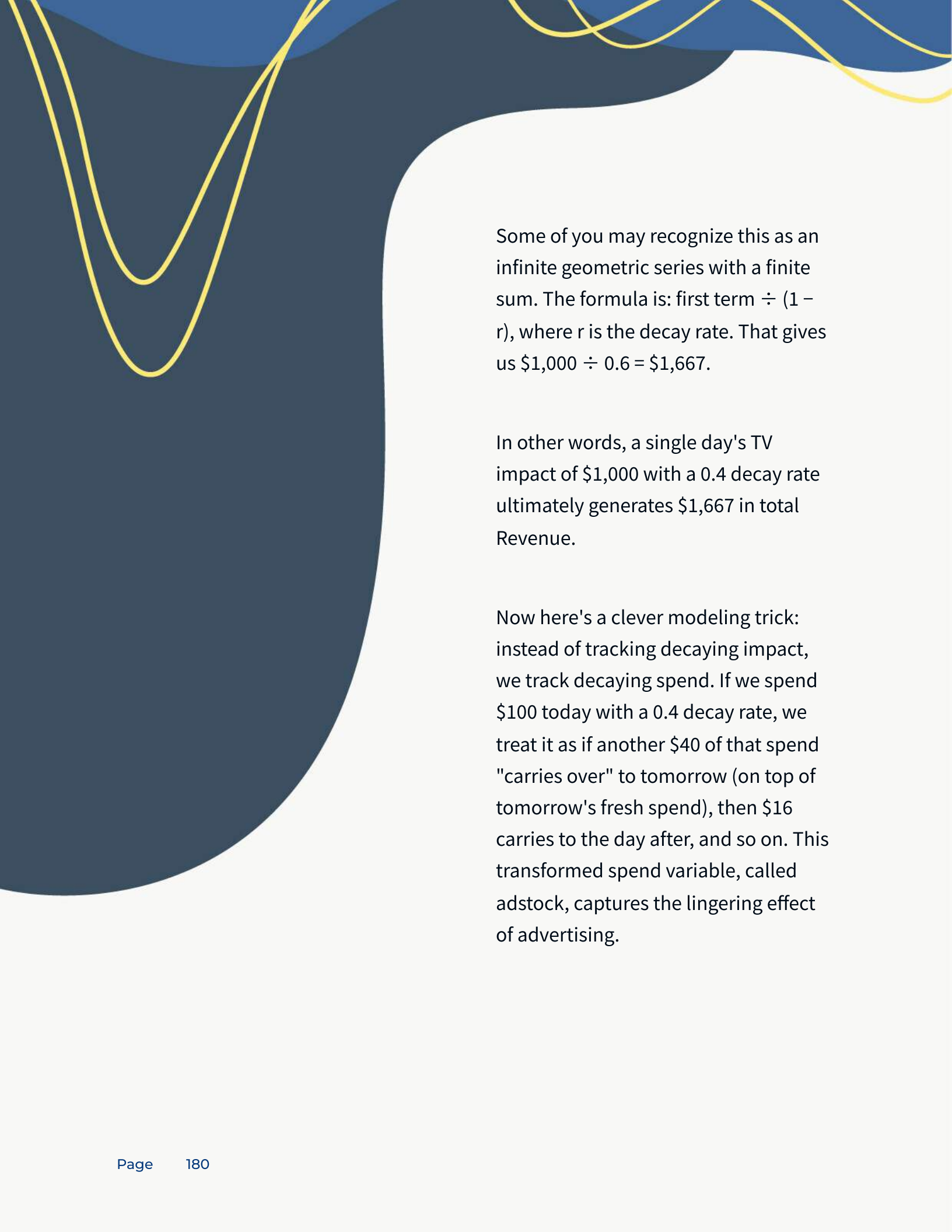


Adding the decay factor (adstock)

So far, our scatter plots and Linear Regression equations assume same-day impact, i.e., whatever effect TV spend has on Revenue happens entirely that same day. But we know that's not true. TV spend today (or Podcast ads today) can drive Revenue not just today, but also tomorrow, the day after, and beyond.

One way to model this is with a decay rate, a multiplier less than 1. Say the same-day impact is \$1,000. If our decay rate is 0.4, the next day's impact is 40% of that: \$400. Day 3 is 40% of \$400: \$160. And so on.

So the total impact becomes: $\$1,000 + \$400 + \$160 + \$64 + \dots$ forever.

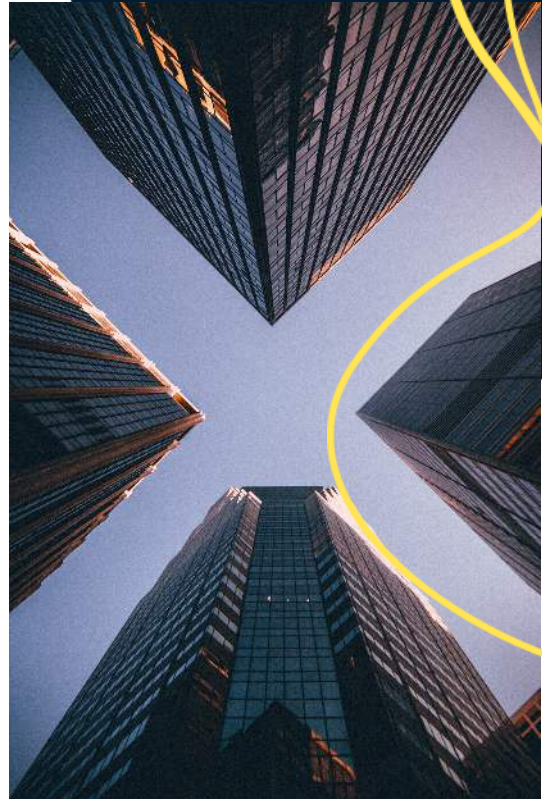
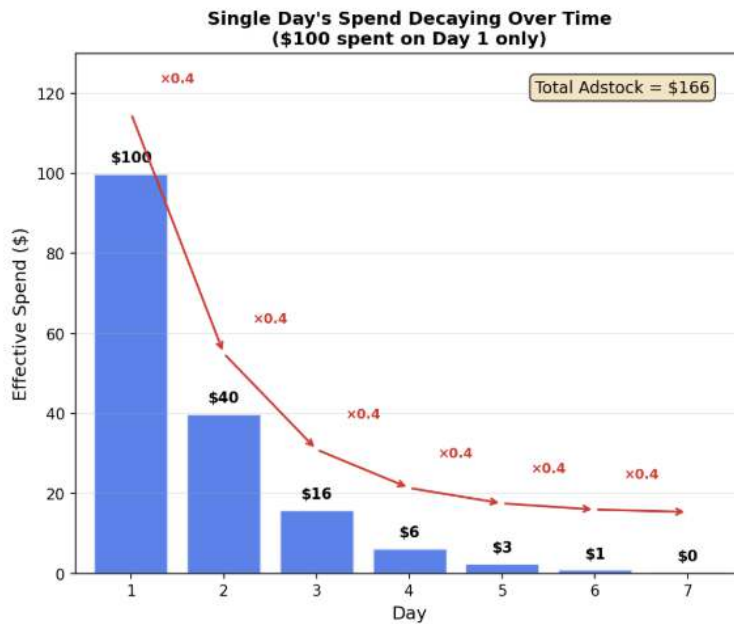


Some of you may recognize this as an infinite geometric series with a finite sum. The formula is: $\text{first term} \div (1 - r)$, where r is the decay rate. That gives us $\$1,000 \div 0.6 = \$1,667$.

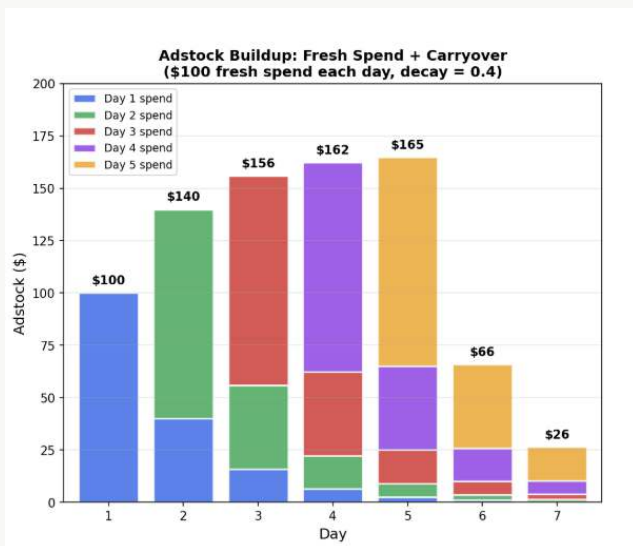
In other words, a single day's TV impact of \$1,000 with a 0.4 decay rate ultimately generates \$1,667 in total Revenue.

Now here's a clever modeling trick: instead of tracking decaying impact, we track decaying spend. If we spend \$100 today with a 0.4 decay rate, we treat it as if another \$40 of that spend "carries over" to tomorrow (on top of tomorrow's fresh spend), then \$16 carries to the day after, and so on. This transformed spend variable, called adstock, captures the lingering effect of advertising.

Note the visual below for an illustration of how a single-day ad spend decays over time.



And below, we assume the full ad stock, where we spend a fresh \$100 from day 1 to day 5, and how the total impact stacks up and then dies out once we stop spending new \$\$ on day 5.



Let's rewrite this more generally:

$$\text{Revenue}_n = a_0 + a_1 \cdot \text{TV Spend}_n + a_2 \cdot \text{Podcast Spend}_n$$

In words: Revenue on day n is a function of TV Spend on day n and Podcast Spend on day n.

Once we include the decay factor, this becomes:

$$\text{Revenue}_n = a_0 + a_1 \cdot \frac{\text{TV Spend}_n}{1 - r_{\text{TV}}} + a_2 \cdot \frac{\text{Podcast Spend}_n}{1 - r_{\text{Podcast}}}$$

So effectively, we're dividing each day's spend by $(1 - r)$ to get its total impact. In our example, r was 0.4.

Let's generalize this with symbols.

Our earlier Linear Regression equation for TV and Podcast spend was:

$$\text{Revenue} = 63.04 + (1.43 \cdot \text{TV}) + (3.49 \cdot \text{Podcast})$$





Notice each channel can have its own decay factor - r_{TV} doesn't have to equal $r_{Podcast}$.

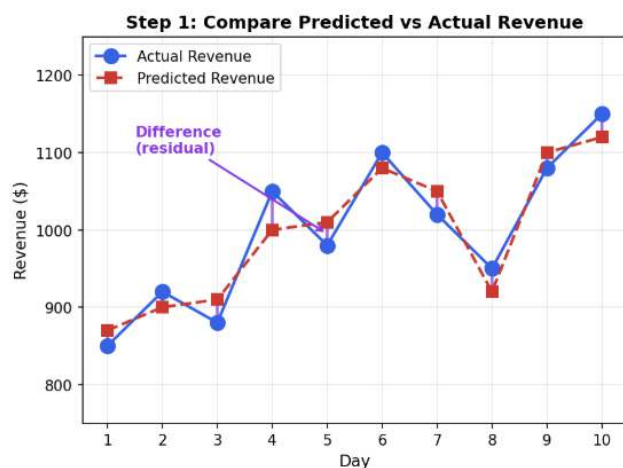
TV ads might have a longer carryover effect than Podcast ads, or vice versa.

Once we have this setup, we can determine the optimal values of a_1 , a_2 , a_3 , r_{TV} , and $r_{Podcast}$ using our data. You can even do this with the Excel Solver.

Here's the intuition. We try different combinations of a_1 , a_2 , a_3 , $r(TV)$, and $r(Podcast)$. For each combination, we calculate what revenue the model predicts for each day. Then we compare that to the actual revenue, take the difference, and square it. We do this every day and sum up all those squared differences.

The "best" values are simply the ones that minimize this sum - the combination where predictions come closest to reality across all days.

See visually below.



Step 2: Calculate Sum of Squared Differences

Day	Actual	Predicted	Diff	Diff ²
1	\$850	\$870	-20	400
2	\$920	\$900	+20	400
3	\$880	\$910	-30	900
...	...	...	...	...
10	\$1150	\$1120	+30	900
			Sum:	8,600

The 8,600 above comes from one selection of a_1 , a_2 , a_3 , r_{TV} , and $r_{Podcast}$. Our goal is to find the selection that minimizes this number.

The Optimization Process

1. Pick values for a_1 , a_2 , a_3 , r_{TV} , $r_{Podcast}$
2. Calculate predicted revenue for each day
3. Compare to actual revenue:
 - Take the difference
 - Square it
 - Sum across all days
4. Try different combinations
5. Keep the combination with the lowest sum of squared errors

Goal: Minimize $\sum (\text{Actual} - \text{Predicted})^2$

While the above is the process, when implemented alongside existing approaches, it is quite efficient, leveraging extensive mathematical theory to quickly converge to a solution. (More on this later).

Note: The $\text{Spend}/(1-r)$ formulation is useful for intuition (total impact of a single day's spend).

However, the actual revenues are for day n . and predicted revenues for day n stretch out over time due to r , the ad stock decay factor.

The correct approach uses the more correct, but harder, recursive adstock formula:

$$\text{Adstock}(n) = \text{Spend}(n) + r \times \text{Adstock}(n-1)$$

This says: the advertising pressure hitting day n is today's spend plus the carryover from all previous days. So:

- Day 1: $\text{Adstock}(1) = \text{Spend}(1)$
- Day 2: $\text{Adstock}(2) = \text{Spend}(2) + r \times \text{Spend}(1)$
- Day 3: $\text{Adstock}(3) = \text{Spend}(3) + r \times \text{Spend}(2) + r^2 \times \text{Spend}(1)$
- And so on...

Now, $\text{Revenue}(n)$ is compared to $\text{Adstock}(n)$ —the accumulated advertising pressure on that specific day.

Next, we move to implementing another very important concept in our MMM: diminishing returns.

Implementing Diminishing Returns

Let's again consider equations of the form (our progress so far):

$$\text{Revenue}_n = a_0 + a_1 \cdot \frac{\text{TV Spend}_n}{1 - r_{\text{TV}}} + a_2 \cdot \frac{\text{Podcast Spend}_n}{1 - r_{\text{Podcast}}}$$





Let's simplify the above by assuming that both r_{TV} and $r_{Podcast}$ are 0, i.e., spending today has an impact on Revenues today and no impact on subsequent days. Let's also assume that a_1 and a_3 are 0. (These assumptions just simplify our analysis for diminishing returns).

So we now have:

$$\text{Revenue}(n) = a_1 * \text{TV_Spend}(n)$$

Let's also assume $a_1 = 5$.

So we have:

$$\text{Revenue}(n) = 5 * \text{TV_Spend}(n)$$

The one issue with this is that there are no diminishing returns built in. If we spend \$100 on TV, the prediction is that it will generate \$500 in revenue. And then if we double the TV spend to \$200, the Revenue prediction also doubles to \$1,000.

But for the most part, diminishing returns do exist, either because to get more volume, we increase unit costs (like click costs), or because we are reaching out to audiences who may not convert as well, etc.

How can we model diminishing returns?

Well, one way is to raise the Spend number to an exponent (power) less than 1.

Let's see what happens if we use 0.95

So now, we have:

$$\text{Revenue}(n) = 5 * (\text{TV_Spend}(n)) ^ 0.95$$

If our TV spend now is \$100, predicted Revenues are: \$397

If we double TV spend to \$200, predicted Revenues are: \$767, i.e, they don't double (in this case, increase by about 93%). And this fall off can be more pronounced if we use a power less than 0.95.

And now we put it all together, as shown below, we have both the lag effect and diminishing returns.

$$\text{Revenue}_t = a_0 + a_1 \left(\frac{\text{TV Spend}_t}{1 - r_{\text{TV}}} \right)^{b_1} + a_2 \left(\frac{\text{Podcast Spend}_t}{1 - r_{\text{Podcast}}} \right)^{b_2}$$

One subtle point above is that we first apply the ad stock (lag effect), then apply diminishing returns to the ad stock (i.e., cumulative "effective" spend).



As earlier, once we have this setup, we can solve for the best values of a 's, r 's, and now b 's, using the data we have. You can even do this in the Excel Solver.

Here's the intuition. We try different combinations of a 's, r 's, and b 's.

For each combination, we calculate what revenue the model predicts for each day. Then we compare that to the actual revenue, take the difference, and square it. We do this every day and sum up all those squared differences.

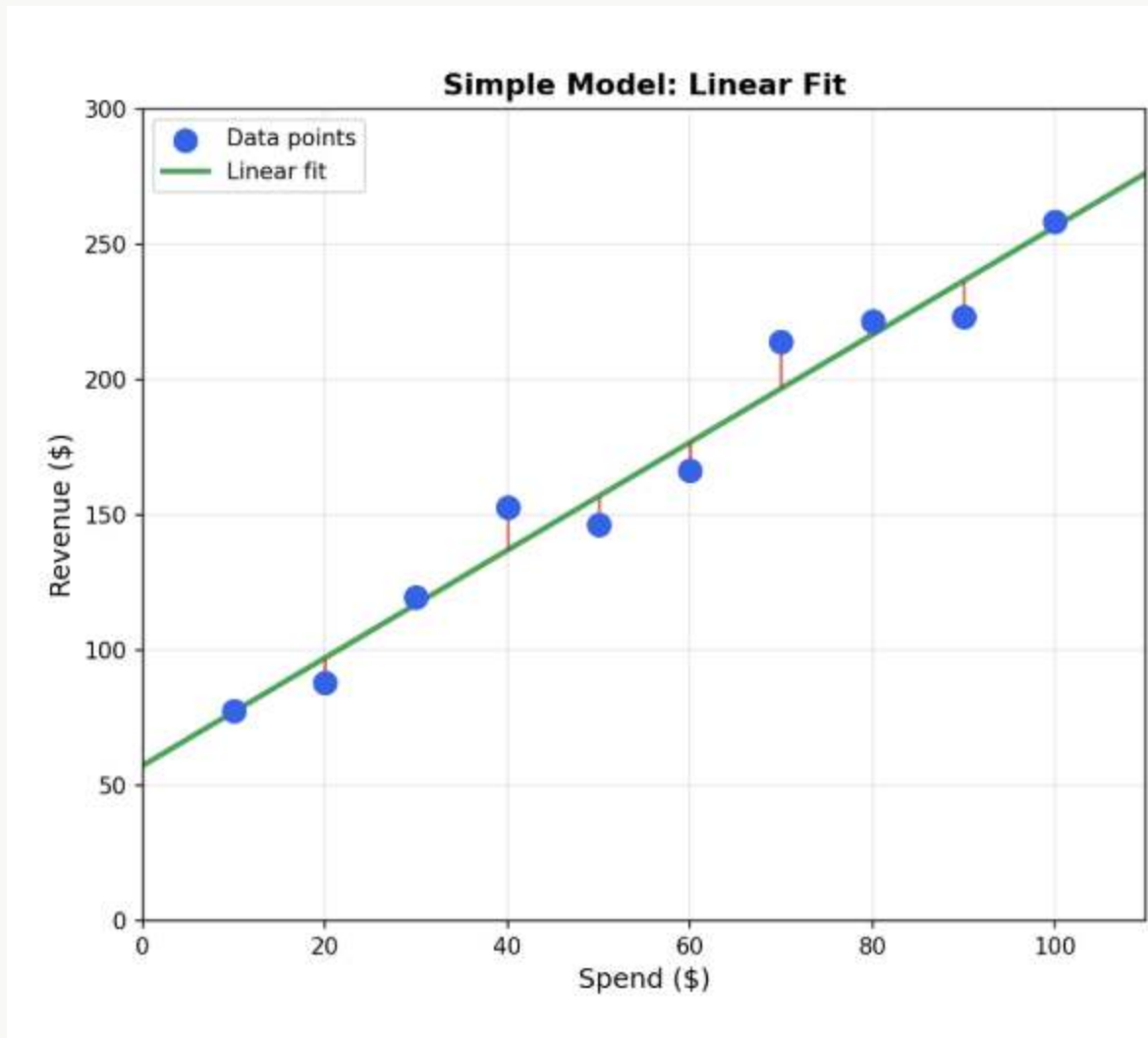
As before, the "best" values are simply the ones that minimize this sum - the combination where predictions come closest to reality across all days.

Wow! We have come a long way in setting up a fairly reasonable MMM and solving it in Excel. This wouldn't be a bad start to get those additional insights you want from an MMM.

But there is still one key ingredient missing - and that is dealing with "overfitting".

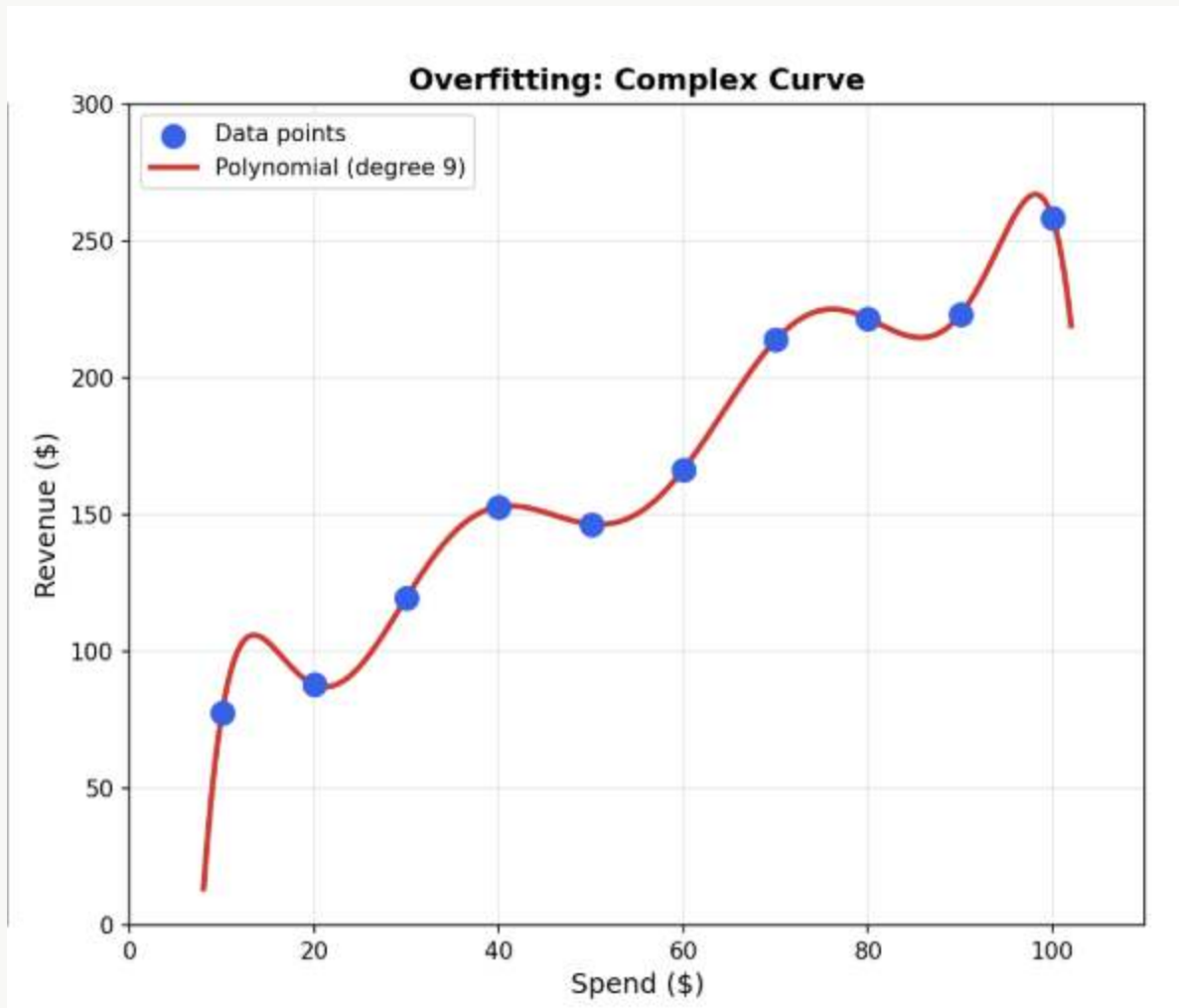
Overfitting the data

To consider overfitting, consider the following data point, and the fairly good curve fitting with a simple straight line.



Could we do better? The answer is yes, in the strict sense of curve fitting.

The curve below is arguably better, as it passes exactly through all the points with no error (the deviations are all 0).



But intuitively, this can't possibly be better than the earlier straight line. For one thing, this is a 9th-degree polynomial, and the last term is actually decreasing. That alone would imply that as we increase spending to \$105, the predicted revenues would decrease, not increase!

We have overfit the data!



This happens when we have many degrees of freedom (flexibility). Here, we allowed ourselves to use a 9th-degree polynomial. With the additional leverage, we were able to drive deviations to 0, i.e., a perfect fit, but it was clearly not an overall good fit.

Now, in our example earlier, with all the a's, r's, and b's, we have a lot of flexibility and are at great risk of overfitting the data.

What do we do?

How to avoid overfitting?

There are two main approaches to avoid overfitting: penalize complexity during training, and test on data the model hasn't seen.

Penalizing Complexity: Ridge Regression

Remember our goal: minimize the sum of squared errors. The problem is, a complex model can always reduce errors by adding more wiggles until it perfectly memorizes every data point, as we saw in the earlier visual.

Ridge regression adds a twist. Instead of just minimizing errors, we minimize errors plus a penalty for large coefficients. The formula becomes:

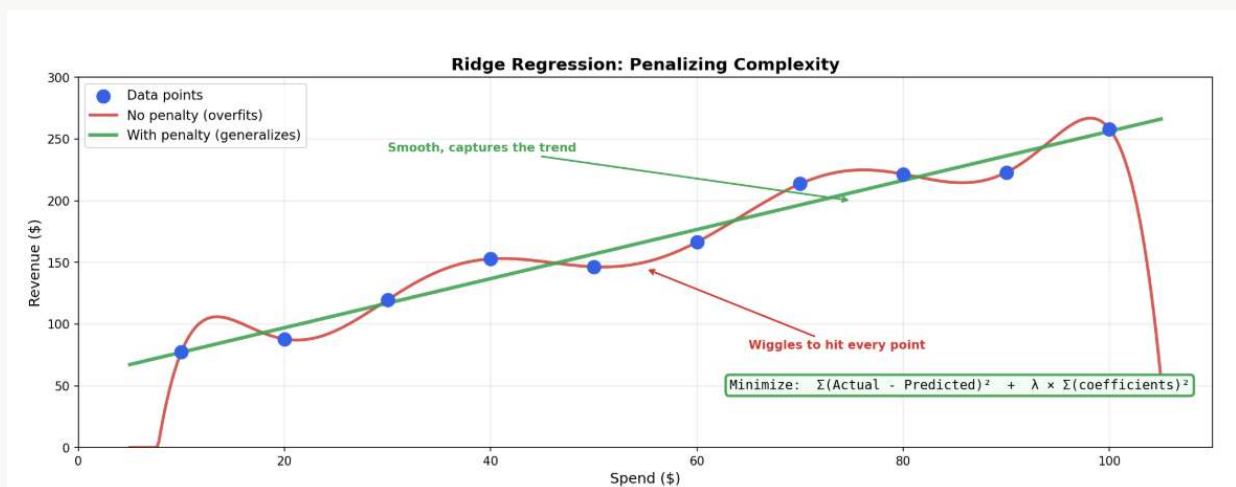
Minimize: (Sum of Squared Errors + λ * (sum of squared coefficients))

That λ (lambda) is a tuning parameter. When λ is zero, you get regular regression. In our case:

$$\text{Revenue}_t = a_0 + a_1 \left(\frac{\text{TV Spend}_t}{1 - r_{\text{TV}}} \right)^{b_1} + a_2 \left(\frac{\text{Podcast Spend}_t}{1 - r_{\text{Podcast}}} \right)^{b_2}$$

As λ increases, the model is penalized more heavily for large coefficients, thereby forcing it to remain simpler.

Think of it this way: the model now has to justify every coefficient. A coefficient that reduces error by 10 but adds 15 to the penalty isn't worth it. This prevents the model from contorting itself to hit every data point.



Above - Ridge Regression: The red curve wiggles to hit every point (overfitting). The green line accepts some error, but stays smooth because of the penalty term. The formula shows the trade-off: minimize errors, plus a penalty for coefficient size.



Cross-Validation: Testing on Unseen Data

The second defense is simpler: don't just measure error on the data you trained on. Hold some data back, train the model on the rest, then see how it performs on the held-out portion.

A common approach is k-fold cross-validation—randomly splitting data into chunks and rotating which chunk is held out. But that doesn't work here.

Why?

Because of adstock, day 15's revenue depends on spend from Days 14, 13, 12, and so on. If Day 14 is in your training data and Day 15 is in your test data, you've got leakage—your "unseen" test data is contaminated.

For MMM, you need time-based splits instead:

1. Train on earlier data (say, Days 1–56 or 8 weeks)
2. Test on later data (Days 57 to 70, or weeks 9 and 10)
3. The model has never seen the test period, so this is a true out-of-sample test.

You also add a gap between training and test to let the adstock decay. If your decay rate is 0.4, the carryover effect is negligible after about a week. So you might train on the first 8 weeks of data, skip week 9, and test on weeks 10 and 11.



If a model has low training error but high test error, it's overfitting: it memorizes the training data but can't generalize.

We will go through more details later when doing this for real in a PRD (product requirements doc).

Ridge Regression and Cross-Validation Together

For MMM, both techniques matter. Ridge regression keeps coefficients reasonable. Cross-validation tells you whether your model will actually predict future revenue, not just explain the past.

The goal isn't zero error. It's a model that captures real patterns without chasing noise.

Here's how to bring the two together in practice.

The time-based split described above also helps you pick the best hyperparameter (λ). In our case, that's λ from the Ridge regression section. This λ sits outside the optimization—the optimization only finds the best a's, b's, and r's for whatever λ you give it.



Here's the thing: for a given λ , the optimization gives you exactly one set of coefficients (a's, b's, and r's). But different λ values produce different coefficient sets. A small λ barely penalizes complexity, so the model might overfit. A large λ penalizes heavily, so the model stays simple but might miss real patterns.

So the process is:

1. Set $\lambda = 0.1$, train on weeks 1–8 → get one set of parameters (a's, b's, and r's) → measure error on weeks 10–11
2. Set $\lambda = 1$, train on weeks 1–8 → get a different set of parameters → measure error on weeks 10–11
3. Set $\lambda = 10$, train on weeks 1–8 → get yet another set of parameters → measure error on weeks 10–11
4. And so on
5. Pick the λ that gave the lowest test error

Once you've found the best λ , train one more time on the full dataset using that λ . This gives you the final a's, b's, and r's.

And now you have your complete MMM: the coefficients (a's), the diminishing returns exponents (b's), and the decay rates (r's) for every channel.



This whole section on avoiding overfitting with cross-validation across folds may not be feasible in Excel (though it may be; I just haven't tried). It would be much better to do this in Python. With Vibe Coding, you won't need to know Python, and we will talk about an approach later. But for now, let's look at some of the output that MMM can produce and why it is so easily actionable to maximize your Profits.

Actionable MMM Output

Once we run the MMMs, we may get an equation like this (assuming 2 channels here to make the explanation a bit easier, without losing any generality).

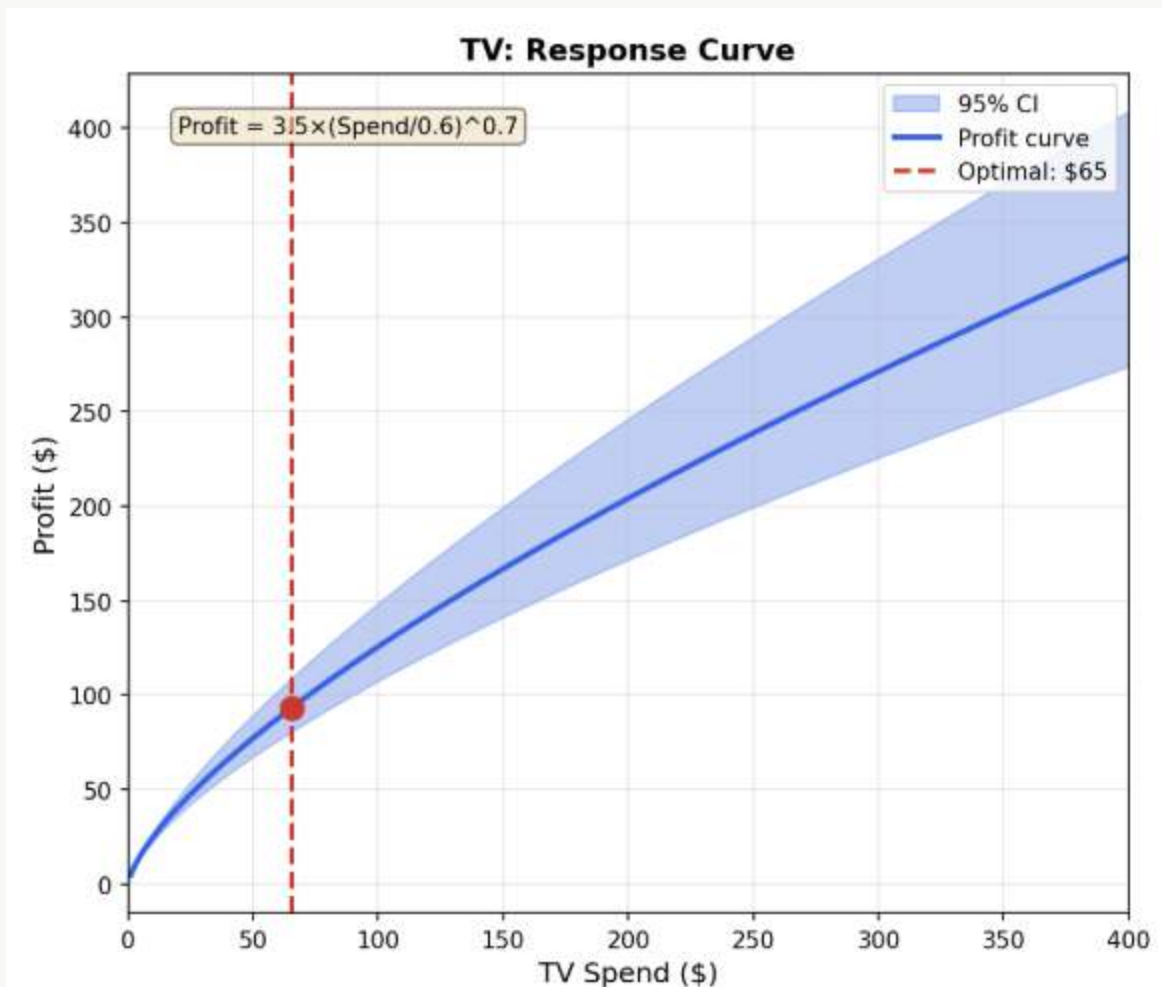
Let's say the "optimized model parameters" are below, with Profit being plotted on the Y axis.

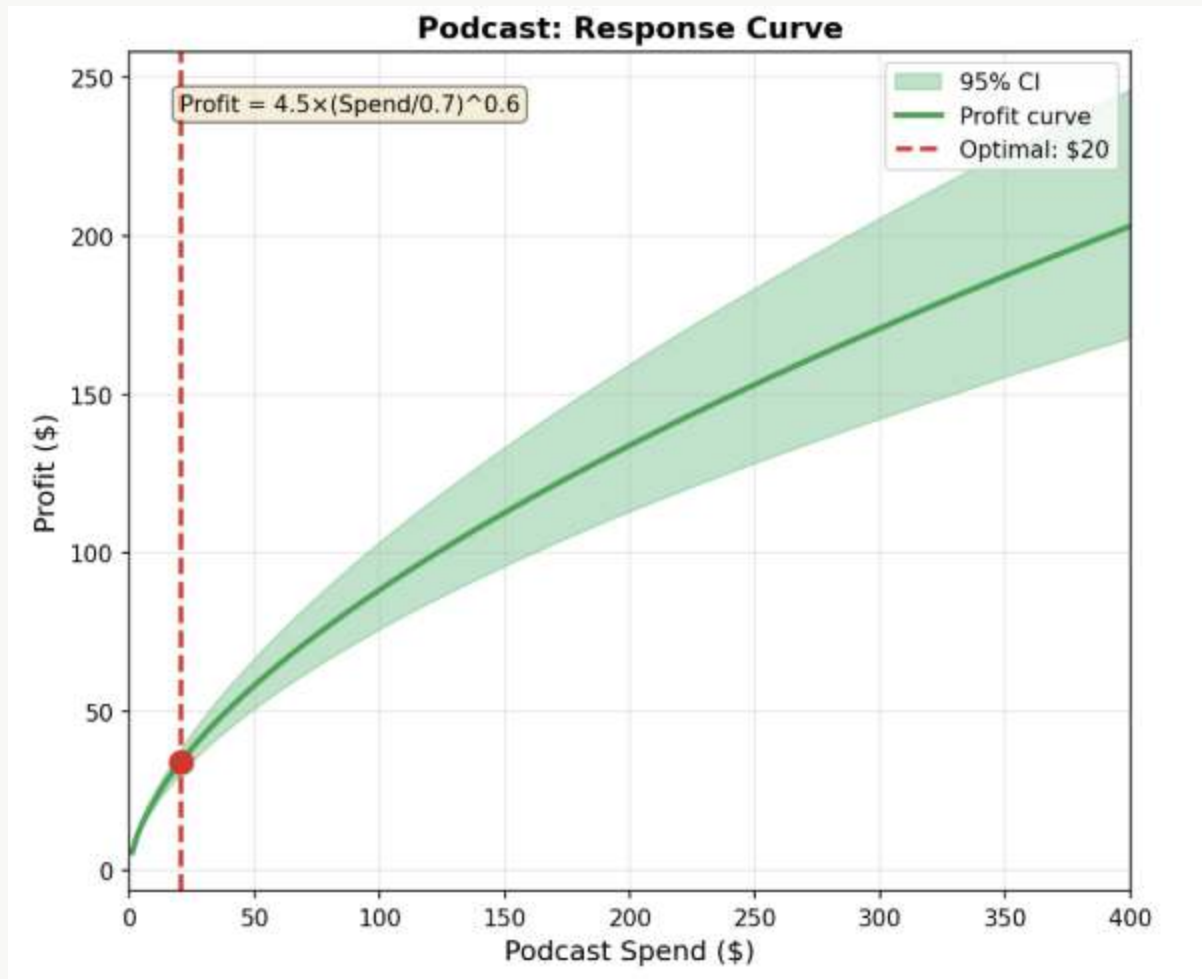
This is Profit before ad costs, but after other variable costs (e.g., cost of goods sold in e-commerce).

Channel	a	b	r
TV	3.5	0.7	0.4
Podcast	4.5	0.6	0.3

Remember that a is the coefficient of the spend, b is the power (exponent for diminishing returns), and r is the memory or lag factor.

Let's draw out the above curves, which the MMM can also output, known as the response curves.





Note that in the curves above, we have also plotted a 95% confidence interval around the curves, i.e., how sure are we about the curves? These can be obtained with a Monte Carlo Simulation in the following way:

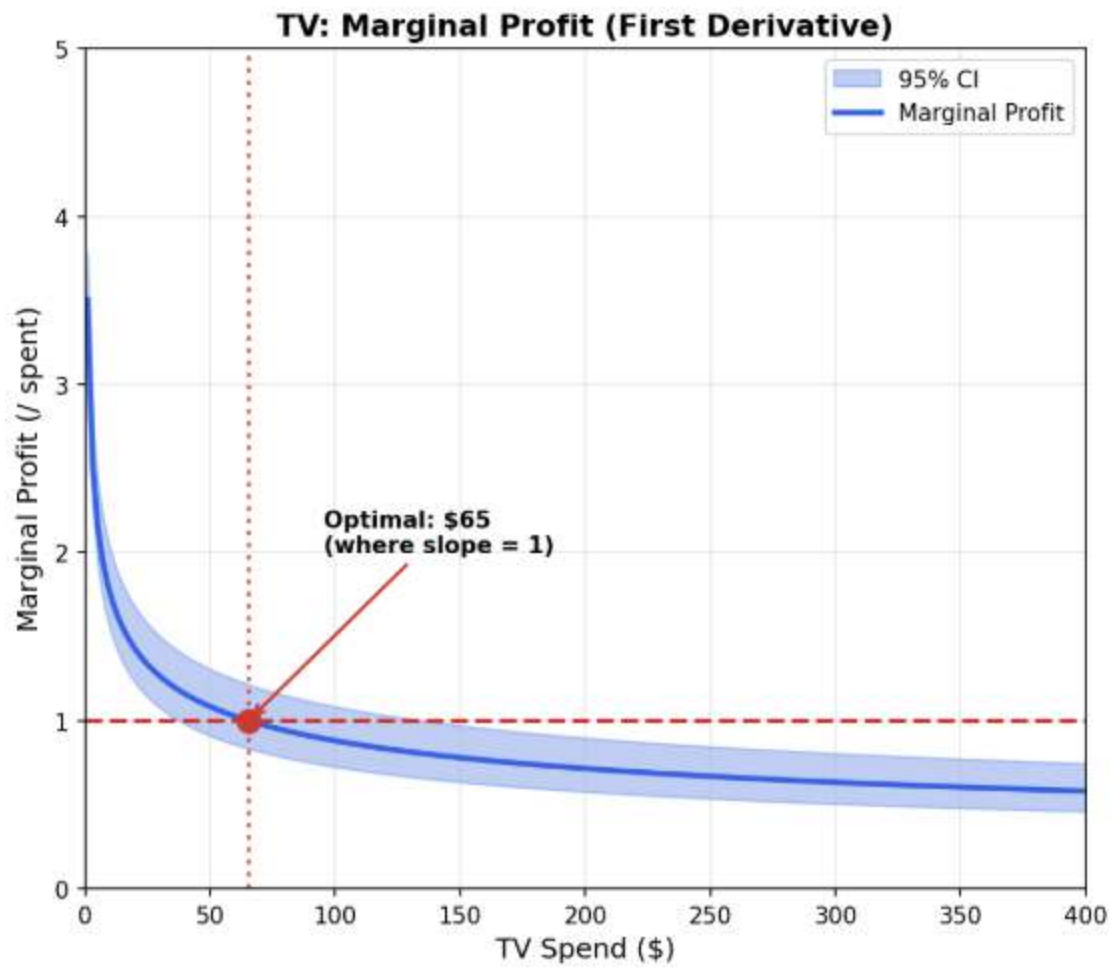
Monte Carlo simulation

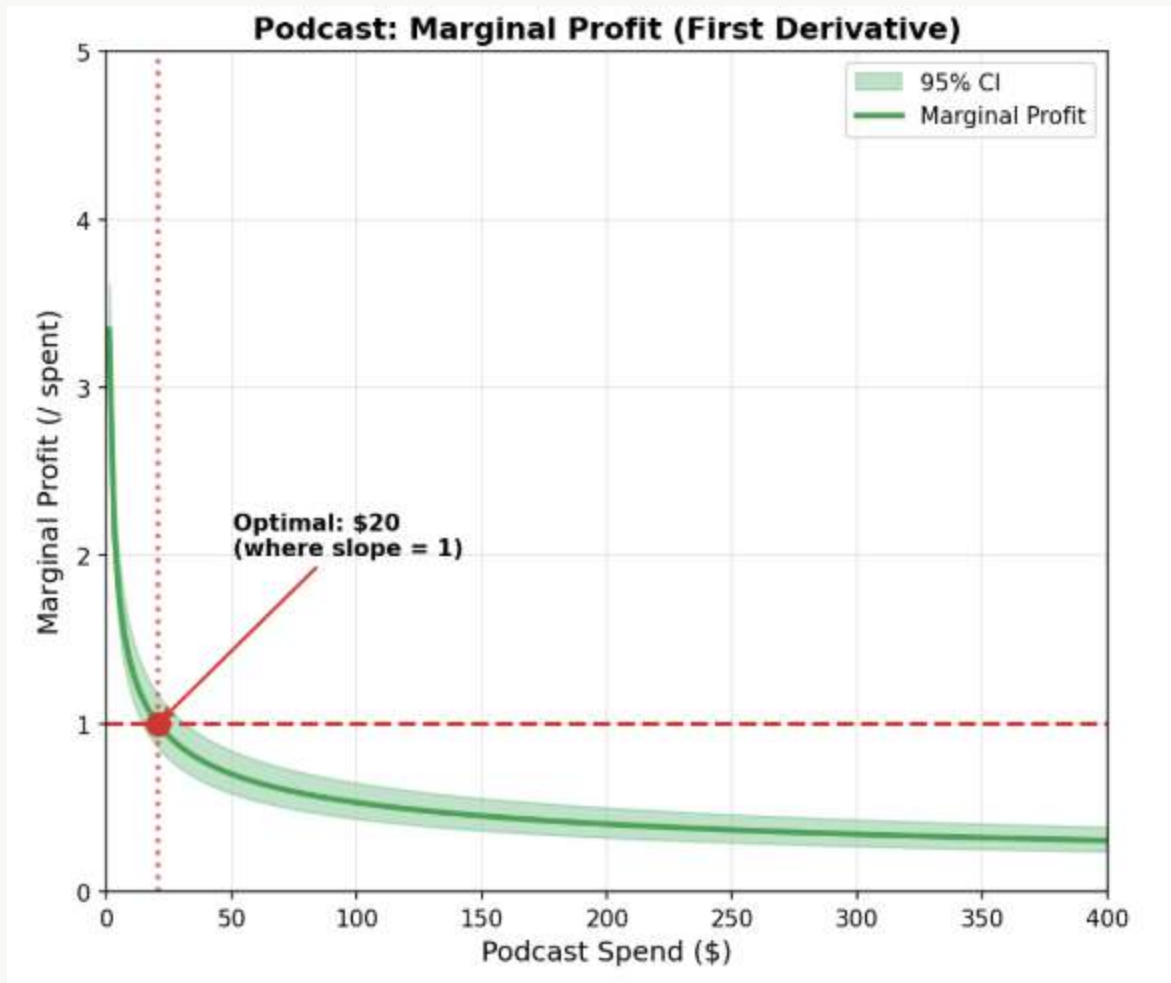
- 
1. Assume each parameter has a point estimate and a standard error (e.g., $a = 3.5 \pm 0.3$). The modeling will produce standard errors, so the 0.3 should be replaced with whatever that is.
 2. Draw random samples from those distributions (e.g., 1,000 draws)
 3. For each draw, calculate the entire profit curve
 4. Take the 2.5th and 97.5th percentiles at each spend level to generate the curves above

Note the red lines in the curves above—these mark the profit-maximizing spend for each channel. You could find this by trial and error, testing various spend levels until profits minus ad costs stop increasing. Intuitively, you keep spending until the extra profit generated by the next dollar no longer covers that dollar.

Those of you who remember high school Calculus will recognize a shortcut: take the derivative of the profit curve. The profit-maximizing point is where that derivative equals 1 (not 0, because we haven't subtracted ad costs yet). At this point, the next dollar of ad spend produces exactly one dollar of profit. Any dollar beyond this yields less than a dollar in return—a net loss.

Below are the Marginal Profit curves (the first derivatives of the curves above). Remember, these profits are before ad costs. I call them "profits" because they're revenues minus cost of goods sold and other variable costs - just not ad costs.





This is the power of MMMs - they can tell you exactly how much to spend in each channel for profit maximization (or whatever KPI you're optimizing toward).

This allows for total budget optimization by channel (or even by campaign, if your inputs include campaign-level data). As a byproduct, you also get forecasting; you have a profit (or any KPI) estimate for each channel at any spend level.



It's important to stress that incrementality and marginal analysis are built into the MMM equations. Here's an example: say a channel generates \$50 in profit at \$100 of spend, and \$53 at \$101 of spend. That extra \$3 in profit is incremental—it came from the additional \$1 in ad spend. Net of that \$1, you're up \$2. And because we're looking at the impact of one additional dollar, this is marginal analysis by definition. The MMM gives you both.

One disclaimer: this is the power of mathematics. The whole setup is still entirely reliant on the quality of your input data, as noted earlier.

MMM with Prior data

MMMs can also incorporate prior data. Say you've run a holdout test for TV and can express the result as a mean and standard deviation - a bell curve. The MMM will blend that with its own TV estimate.

The blending logic is intuitive: whichever estimate is more precise gets more weight. If the holdout test has a tighter standard deviation than the MMM's confidence interval, the blended result pulls toward the holdout. If the MMM's confidence interval is tighter, the blend pulls toward the MMM. You're keeping both sources of data and weighting them by each source's confidence level.

One caveat: a holdout test gives you lift at a single spend level - one point on the curve. The MMM estimates an entire curve with diminishing returns. So the blending typically anchors the overall scale of the channel's contribution while trusting the MMM for the shape of diminishing returns.



MMM with Non-Spend Variables

Beyond Media: Accounting for Business Context

Up to this point, we have treated channel spend as the sole driver of revenue. In a perfect world, "more spending equals more revenue." However, your revenue is constantly being pulled and pushed by factors that have nothing to do with your ad budget: seasonal promotions, sudden price changes, or aggressive competitor moves.

To build a more accurate MMM, we must account for these "Externalities." We do this by adding Categorical or Dummy Variables to the model. For example:

- Promotions: A "0" on normal days and a "1" on promotion days.
- Price Shifts: A "0" when the price is at baseline and a "1" when a price increase is in effect.

By including these, the model can "strip away" the noise. It realizes that your Google Search spend didn't cause a spike in sales on Tuesday, but the 20% off site-wide sale you were running did. This prevents the model from over-attributing success to your media channels.



Section Summary: Media Mix Models

Attribution tells you who gets credit. Incrementality testing tells you what's causal. Media Mix Models bring it all together. They are the framework that can simultaneously estimate incrementality, model diminishing returns, incorporate carryover effects, and tell you exactly how much to spend in each channel to maximize profit.

The section builds an MMM from the ground up, starting with something anyone can grasp: a simple scatter plot and trend line in Excel. From that foundation, it layers on each concept that makes an MMM powerful. First, multiple channels showing that analyzing channels in isolation produces misleading coefficients (TV looked like a 4.15 ROAS on its own, but dropped to 1.43 once Podcast spend was properly accounted for). Then adstock - the decay factor that captures the reality that today's ad spend doesn't just drive today's revenue but carries over into tomorrow and beyond. Then diminishing returns - modeled by raising spend to an exponent less than 1, so doubling your budget no longer doubles your predicted revenue. Each addition makes the model more realistic and more useful.



Overfitting is a serious risk: with enough flexibility, a model can perfectly memorize your historical data while producing nonsensical predictions. The chapter explains two defenses: 1) Ridge Regression (penalizing overly large coefficients) and 2) time-based cross-validation (testing on data the model has never seen). This section then shows why standard random cross-validation breaks down for MMMs due to adstock leakage. It also emphasizes that the quality of your input data matters enormously: if your channel spend is highly correlated, no algorithm can reliably separate their effects.

Where the chapter gets most exciting is the actionable output. A well-built MMM produces response curves for every channel, complete with confidence intervals from Monte Carlo simulation showing exactly how profit changes as spend increases. From those curves, you can derive marginal profit curves that pinpoint the profit-maximizing spend level for each channel. Incrementality and marginal analysis are mathematically embedded in the model's equations. The chapter also shows how MMMs can incorporate prior data from holdout tests, blending causal evidence with the model's own estimates, weighted by confidence, to get you closer to the truth than either source alone. And non-spend variables, such as promotions, price changes, and competitive activity, can be added to account for factors beyond your ad budgets.

Why should you care?



Because MMMs are the only tool that can answer all the key marketing budget questions in a single, unified framework: How much should I spend in total? How should I allocate across channels? What's my expected return at different spend levels? Where am I hitting diminishing returns? What's the incremental value of the next dollar? Every other method covered so far — attribution, holdout tests, geo tests, marginal analysis answers one piece of the puzzle. MMMs attempt to solve the whole thing. They're not perfect as they depend on data quality, require enough variability in your spend patterns, and produce ranges rather than certainties. Still, when built well, they are the most effective tool in your arsenal that brings everything else together. And with modern tools making them far more accessible than they were even a few years ago, there's less and less reason not to build one.

09

Other MMM equation options used in industry

In our MMM section, we used this equation:

$$\text{Profit}_t = a_0 + a_1 \left(\frac{\text{TV Spend}_t}{1 - r_{\text{TV}}} \right)^{b_1} + a_2 \left(\frac{\text{Podcast Spend}_t}{1 - r_{\text{Podcast}}} \right)^{b_2}$$

This works well for explaining MMMs and the math behind them. But it isn't the only choice. In fact, most industry implementations use a Hill function instead - including Google Meridian and Meta Robyn (though not in identical ways).

Let's take a deeper look at how the Hill function works and how it models diminishing returns.



The Hill Function

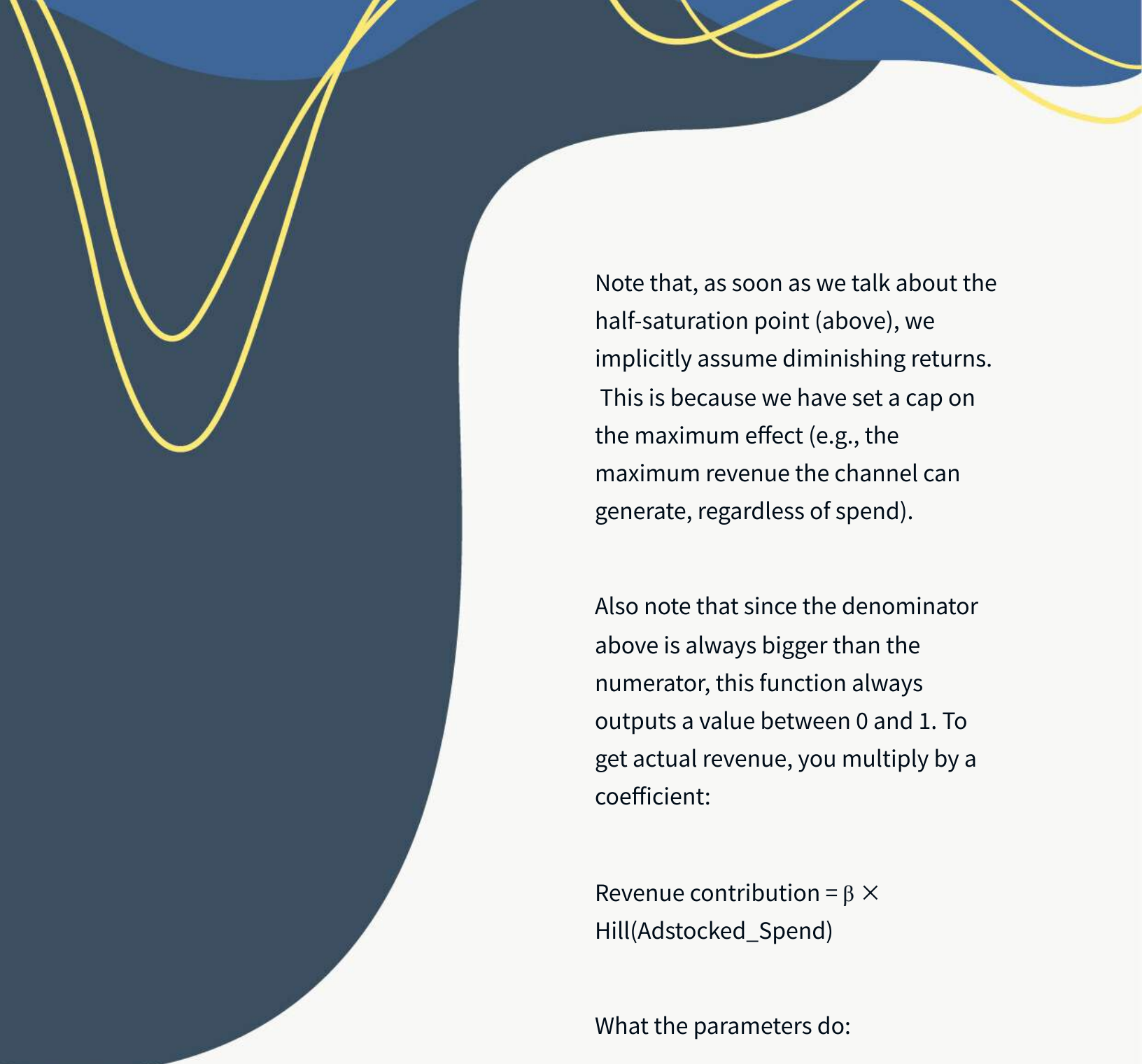
The Hill function comes from biochemistry - it was originally used to model enzyme saturation. Marketing borrowed this idea because ad response behaves similarly: at some point, more spending stops producing results in proportion.

The basic form is:

$$\text{Hill}(x) = x^\alpha / (x^\alpha + \gamma^\alpha)$$

Where:

- x = spend (after adstock transformation)
- α (alpha) = controls the shape of the curve
- γ (gamma) = the half-saturation point (the concentration level where you get 50% of maximum effect)



Note that, as soon as we talk about the half-saturation point (above), we implicitly assume diminishing returns.

This is because we have set a cap on the maximum effect (e.g., the maximum revenue the channel can generate, regardless of spend).

Also note that since the denominator above is always bigger than the numerator, this function always outputs a value between 0 and 1. To get actual revenue, you multiply by a coefficient:

$$\text{Revenue contribution} = \beta \times \text{Hill}(\text{Adstocked_Spend})$$

What the parameters do:

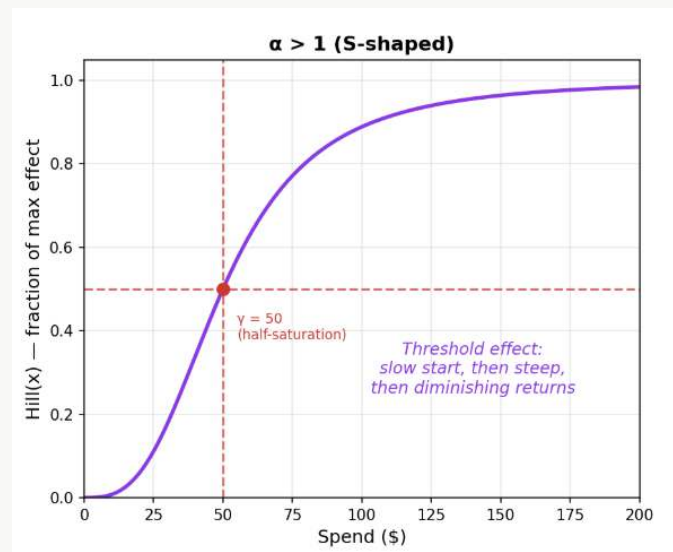
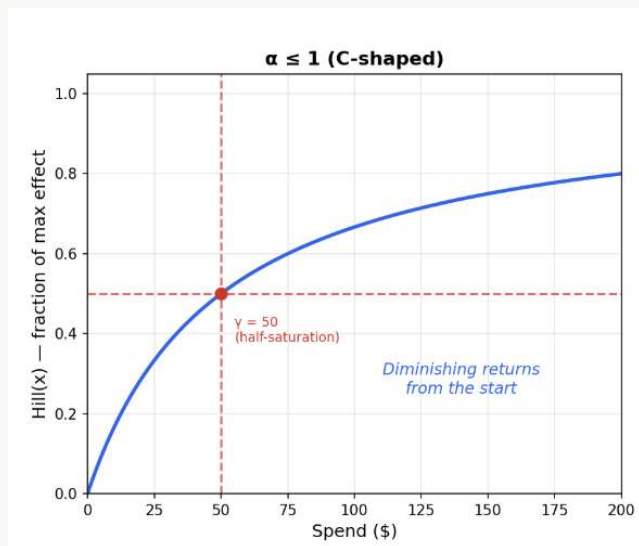
The γ (gamma) parameter is intuitive; it's the spend level where you hit half your potential. If $\gamma = \$50$, then at \$50 of spend, you're getting 50% of the maximum possible effect from that channel. Spend more, and you approach the ceiling; spend less, and you're in the steep part of the curve.

The α (alpha) parameter controls the curve's shape:

- $\alpha \leq 1$: Diminishing returns from the start (C-shaped)
- $\alpha > 1$: Threshold effect, then diminishing returns (S-shaped). Slow start, then steep growth, then saturation. This models where you need to hit a threshold before the spending kicks in, where you need enough frequency to register at all.

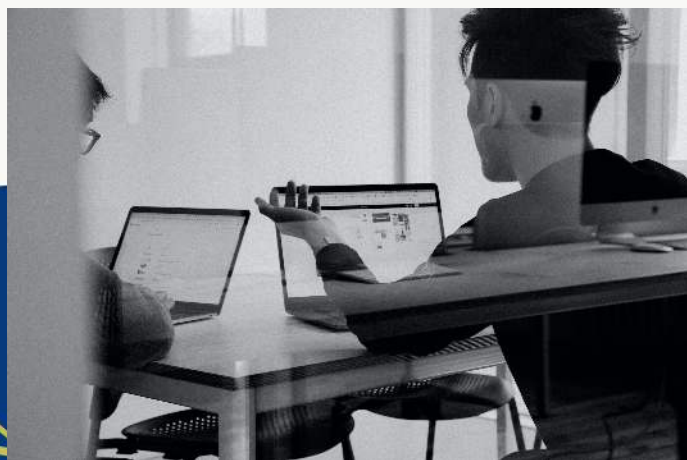
See visuals below.





How does it differ from our power function?

Our equation used Spend^b for diminishing returns. That's a one-parameter approach that only produces C-shaped curves. The Hill function uses two parameters and can produce both C and S-shaped curves. It also has a natural ceiling - the curve flattens toward a maximum; while the power function keeps growing forever (just more slowly).





For most channels, both approaches give similar results in the typical spend range. The Hill function is more flexible, which is why it's become the industry standard. But flexibility comes with a cost: two parameters, instead of one mean more estimates from the same data.

My suggestion: run your data against both and see how much the results differ. If they're similar, it doesn't really matter which you choose. If they differ meaningfully, compare cross-validation error, predicted vs. actual revenue (or profit or conversions), and go with the lower one.

You may also want to try Google Meridian or Meta Robyn directly and see how they perform on held-out data. Both can be implemented with Vibe coding, even if you've never written code before.

Industry Note: This Hill function approach is the engine behind the industry's two most prominent open-source libraries: Meta's Robyn and Google's Meridian. Both models rely on the Hill function's ability to switch between "C-curves" and "S-curves" to accurately reflect how different media, from high-intent Search ads to high-frequency Television spots, actually saturate in the real world.

10

Bringing It All Together: The "And" in Action



You've now seen three approaches to marketing measurement: attribution, incrementality testing, and media mix models. Each has strengths. Each has blind spots. The question is: how do you actually use them together?

This is where art meets science.

What Each Source Gives You

Let's recap what you're working with:

Multi-touch attribution gives you a full picture of the customer journey - every touchpoint, every click, every assist. It's comprehensive and always-on. But it only sees what it can track (mostly digital), can't prove causation, and it's biased toward channels that appear late in the funnel.

Holdout tests give you causal truth. If you ran a clean test, you know the incremental impact of that channel during that period. But you can test only one or two channels at a time, tests are expensive, and results may not generalize across different spend levels or time periods.

Media mix models give you a complete system—every channel, with diminishing returns and carryover effects, producing response curves you can use for optimization. But MMMs are correlational, sensitive to model assumptions, and struggle when channels are highly correlated.

No single source is sufficient. But together, they're powerful.

A Framework for Triangulation

Here's how I think about bringing them together:

Start with what you trust most.

If you have a clean holdout test for a channel, that's your anchor. It's casual. Use it as a prior when building your MMM.

Let the model know that TV, for example, should be in the neighborhood of \$3.50 return per dollar, not \$7.00, at some known level of spend. The MMM can refine the estimate, but the holdout keeps it honest.

If you don't have holdout data, your MMM becomes the primary source for that channel. But flag it as lower confidence. The estimate is only as good as your model assumptions.

Use data-driven attribution as a sanity check, not a source of truth.

Attribution tells you something; channels that show up frequently in conversion paths are doing something. If your MMM says a channel is worthless, but attribution shows it assists on 30% of conversions, that's worth investigating. The MMM may be missing something. Maybe attribution is overcounting. Either way, the discrepancy is a signal.

Use the disagreement to ask better questions.

Respect the confidence intervals.

Every estimate has uncertainty. Your holdout test said, "TV returns somewhere between \$2.80 and \$4.20, and \$3.50 is our best guess."

When intervals overlap, you can be more confident. When they don't, something is off; either one source is wrong, or conditions changed between measurements.

Weight by recency and relevance.

A holdout test from two years ago is less reliable than one from last quarter. An MMM trained on 3-year-old data may not reflect current behavior. Attribution during a holiday period may not generalize to the rest of the year.

More recent data gets more weight. Data from similar conditions gets more weight. This is a judgment.

An Example: Reconciling Conflicting Signals

Say you're looking at your Paid Social channel. Here's what your data shows:

- Attribution: Paid Social gets credit for 25% of conversions (last-touch) and 35% of conversion paths (any-touch).
- *Holdout test* (6 months ago): Showed a 2.5x return on ad spend during a 4-week test.
- *MMM*: Estimates a 1.8x return, with a 95% confidence interval of 1.4x to 2.2x.

These don't match. What do you do?



First, check if the holdout and MMM are measuring the same thing. The holdout was 6 months ago at a specific spend level. The MMM is estimating the average return across the full data period at varying spend levels. If spending has increased since the holdout, diminishing returns could explain the gap - 2.5x at \$50K/month might become 1.8x at \$80K/month.

Second, look at the confidence intervals. The holdout had its own uncertainty - say, 2.0x to 3.0x. The MMM says 1.4x to 2.2x. These ranges overlap in the 2.0x-2.2x zone. That's your best estimate: probably around 2x, could be a bit higher.

Third, check attribution for plausibility. If Paid Social is touching 35% of conversions, a 2x return seems reasonable. If it were only 5%, you'd be more skeptical.

You don't get a single precise answer. You get a range, 1.8x to 2.2x, with the most confidence around 2x. That's enough to make a budget decision.





When Sources Agree

This is the easy case. If your holdout says 3x, your MMM says 2.8x, and attribution shows the channel is heavily involved in conversions, you can be confident. Not certain - never certain - but confident enough to act.

Agreement across independent methods is the strongest signal you can get.

When Sources Disagree Badly

Sometimes the gap is too big to reconcile.



Your MMM says Display is your best channel. Your holdout test says it's barely incremental. Attribution shows it touches lots of conversions, but always late in the journey.

This is uncomfortable, but it's also valuable. It tells you something is wrong with at least one of your measurements. Display may be taking credit for conversions that would have happened anyway (attribution and MMM are fooled; holdouts reveal the truth). Maybe your holdout was flawed - too short, contaminated, or run during an atypical period.

Don't force agreement. Investigate. Run another test if you can. Adjust your MMM priors and see if results stabilize. In the meantime, treat that channel as high uncertainty and be conservative with budget increases.

The Hierarchy of Trust

If I had to rank these sources by reliability, it would be:

1. Clean holdout test (recent, well-designed, sufficient duration)
2. MMM calibrated with holdout priors
3. MMM without priors (but validated on held-out time periods)
4. Attribution (useful for pattern recognition, not for budget decisions)

This isn't absolute. A sloppy holdout test is worse than a well-built MMM. A sophisticated, data-driven attribution model might catch things your MMM misses. But as a starting point, this hierarchy reflects the sources of causal confidence.





The Art

I've used the word "judgment" a lot in this chapter.

No framework will tell you exactly how to weigh a 6-month-old holdout against a fresh MMM. No formula will reconcile a 3x estimate with a 1.5x estimate. You have to look at the data, understand the context, and make a call.

This, then, is the Art. It's built on experience - yours and your team's. It gets better over time as you see which estimates hold up and which don't. It's informed by science, but it's not reducible to it.

The goal is to make decisions that are robust to uncertainty. If your budget allocation makes sense, whether the true return is 1.8x or 2.2x, you're in good shape. If it only works at exactly 2.5x, you're taking on more risk.

The Payoff

When you bring all three sources together, you get something none of them provides alone: a self-correcting measurement system.

Data-Driven Attribution flags channels that are active in the funnel. Holdouts test whether that activity is causal. MMMs estimate the full response curve and guide optimization. Each one challenges the others.



Over time, you build a picture of your marketing performance that's grounded in data, calibrated by experiments, and refined by judgment. You're not over-trusting any single model. You're triangulating.

That's how you know what's working.

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Your Roadmap: From Data to Decisions

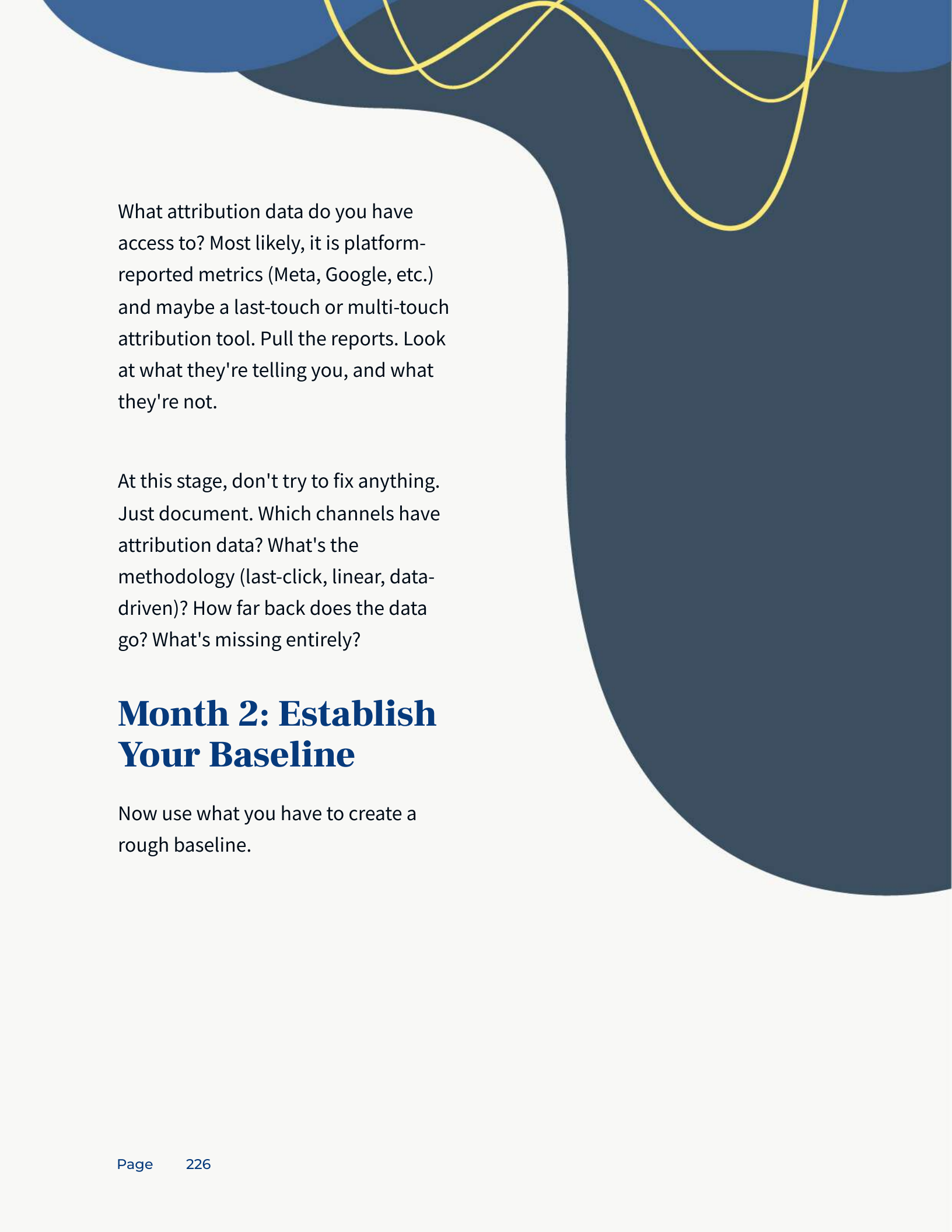
You've made it through the concepts. Now let's put them into practice.

What follows is a 6-month roadmap (with overlapping time periods for different tasks) for building a marketing measurement system from scratch. If you're already partway there, you can skip ahead. The goal is the same either way: get to a place where you can confidently answer "how much should we spend, and where?"

Month 1: Know What You Have

Start by auditing your current measurement setup.





What attribution data do you have access to? Most likely, it is platform-reported metrics (Meta, Google, etc.) and maybe a last-touch or multi-touch attribution tool. Pull the reports. Look at what they're telling you, and what they're not.

At this stage, don't try to fix anything. Just document. Which channels have attribution data? What's the methodology (last-click, linear, data-driven)? How far back does the data go? What's missing entirely?

Month 2: Establish Your Baseline

Now use what you have to create a rough baseline.



Even imperfect attribution data tells you something. Calculate the approximate ROI for each channel using the available data. Flag the channels where you trust the numbers and the ones where you don't. Be honest about the gaps.

This baseline is your starting point. You'll refine it over the coming months. But you need something to compare against, and "what our current attribution says" is as good a starting point as any.

Months 2 to 5: Design and Run a Holdout Test

Upgrade to a data-driven attribution system if you don't already have one.

This will be more insightful and a better estimate of incrementality than any other attribution system.

If you have the budget and organizational buy-in, run a holdout test on at least one channel. (More, if possible).

Pick a channel where the stakes are high enough to matter but not so high that a test is politically impossible. Design the test with proper holdout geography or audience. Let it run long enough to see a real signal - typically 4 to 8 weeks depending on your sales cycle.

This is your first piece of causal data. It won't cover every channel, but it gives you ground truth for at least one of them.

Months 4: Prep for MMM

While you hold out for test runs, start preparing for your media mix model.

Start varying your spend by channel so that you can give good data to the MMM for the 3 months.

Decide whether to build or buy your first MMM.



If building, start building. You will need machine learning engineers and software engineers, but Vibe coding can also do most of the heavy lifting. You can build your own using the PRD in this book (as a starting point only) and AI-assisted coding. Or you can hire a consultant.

You can also use Google Meridian or Meta Robyn platforms, which will make it easier to build, but are only partly customizable. Both are well-documented.



If you decide to buy an MMM, go through RFPs and decide who you want to work with.

Months 5 to 6: Build or Buy your first MMM

Whichever path you choose above, the mechanics are the same: gather your spend data by channel by day (or week), gather your revenue or conversion data, and fit a model that accounts for adstock and diminishing returns.

Your first model won't be perfect. That's fine. The goal is to get something running so you can start learning.

Validate and Calibrate

Now you have multiple data sources: your attribution baseline, your holdout test results (for at least one channel), and your MMM output.

They won't all agree. That's expected.



Start comparing. Where does the MMM line up with the holdout? Where does it diverge? If your holdout test said TV drives \$3.50 in profit per dollar spent, and your MMM says \$5.00, that's a signal: either the model needs calibration, or there's something the holdout didn't capture.

Use your holdout results as priors to anchor the MMM. This blending - science plus judgment - is where the real value emerges.

Build Your Response Curves

With a calibrated model, you can now generate response curves for each channel.

These curves show you profit at every spend level—and more importantly, the marginal profit. Where is the next dollar still productive? Where have you hit diminishing returns? Where is the optimal spend for each channel?

This is the payoff. You're no longer guessing at budget allocation. You have curves that tell you: "Spend up to \$X on TV, \$Y on Podcast, and here's why." Of course, keep in mind the uncertainty intervals.

Month 7 and beyond: Operationalize and Iterate

Measurement isn't a one-time project. It's a system.



Set up a cadence: refresh your MMM quarterly (or monthly if your data supports it). Run a new holdout test quarterly on a different channel. Compare each new model to the last. Track whether your budget shifts actually produced the expected results.

Over time, you'll build institutional knowledge. You'll know which channels' estimates are stable and which are volatile. You'll get better at spotting when the model output doesn't match reality. You'll develop intuition grounded in data.

That's the art, built on top of the science.

The Short Version

If 6 months (and beyond) feels like a lot, here's the condensed path:

1. Audit what you have (2 weeks)
2. Establish a baseline with current attribution (2 weeks)
3. Run one holdout test (6 weeks)
4. Build a basic MMM (4 weeks)
5. Compare, calibrate, and generate response curves (4 weeks)

That's roughly four months if you move with urgency. The remaining time is iteration and refinement, which never really ends.

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Conclusion

If you've made it this far, you now have a framework that most marketers don't.

You understand that attribution is a good starting point. And data-driven attribution is the best attribution system. Holdout tests give you causal truth, but only for the channels you can afford to test. MMMs can fill gaps but come with their own challenges: correlated spending, assumptions about decay and diminishing returns, and confidence intervals that are too wide to be materially meaningful.

And you understand that the real work isn't choosing one approach over another. It's bringing them all together.

This is the "and" I want to continue to emphasize. Multi-touch attribution, holdout tests, and media mix models. Each has blind spots. Each has strengths. Your job is to triangulate - to look at all the data, weigh the confidence behind each input, and make a judgment call.

That judgment is the art.

The science provides the reports, response curves, and optimal spend calculations. The art is knowing when the model output doesn't pass the smell test, when correlated channels make the results unreliable, and when last year's holdout test may not apply to this year's market. When to trust the confidence interval and when to widen it based on what you know about your business.

No model will make these calls for you. But you have come a long way and now have mastered the tools to make these important calls yourself.

What's Next

The barrier to entry has never been lower.

A few years ago, building a media mix model required a team of software engineers and data scientists. Today, with AI-assisted coding, you can build one yourself - even if you've never written a line of code. The PRD in this book is designed to help you practice with vibe coding and to start small with an MMM. After that, start iterating and keep building. You will improve and build more complex models as needed.

Google Meridian and Meta Robyn are open-source MMMs and well documented. Bayesian methods that once required PhD-level statistics are now accessible through libraries that handle the complexity for you. The "vibe coding" approach means you can describe what you want in plain English and get working code back. And if you'd like to understand more of the machine learning in Google Meridian or Meta Robyn, just ask ChatGPT or Claude AI to explain in simple terms.

This is a shift. Marketing measurement is no longer the domain of technical specialists. The constraint now is understanding - knowing what to build, what questions to ask, and how to interpret the output. That's what this book aimed to provide.



Why I Wrote This

I've spent 20 years watching smart marketers struggle with budget decisions using incomplete data, or, just as often, with good data and not enough confidence to act on it.

Both are costly.

The first leads to wasted spend and missed opportunities. The second leads to paralysis, endless debates about methodology, and defaulting to last year's budget because no one can agree on what the data says.

I am well aware of the challenges, as I was there, learning from all the painful mistakes along the way.

I hope this book helps you avoid both traps so you can walk away able to look at your attribution reports, your test results, and your models and see a coherent picture. And where there are conflicting signals, how to resolve them. That you can defend your budget recommendations and financial forecasts with honest uncertainty, sound reasoning, but most of all, with confidence in making the best actionable decisions you can.



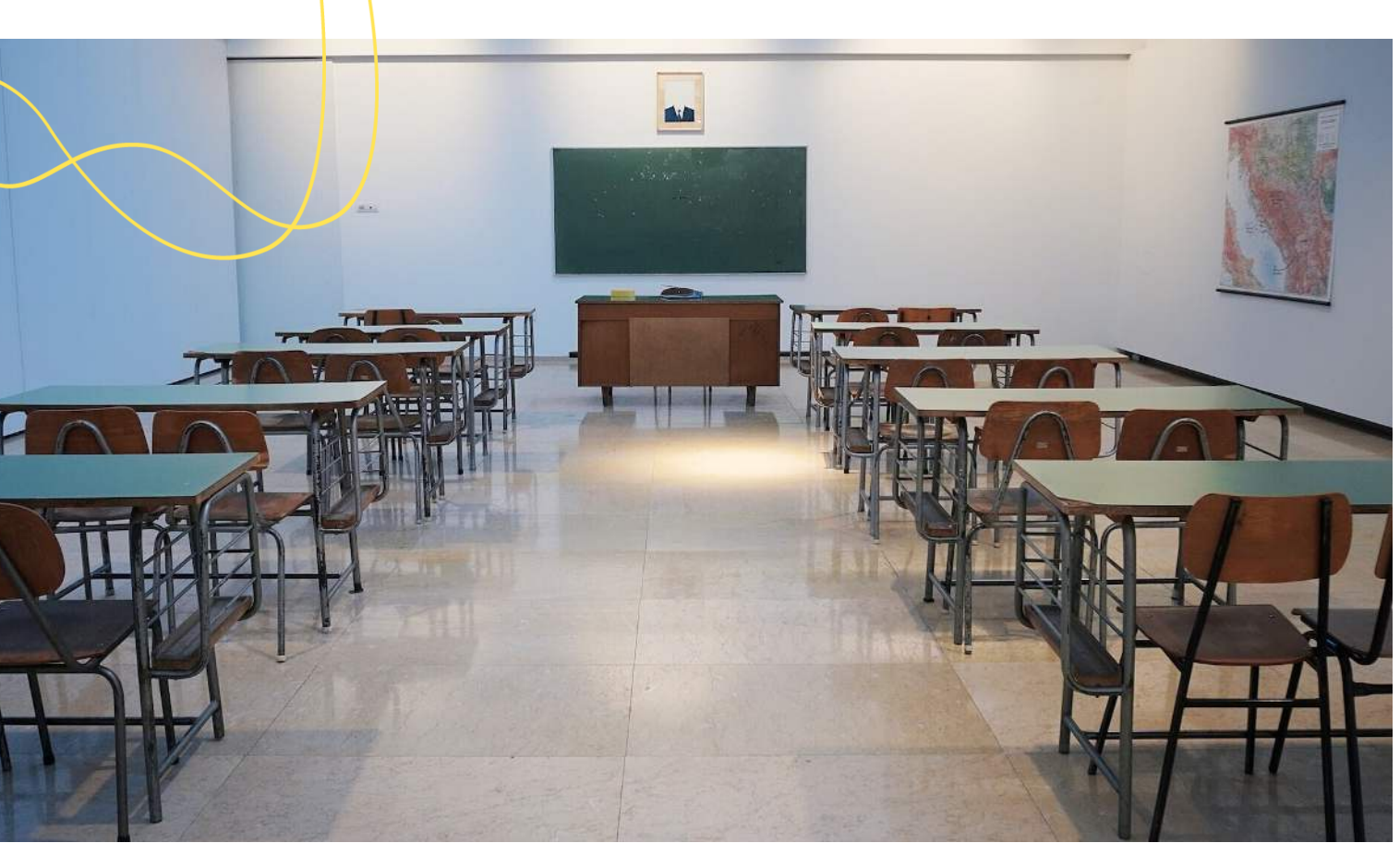


Marketing measurement is probabilistic. Budgets are bets. But they can be informed bets, grounded in science, sharpened by art, and ultimately, yours to make. You may still feel some uncertainty. We all do. But understand that you are now best positioned to make the difficult budgeting and forecasting decisions.

So go make them.

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Appendix 1 - Bid Optimization



Even though bid optimization (on platforms such as Google and Meta) isn't strictly measurement, I wanted to share some insights that I think might be useful to many of you.

When there is a lag between click and conversion

If your business is of the sort where there can be a significant lead time between the click and conversion, you will need to optimize based on the day/time of the click, not the day/time of the conversion. (The engines will do this for you, as long as you are using their own bid optimization).



For instance, say you are advertising with Google Ads. If people are searching/clicking on Sunday, but converting on Monday, then bids should be increased on Sundays, not Mondays. In general, if there is a lag between a click and a conversion, for optimization purposes, the conversion should be attributed to the time of the click.

Note that after you increase bids on Sunday in this example, Sunday may not look good from an ROI perspective because of all those click costs without conversions, while Monday may look great because of all those lagging conversions.

Furthermore, bid adjustments at any time of day should also take into account the time the click occurred, not the time the conversion occurred. After all, your bids should reflect the day & time people are searching and clicking, and if it takes them a couple of days to convert, so be it.

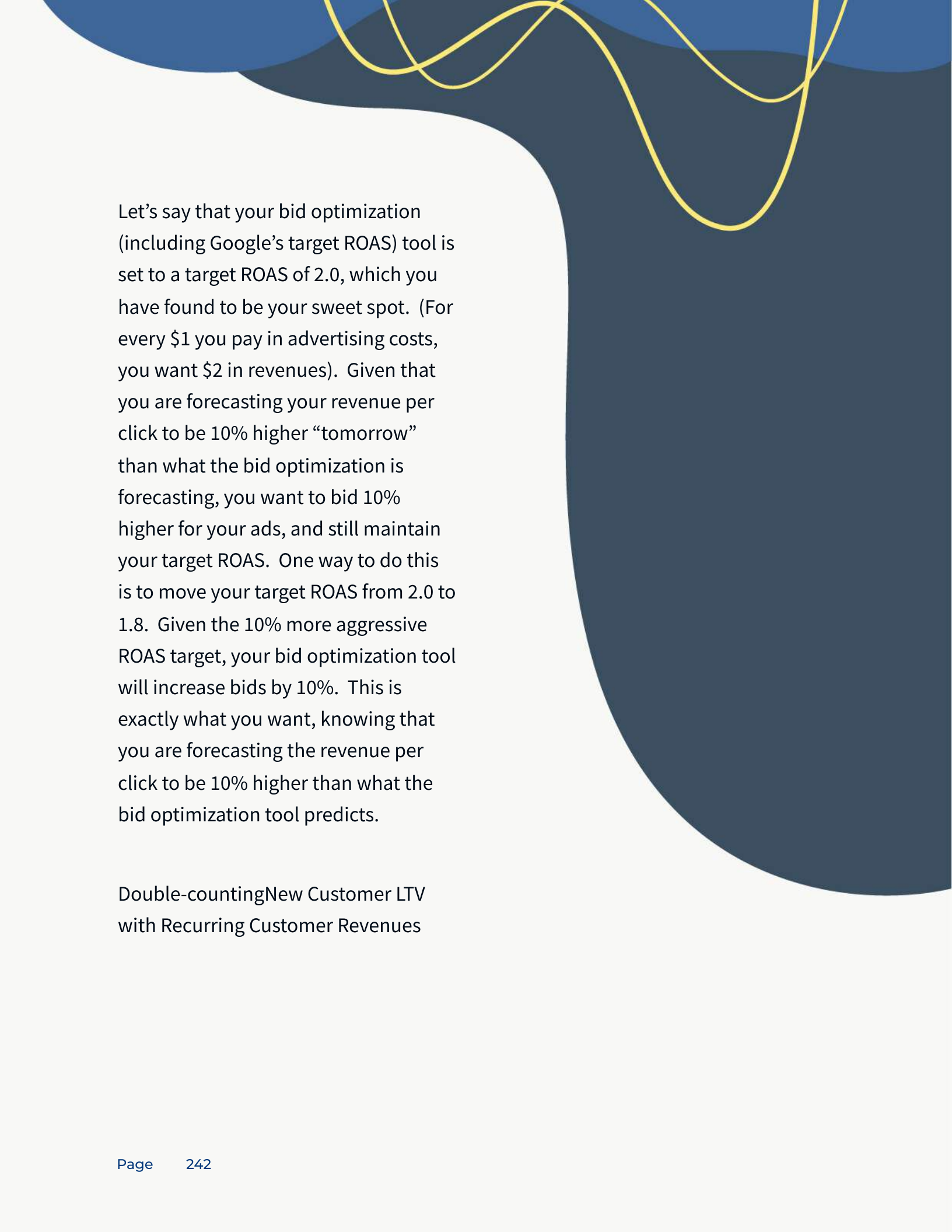
Looking at old data

Most bid optimization tools (including Google and Meta) optimize only on recent data, perhaps even 90 days old. But what if yours is a very seasonal business, e.g., the holiday season, then you really do need to look at what happened last year, e.g., on Black Friday or Cyber Monday. How should you do this?



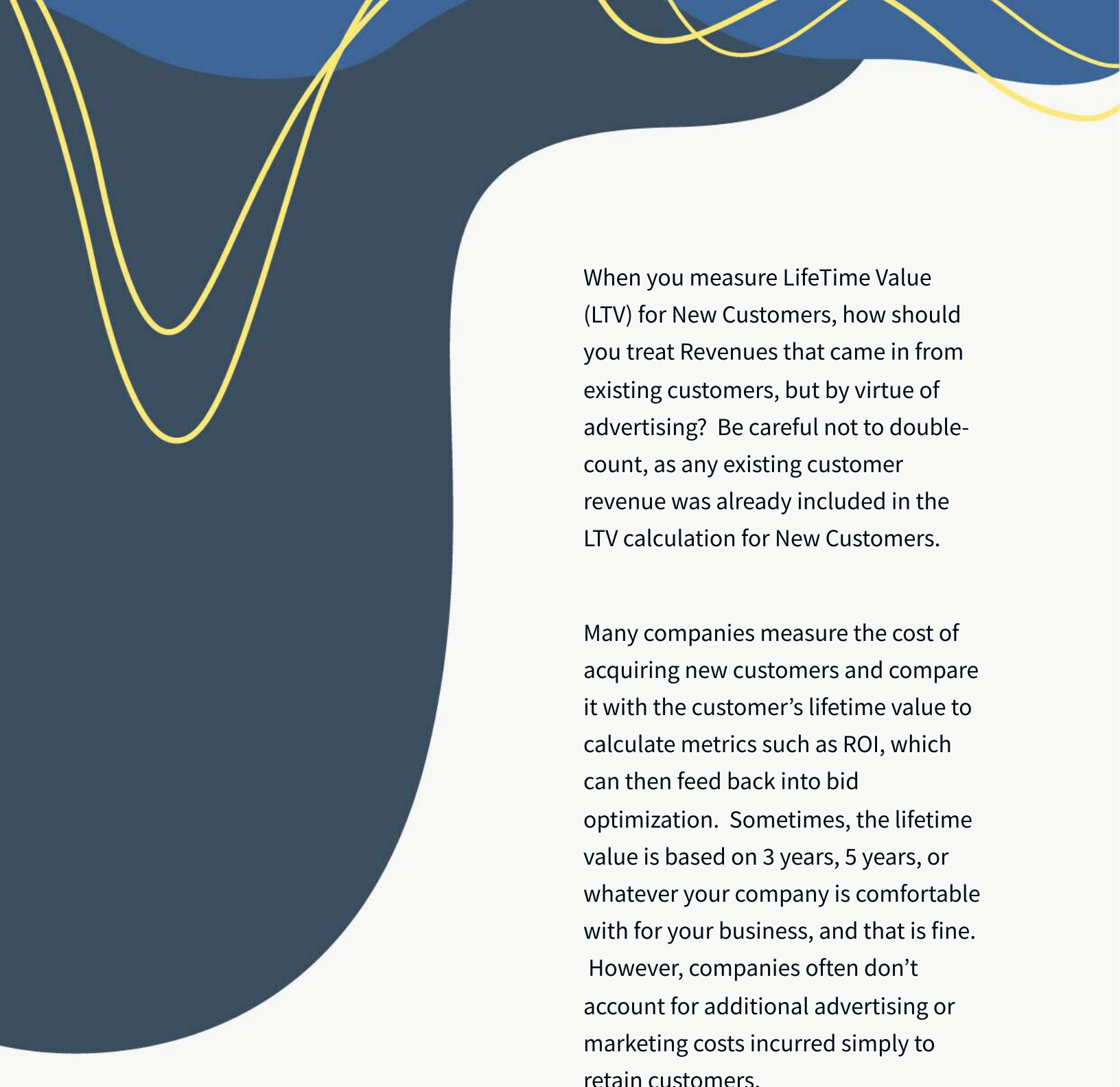
Let's say that you are an e-commerce business, and you know that on Black Friday, your Revenue per Click increases by 10% compared to the prior week. (You know this based on your data from last year that you have. You also know that your bid optimization tool only looks at the past seven days (for example). So you can predict that your bid optimization tool will under-forecast revenue per click by 10% for Black Friday. Armed with this information, you can “game” your bid optimization tool to do the right thing.





Let's say that your bid optimization (including Google's target ROAS) tool is set to a target ROAS of 2.0, which you have found to be your sweet spot. (For every \$1 you pay in advertising costs, you want \$2 in revenues). Given that you are forecasting your revenue per click to be 10% higher "tomorrow" than what the bid optimization is forecasting, you want to bid 10% higher for your ads, and still maintain your target ROAS. One way to do this is to move your target ROAS from 2.0 to 1.8. Given the 10% more aggressive ROAS target, your bid optimization tool will increase bids by 10%. This is exactly what you want, knowing that you are forecasting the revenue per click to be 10% higher than what the bid optimization tool predicts.

Double-counting New Customer LTV
with Recurring Customer Revenues



When you measure LifeTime Value (LTV) for New Customers, how should you treat Revenues that came in from existing customers, but by virtue of advertising? Be careful not to double-count, as any existing customer revenue was already included in the LTV calculation for New Customers.

Many companies measure the cost of acquiring new customers and compare it with the customer's lifetime value to calculate metrics such as ROI, which can then feed back into bid optimization. Sometimes, the lifetime value is based on 3 years, 5 years, or whatever your company is comfortable with for your business, and that is fine. However, companies often don't account for additional advertising or marketing costs incurred simply to retain customers.



A good example of this is with Paid Search. Consider a customer who was acquired a few months ago and returns to Google to search for keywords related to other products your company already sells. For instance, assume that your company sells shoes and socks. A customer is acquired when they buy shoes, but a month later, they search for socks on Google. They see your company's Paid Search ad, click, and convert at a certain rate. If this kind of customer behavior has happened often enough historically, then just taking the customer lifetime value and comparing it to the cost of customer acquisition isn't enough. One has to include the cost of customer retention to get a better estimate of ROI, which, too, should be obtainable from historical data if all the right tracking and measurement are in place within your backend databases.

Then the bid optimization should be set with an aggregated view of costs and revenues.

If you don't measure LTV and are only looking at immediate revenues, you risk undercounting, since there may be additional revenue from the customer without additional advertising costs (e.g., via your "free" email channel).

Dealing with Referrals

In Marketing, there can be a strong element of Referral value that is difficult to measure, so most organizations estimate it at the default value of 0, which can markedly under-represent the true value of advertising or marketing.



The volume of referrals or word of mouth is very hard to measure, and we know that. We can measure in certain cases by tracking whether customers are forwarding emails and whether the recipients are engaging/converting. Or we can ask customers to share specific discount codes and see if they are being used. By and large, referrals that occur when a customer tells their friend about your business in person or on the phone, and that friend then transacts based on that referral, will be hard to measure (although Media Mix Models discussed later can take a crack at it).

The main thing to be aware of here is that, since we can't capture such a referral value, we ignore it for the most part, which is mathematically equivalent to assigning it a value of ZERO. Well, can we do better than giving it a value of zero? For instance, in aggregate, are there ways to estimate the percentage of our business coming from referrals, and then use that to inform how we think about advertising spend?

For instance, what if we measured that 20% of our new customers came directly to our website and transacted with us, without any detectable marketing touches?

Unless you are Amazon, it is fairly safe to assume that people don't just get up one fine morning and go to a website they have never heard of to transact. Something (that we cannot measure) caused these new customers to visit our website and do business with us. It certainly wouldn't hurt to assume that half of the 20% (10%) came in through customer referrals. Now, this may sound like a wild guess, and it probably is. BUT it is better than the other wild guess (by default), of 0%, which is what it becomes if we ignore this completely.



Once we decide on that 10%, we can layer it into customer lifetime value to inform advertising bids and budgets. In other words, the lifetime value of any new customer obtained, say, from Google Ads, is both the forecasted Repeat business and now, plus the forecasted (though hard to measure) Referral business as well. This then feeds into bid optimization.

Section Summary: Bid Optimization

Bid optimization sits at the intersection of measurement and execution. Getting your measurement right is only half the battle; this appendix covers the practical decisions that determine whether good measurement actually translates into good outcomes when you're setting bids and targets on platforms like Google and Meta.

The section opens with a simple point: when there's a lag between a click and a conversion, you must optimize based on when the click happened, not when the conversion happened. If people click on Sunday and convert on Monday, Sunday is the day that deserves higher bids, even though your Sunday reports will look terrible and Monday will look like a star. This applies equally to time-of-day adjustments.

It then tackles the problem of seasonality and backward-looking bid tools. Most platforms optimize using recent data, often just the last 7 to 90 days. If you're heading into Black Friday and your historical data shows revenue per click jumps 10%, you can "game" the system by temporarily lowering your target ROAS (say, from 2.0 to 1.8) to force bids up by the right amount. The math is straightforward, but the insight is that you sometimes need to override your tools with knowledge they can't see.



The section also raises a subtle but costly accounting trap around Lifetime Value. Many companies calculate LTV for new customers and compare it against acquisition cost, but forget to include the ongoing marketing costs required to retain those customers. If a customer acquired through shoes later comes back through a paid search click for socks, that retention cost needs to be factored in. Otherwise, your ROI calculations are inflated, and your bids are too aggressive.

Finally, the chapter makes the case against ignoring referral value. Most organizations default referral value to zero because it's hard to measure, but that default is itself a wild guess. If 20% of new customers arrive with no detectable marketing touch, and even half of those came from word-of-mouth referrals, that likely gives you a better estimate than zero. Layering that referral value into your LTV calculation changes how aggressively you should bid, and in some cases, it means you've been systematically underspending.

Why should you care?

Bid optimization is where your measurement insights meet real dollars. You can have perfect attribution, clean incrementality tests, and a well-built MMM. Still, if your bids are set using stale data, incomplete LTV calculations, or tools you've never audited, you're leaving money on the table. This appendix won't make you a bid management expert. Still, it will help you catch the most common and most expensive mistakes: optimizing on conversion timing instead of click timing, ignoring seasonal signals your tools can't see, double-counting or undercounting customer value, and accepting average performance when the underlying variance tells a very different story.

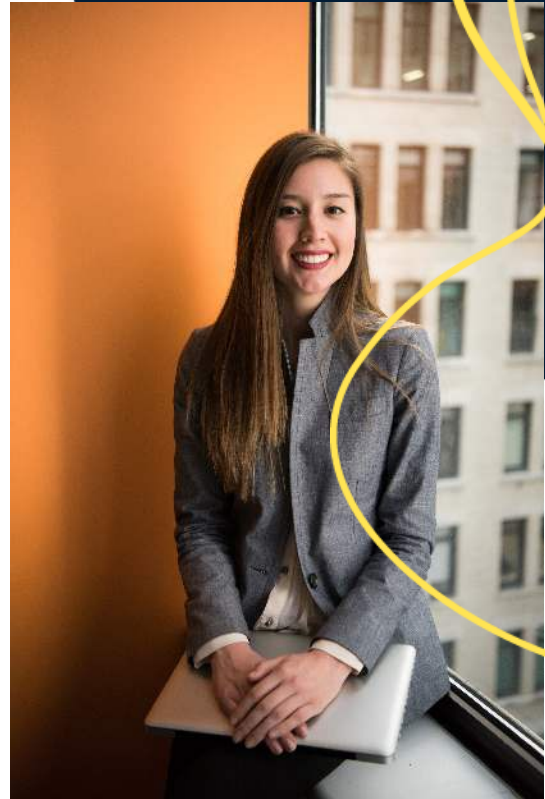
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Appendix 2 - Measurement Through Surveys

TV (and Radio) ads' ROI is perhaps the hardest to measure. The vanity URL approach (where a specific URL is mentioned in the TV ad, making it easy to track users who convert after landing on that URL) doesn't quite work, since only a small percentage of users use it. A large percentage of users will ignore the vanity URL and go directly to the home page.

One could test TV ads using the Control Geos approach mentioned earlier. The idea would be to run the TV ads in the Control Geos and measure the overall Conversions from TV ads in the Treatment Geos to calculate ROI. This can work fairly well, but there are 2 caveats:

1. This approach will only work if Conversions from TV ads are a significant percentage of your overall Conversions; otherwise, Conversion lift from TV ads may not be able to provide a signal, as it will get "drowned out" with the normal variance of Conversions based on Geos.
2. If you run TV ads in only part of the country rather than the full country, you may not qualify for a pricing discount available only for National Buys.



With that in mind, an approach that could also work is using surveys on your website.

These surveys should be presented to users only post-conversion, e.g., on the Thank you page or in subsequent emails (or both).

The survey can simply ask “How did you hear about us?” and offer 8 or 9 options, eg,

Google

Meta

YouTube

TV

Radio

Direct Mail

Referral from someone

Third-party website

Other (please fill in)

If you find that too many people are selecting “Other,” consider omitting it.





You can either allow the user to select multiple options or just one. In either case, you can do the Math to figure out the percentage of conversions coming from TV, since TV will be one of the options. You then apply this percentage to the business as a whole, since the number of survey respondents will likely be less than 25%.

Two things to note:

1. Are the survey responders somehow different from the non-responders, even though they only see the survey after they convert? For instance, what if those who responded were somehow less likely to have seen the TV ad? (Maybe they spend more time online, and hence, are more likely to respond to a survey). Just something to keep in mind, as you present your data.
2. The survey design itself should randomize the order in which the 8 or 9 options are presented. For instance, users may be more likely to select the first option, or the last option, and less likely to select any of the intermediate options. To counteract this bias, the ordering of the options can be randomized, so in some instances TV is the first option, in others it may be the last, and in still others it may be an option somewhere in the middle.

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Appendix 3 - Statistics Overview & A/B Testing

This Statistics overview is optional for those who want to understand the material only at a high level.

If you have any background in Statistics, some of the content discussed in this book will be much easier to follow. But if not, you can still follow this book, but perhaps without going into all the details. In any case, let's do a quick review for those interested in understanding the nitty-gritty of Statistics used in holdout tests, geo-based tests, etc. Note that even if you don't want to bother with the Statistics, just reading through this will make you dangerous and enable you to ask all the right questions as you pursue that goal of getting the most from your advertising dollars.

I have also included details on A/B testing of web pages and ad copy, as one needs to understand more of the Statistics to get a good sense of the insights.

The main concept in Statistics to understand is Hypothesis Testing. Let's briefly explore the world of Medicine to understand this concept before returning to the world of Marketing.

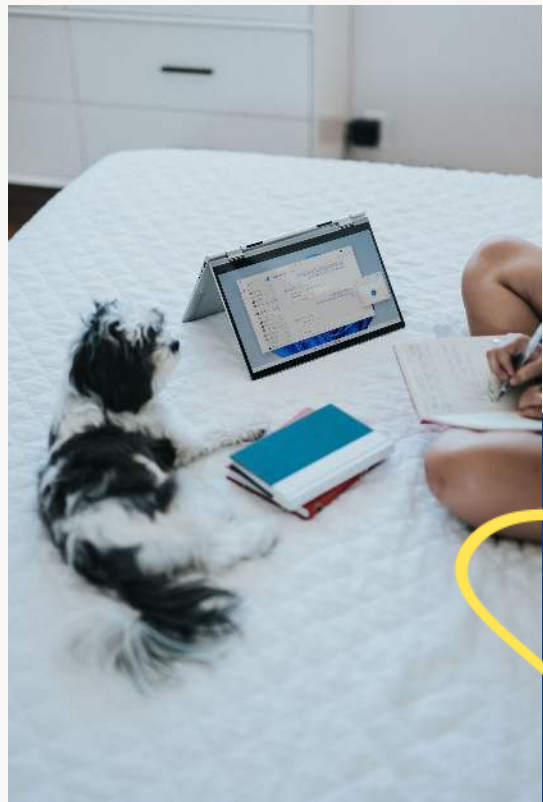
In Medicine, when a potential new drug is tested, the presumption is that it is no better than a placebo. Then patients are divided into two groups: the Control group, who do not receive the drug (and will be given a placebo instead), and the Treatment group, which, as the name suggests, receives the drug. Then, both groups are observed to measure recovery rates or recovery times, to see if the drug is effective. Patients do not know which group they are in.

Hypothesis Testing:

Let's say there are 1,000 patients in each of the Control and Treatment groups, and the 2,000 patients are completely randomly distributed. Note that a perfectly random distribution of patients across the two groups is essential. As an extreme counterexample, if all healthy 25-year-olds were in the Treatment group and all senior adults were in the Control group, the results of the experiment would clearly be tainted. After all, if recovery rates within a specific time period are much higher in the Treatment group, it may have nothing to do with the drug's efficacy, but rather with the fact that healthy 25-year-olds tend to recover faster.

Now, assuming a perfect (or close to perfect) random distribution of patients in the Control and Treatment groups, let's assume that 500 patients (out of 1,000) in the Treatment group recover within a week, but only about 480 patients (out of 1,000) in the Control group recover completely within a week. Is the drug really effective, or was this just a random chance?

Without going through the Math (Statistics), one can't tell. If, however, we were to say that all 1,000 patients in the Treatment group recovered within a week, and only 480 in the Control group, your intuition would tell you, even without going through the Math, that the drug is likely effective.



But in general, how can you be more sure?
More confident one way or the other?

Continuing with this example, in Hypothesis Testing we set up a Null Hypothesis (H_0) as our base belief. In this case, our base belief would be that the drug is not any more effective than a placebo. The Alternate Hypothesis (H_1) is that the drug is effective. We want to gather evidence to see if we can reject the above Null Hypothesis in favor of the Alternate Hypothesis and thereby deem the drug effective.

After gathering data, we can say that if the Null Hypothesis were true (that the drug is not effective), the result we observed would occur only 1% of the time. (The Statisticians amongst you will tell that I am not being mathematically rigorous here, but I am just trying to give a sense of what is going on). With this “1% p-value”, we can reject the Null Hypothesis and conclude that the drug is effective. Of course, by doing so, we need to be aware that we will be wrong 1% of the time.

So, how is this related to the world of Marketing?

Holdout testing





In holdout testing (or incrementality testing), as described in the Holdout section of this book, we are conceptually doing exactly the Hypothesis testing described above. In this case, the placebo group is the holdout group (which does not receive the medicine or treatment). The treatment group is the group that gets the treatment or at least “eligible for treatment,” i.e., eligible to see the ad. We then start with the Null Hypothesis (H_0) that the ads are not effective, just as in medicine, we start with the Hypothesis that the medicine isn’t effective. In either case, we run the test and see whether we have enough evidence to reject the Null Hypothesis and conclude that the ad is effective (i.e., the medicine works).



In actual Medical testing, we need a high degree of confidence (at least 95%, or even 99%) that the medicine works based on the data. We may not need the same high degree of confidence that the ad is effective. However, keep in mind that the ad costs money, so it can't just be effective; it has to be effective enough to be profitable. In other words, the Ad generates revenues that exceed its costs.

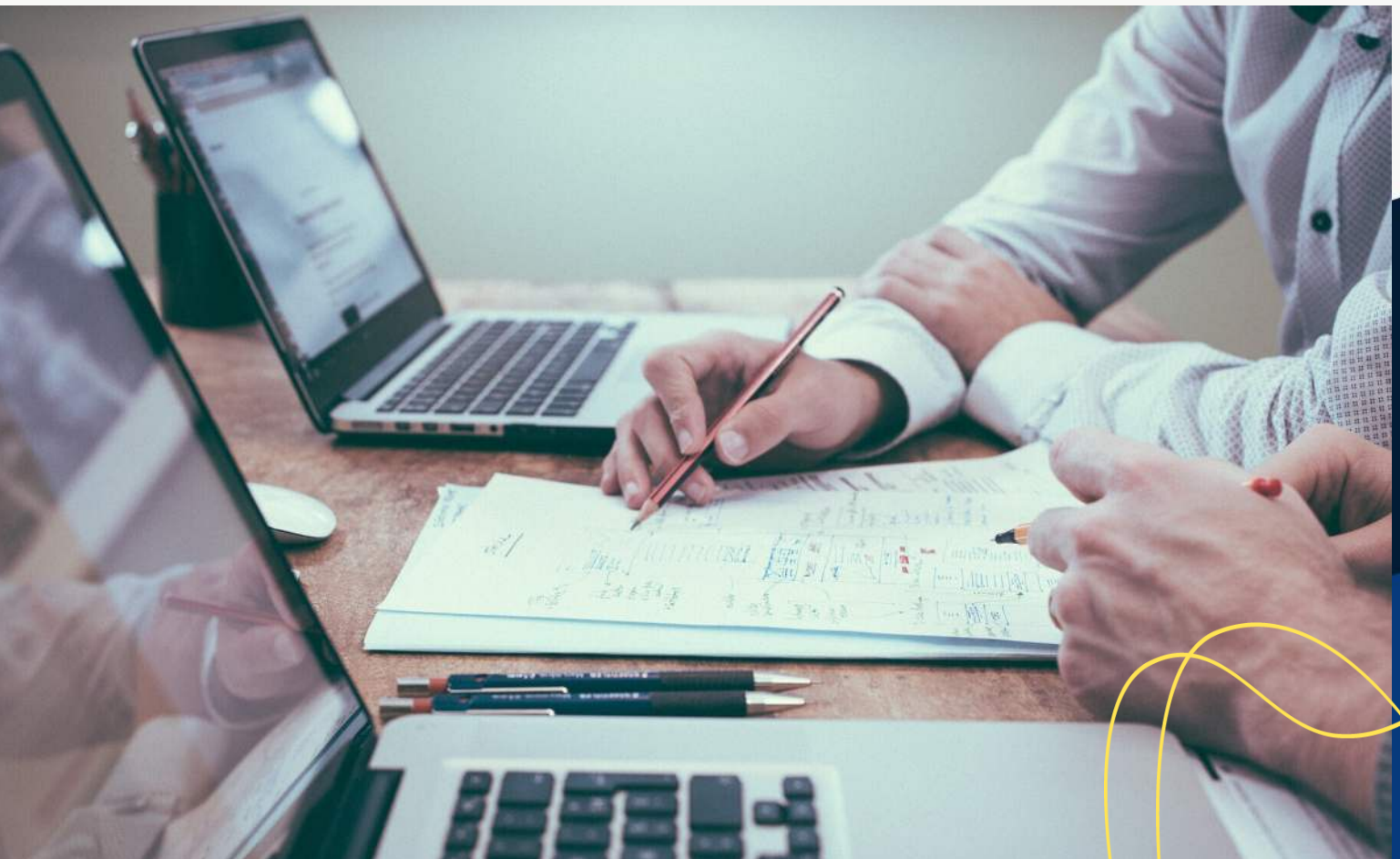
A/B testing for web pages?

In A/B testing, where we may test our current webpage (or landing page, as it is usually called, where visitors land on frequently from clicking an ad; this can be the home page, but usually is not) against a new version of the webpage (known as Challenger) that we believe will convert users at a better rate.

In such a test, let's say that the A version of the page is the original, i.e., the current webpage, and the B version of the page is a new page to be tested. Our Null Hypothesis here, H_0 , would state that the conversion rate of B is less than or equal to the conversion rate of A. (As a side note for the advanced reader, there is debate over whether this should be a 1-sided or 2-sided test.) Thus, the burden of proof is on the new page B to see if it can convert better.



In other words, (loosely speaking), can we gather enough data to have evidence that if this Null Hypothesis were true, the results we see in terms of conversions of A, B have less than a 5% of occurring. If so, we may reject the Null Hypothesis in favor of the Alternate Hypothesis and deem that the conversion rate of B is better than the conversion rate of A. In doing so, B would become the new default landing page. Again, we may be wrong 5% of the time, but something we could live with, given that we can now (likely) increase our conversion rates with page B. Once we achieve this level of evidence that page B is better than page A, we quite often use the term that our results are “Statistically Significant” or “Stat Sig”.

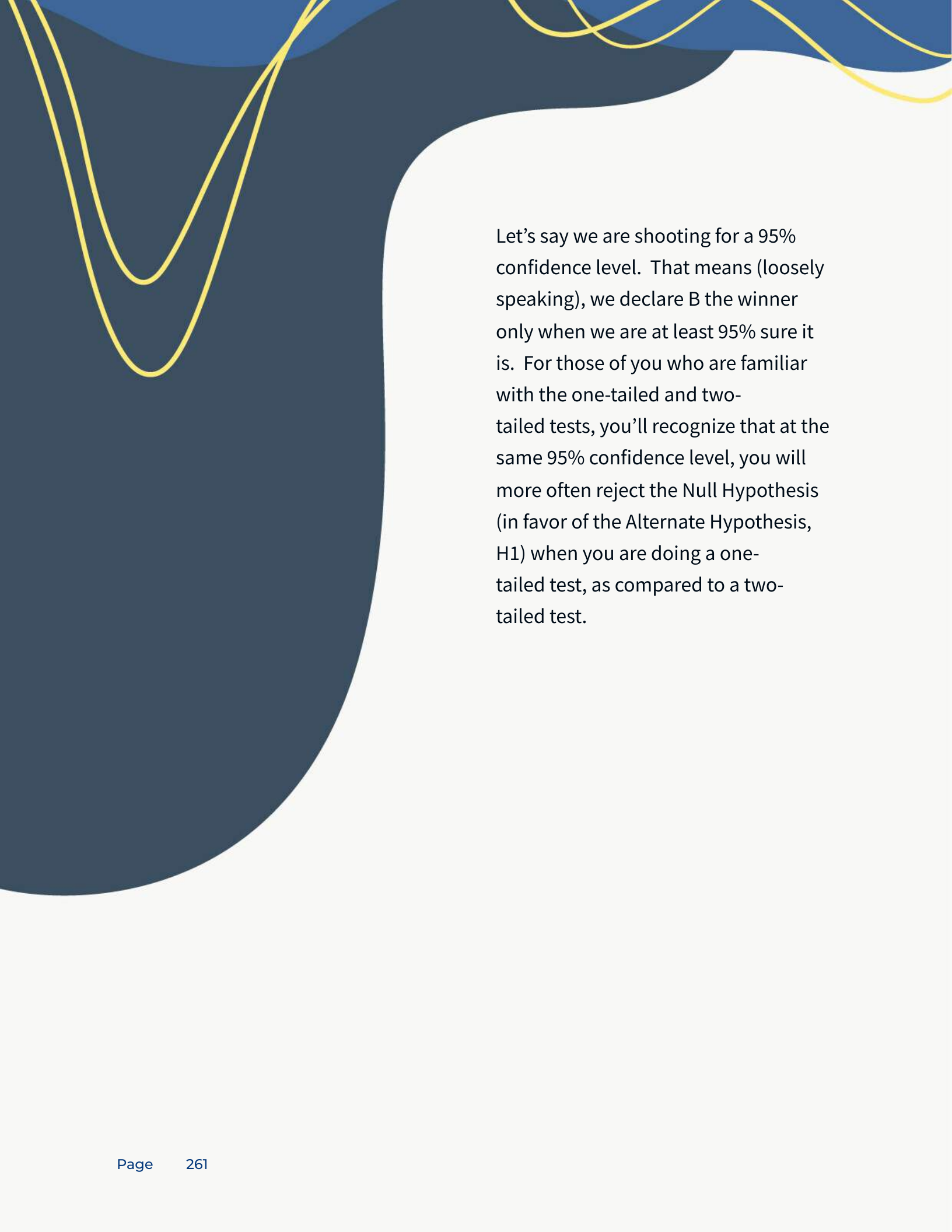


Type 1 and Type 2 errors

Let's now discuss what a Type I error and a Type II error would look like here in the one-tailed A/B test.

Type I error (false positive) is declaring the challenger, B, to be the winner, when in fact, it is not. (i.e., incorrectly rejecting the Null Hypothesis).

Type II error (false negative) is declaring the challenger B not to be a winner, when in fact, it is. (i.e., failing to reject the Null Hypothesis when we should have).



Let's say we are shooting for a 95% confidence level. That means (loosely speaking), we declare B the winner only when we are at least 95% sure it is. For those of you who are familiar with the one-tailed and two-tailed tests, you'll recognize that at the same 95% confidence level, you will more often reject the Null Hypothesis (in favor of the Alternate Hypothesis, H_1) when you are doing a one-tailed test, as compared to a two-tailed test.

Data sufficiency in A/B testing and Incrementality testing

Before you begin any testing, always check for data sufficiency, known as the power test, to see if you have a shot at picking up the signal. Such a “pre-test” requires you to guess the lift, and you can then decide whether or not you will pick up a signal at your desired statistical significance.

So let's try to phrase this in our Type I and Type II error terminology.

Let's say that in an A/B test (A is Control, B is New Variation), the Null Hypothesis, H_0 , is that the conversion rate of $B \leq$ the conversion rate of A.

The alternate hypothesis, H_1 , is that the conversion rate for B is greater than the conversion rate for A.

As before, Probability of Type I error, say α = probability of incorrectly declaring B to be the winner, i.e., incorrectly rejecting the Null Hypothesis. (False Positive rate)



Probability of Type II error, Beta, = probability of failing to declare B a winner, even if it really has a higher conversion rate than A, i.e., incorrectly failing to reject the Null Hypothesis (False Negative rate).

Then, what is $1 - \text{Beta}$?

If you think about it, $1 - \text{Beta}$ is the probability of correctly rejecting the Null Hypothesis, i.e., correctly declaring B as the winner. (True Positive rate).

Thus, Power is defined as $1 - \text{beta}$, because what we really want to know is the probability that the test will declare B as the winner when, in fact, it has higher conversion rates than A.

We can estimate A's conversion rates using actual data.

We also have to estimate the conversion rate from B to calculate the Power and the sample size required.

So, for instance, if we know the conversion rates of A (as measured) is 3.0%, and then we estimate conversion rates of B to be 3.5%, we can calculate the number of conversions and traffic that will be required to achieve Power ($1 - \text{Beta}$), of say 0.9. (90% chance that we will be able to declare B as a winner, when in fact, it does convert better than A.)





Now, this clearly leads to a question: the reason you want to test B against A is to see if B converts better than A. Yet you are being asked to estimate B's conversion rates before even beginning the test. Seems a bit like a Catch-22. Yes, it is a bit of a Catch-22, but it's still useful to try to estimate how much better B converts, because that will give you insights into whether you have a shot at picking up the signal (that B converts better than A). For instance, you may find that with your estimates of conversion rates for A and B, you will need to run the experiment for 1 year to achieve 90% power, which may lead you to call off the test entirely before you begin.

See example in the table below:

If you estimate B at...	Expected Lift over A (3.0%)	Sample per Group	Total Sample Needed	At 10K visitors/week
3.5%	0.5% (17% relative)	~21,500	~43,000	~4 weeks
3.3%	0.3% (10% relative)	~58,000	~116,000	~12 weeks
3.1%	0.1% (3% relative)	~505,000	~1,010,000	~2 years

When should you call off an A/B test and instead go on to the next test

If you are running an A/B test to see which page (A or B; where A is Control and B is Variation) is leading to higher conversion rates, when should you call the test off if you are not able to reject the null hypothesis, H_0 , at a predetermined confidence level?

For the most part, this is a judgment call, but there are a couple of things to think about to guide your decision. The only reason that you can't end the test quickly (and reject or accept the Null Hypothesis) is that the conversion rates of A and B are close to each other. If the conversion rates weren't even close, you could end the test and call a winner. On the other hand, since conversion rates of A and B are close to each other, what harm is there in continuing to run the test? After all, your business isn't affected by keeping B around, since it converts at rates close to A. The real cost of keeping B around and continuing with the test is an "opportunity cost", since if you ended this test, you could run an entirely different A/B test instead.

You could also end the test after a predetermined sample size and/or number of days you are going to run the test, and if you can't reject the Null Hypothesis within that sample size or timeframe, you simply go on to the next test.

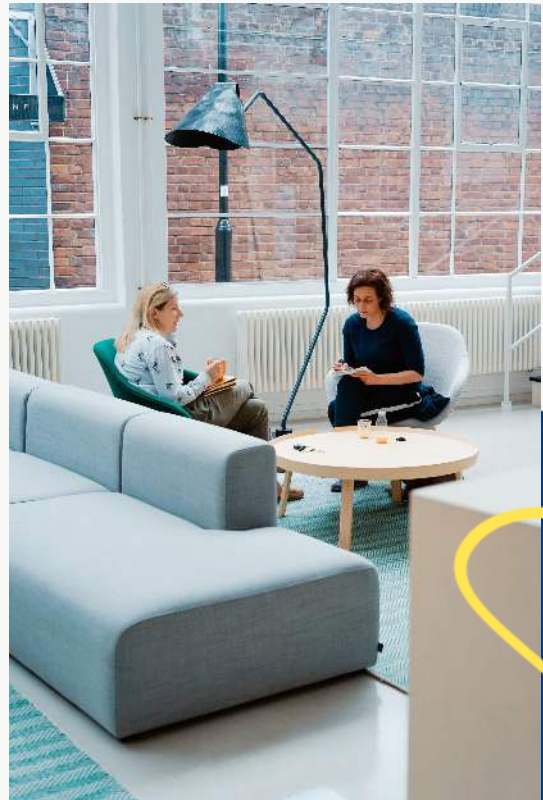
Don't peek and end tests early before the pre-determined sample size. If you keep looking every day to see if statistical significance has been reached, you will get more false positives than the 5% threshold you may have set.

Multi-armed bandit approach to optimize even while you A/B test

In A/B tests, say, traffic for conversion rates is split 50/50. This is the best way to start. But what if you see one version, say B, winning? Should you still send traffic 50/50, or can you do better?

This is the well-known “Multi-armed bandit” problem. Imagine a casino where each slot machine has different odds of winning. A gambler (here, you) wants to maximize your winnings in some amount of time. To do so, you want to get information on the odds of winning in specific slot machines. But as you go along, you also want to weigh that with what you have already uncovered, e.g., certain slot machines seem to have better odds, and you therefore want to “pull those” more often.

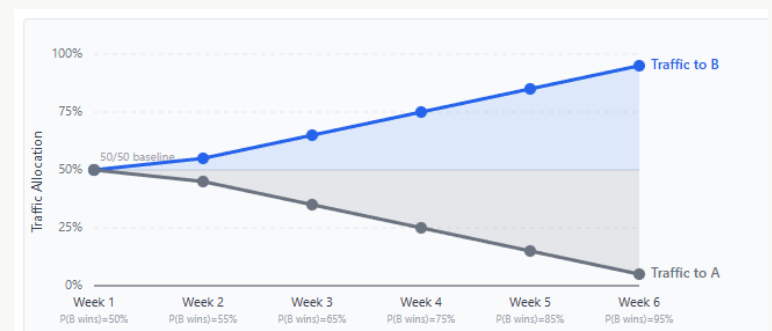
The same is true here in the A/B test. If version B is winning, you want to start sending it more than 50% of the traffic because, if its conversion rates are higher than A's, you want a higher number of conversions for your business, even while testing. But you don't want to send 100% of the traffic to version B, since you are still learning and you could be wrong about B having higher conversion rates. After all, in the data so far, B could have simply gotten lucky.



The above (sending more traffic to B) is the right way to go, but be very careful of Simpson's paradox (discussed below). In short, when you change traffic allocation mid-test, you must be careful when analyzing the data. Naively combining data from different allocation periods can lead to Simpson's Paradox - where aggregated results tell a different story than the underlying segments.

The exact math of this is complex, but roughly, here is what you can do. If there is a 70% chance that B converts better than A, given the data you have, then send 70% of the traffic to B, and only 30% to A. You don't want to send 100% traffic to B, even though it seems to have the higher conversion rate, because there is a 30% chance that you could be wrong, and you will never know if you exclude A altogether. (If you are serious about implementing Multi-armed bandit approach, then please do a deep dive into exactly how to implement as what is being described here is directional for illustrative purposes only).

As confidence in B grows, traffic shifts from 50/50 toward the likely winner (below).



Statistically Significant



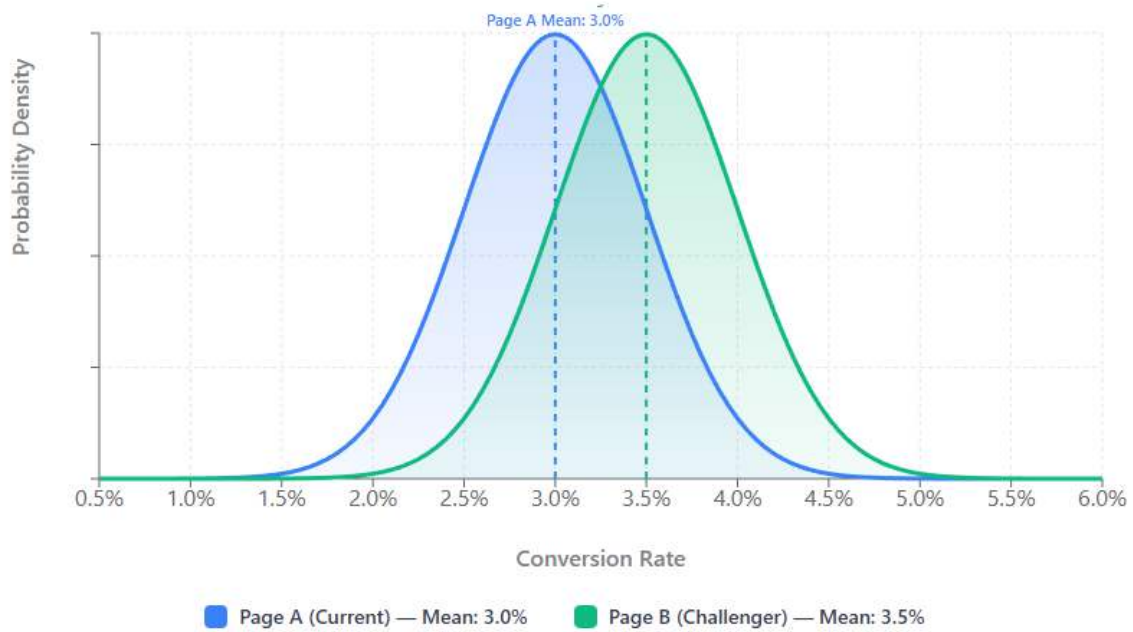
Please see below where Page B converts better than Page A, but not at a statistically significant level. Important to note that the 3% conversion rate for Page A and the 3.5% conversion rate for Page B are point estimates based on our test-period sample. The true conversion rates are distributions around these point estimates, based on bell-shaped curves (Normal distributions). In other words, if we ran the same test again, we would not get the same 3.5% and 3% numbers, but rather somewhat different numbers whose probabilities would be determined by the bell-shaped curves.



Given the high degree of curve overlap, it is possible that page B isn't really better than page A, but just got lucky for this particular test. Hence, we don't have statistical significance.

Distribution of Estimated Conversion Rates

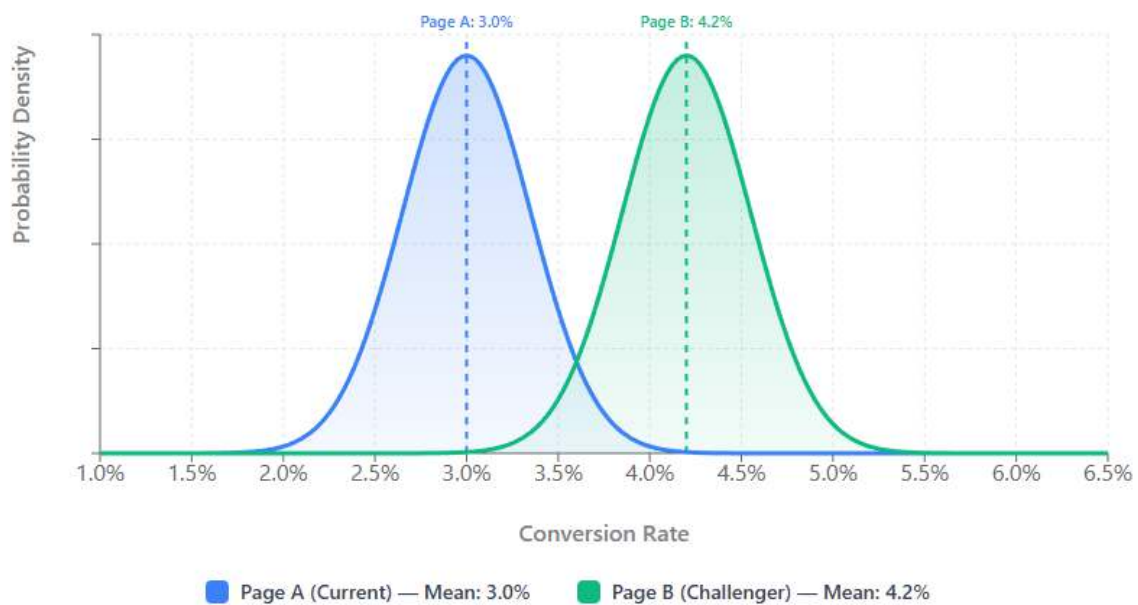
Each curve represents the range of likely true conversion rates given our sample data



In contrast to the above, for the below, we have statistical significance. The curves overlap much less, and now it is very unlikely, given the much smaller overlap, that the true conversion rate of A is greater than that of B. (The true conversion rate of A would have to be to the far right of the blue curve, and also, the true conversion rate of B would have to be on the far left of the curve. Thus, while the conversion rate of A may be better than the conversion rate of B, given the data, we conclude that this is unlikely enough for us to reject the Null Hypothesis. We therefore conclude (from the data below) that page B is the winner and the results are Statistically Significant.

Distribution of Estimated Conversion Rates

Minimal overlap = high confidence that B truly outperforms A





Now we might question where the 95% number (the probability of being right) comes from. If we are testing medicine, we may want something even tighter before releasing it to the public. However, arguably, with A/B testing of landing pages, we may not need to be that tight. For instance, if we have no cost in switching page A with the new page B, then why not pick it as the winner regardless of level of confidence, as long as page B converts even slightly better than page A during the length of our experiment? I know this is very contentious but certainly worth discussing in your organization.

Null Hypothesis and Rejection Regions

In our example, again look at the first visual above, where we did not have Statistical Significance at the 95% level. But if it is just easy to make page B the new landing page, why not do it? After all, if someone gave me even odds, I'd bet on page B being better, even though the test was limited and didn't show B as a clear winner.

Typically, in A/B tests, we set up the Null Hypothesis that Page B is not driving increased conversions, and then collect data to see if we have enough evidence to reject it. On this rejection Region, then, the question we should ask is: what is so special about the 95% confidence level, anyway? Why can't it be 90% or even 80%?

As noted, the 95% confidence interval has been used traditionally and historically. But there is really no reason to stick with it, and you should carefully analyze your business situation before determining what it should be for you.



Let's say you go with the lower threshold of 90% for your A/B testing. So here is your Null Hypothesis, H_0 : landing page B is no better than landing page A for conversion rates.

The Alternate Hypothesis, H_1 , is that landing page B has higher conversion rates than landing page A.

Now, let's consider what it means to have a 95% confidence level threshold vs. a 90% confidence level threshold. At a 90% confidence threshold, you are more likely to reject the Null Hypothesis (compared to a 95% confidence threshold) and declare landing page B to have a better conversion rate than landing page A. At a 90% confidence level, you could be wrong 10% of the time in reaching that conclusion, whereas for a 95% confidence level, you are wrong only 5% of the time. This is what is known as a Type 1 error (incorrectly rejecting the Null Hypothesis).

However, consider the Type II error as well, which is “failing to reject the Null Hypothesis, when you should have”.

At a 95% confidence level, the bar is so high that you are more likely to conclude (compared to the 90% confidence level) that landing page B is not converting at a better rate than landing page A, even though it might be.

In deciding what confidence level to pick in your experiments, first have a detailed discussion around how you want to trade false positives with false negatives.



On a tangential note, I do want to warn you about the high level of “false positives” when you do a lot of testing. Even at the “95% sure” standard, that means our false-positive rate could be 5%. In other words, on average, we could be wrong 1 out of 20 times we declare page B the winner. With A/B testing of landing pages, we usually can live with this.

However, here is an experience I had back in 2017. We were A/B testing ad copies in Google Paid Search (not landing pages, but same concept). The issue is that we were using a tool to do such tests in high numbers - 100s of ad copies being simultaneously tested. With so many tests running in parallel and us being okay with a 5% false-positive rate, we found many winners. Quite likely, many of those winners were just false positives given the large number of tests we were running. This could get needlessly expensive if we pay an external agency per test won, which we were doing at the time.

Simpsons Paradox

Back in 2016, I was looking at click-through rates for our Paid Search ads and comparing year-over-year for the same month of November (so November 2016 compared to November 2015, given that Shutterfly was such a seasonal business). Our click-through rates had dropped. I wanted to see which device was the culprit. So diving deeper, I compared click-through rates by device year over year for November. And across desktop and mobile, 2016 had higher click-through rates than 2015, but in aggregate, 2015 was better.

What was going on?

While I don't recall the exact numbers, consider the following made-up example.

Click-Through Rates: 2015 vs 2016

Year	Device	Impressions	Clicks	CTR
2015	Desktop	500	50	10%
2015	Mobile	500	10	2%
2016	Desktop	100	12	12%
2016	Mobile	900	27	3%

Within each device, 2016 wins on click through rates:

- Desktop: 12% vs 10%
- Mobile: 3% vs 2%

But in aggregate, 2015 wins:

- 2015 overall: $60 / 1,000 = 6\%$
- 2016 overall: $39 / 1,000 = 3.9\%$

How is this possible?



What is happening is that while click-through rates increased on both devices in 2016 compared to 2015, many ad impressions shifted from desktop to mobile, and mobile click-through rates were much lower than desktop. In essence, 2016 was hamstrung by the large mobile volume, which had a lower click-through rate than desktop.

Now, what does this have to do with A/B testing of landing pages?

Well, consider the following:

Landing Page A vs B: Weekday and Weekend Traffic				
Page	Traffic Type	Visitors	Conversions	CVR
A	Weekdays	500	50	10%
B	Weekdays	500	60	12%
A	Weekend	100	4	4%
B	Weekend	900	45	5%

Say we begin an A/B test on Monday. Through Friday, we've seen 500 visitors to each page A and B, with B performing better, with a conversion rate of 12% compared to A's 10%.



As a result, we decided to take advantage of B's stronger conversion rate and move the traffic split to 90% to B and only 10% to A. Again, B does better than A for conversion rates. Just that in general, weekend conversion rates are lower than weekday conversion rates. But B has won over A on both weekdays and weekends.

So we calculate in aggregate and surprisingly, find that Page A wins:

- Page A overall: $54 / 600 = 9\%$ conversion rate
- Page B overall: $105 / 1,400 = 7.5\%$

Simpson's paradox again. Page B wins both on Weekdays and Weekends separately. Yet in aggregate, Page A wins.

Why?

This is because we burdened page B with 90% of the poor-performing weekend traffic (and page A got only 10% of the poor-performing weekend traffic).

Two lessons here:

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1. If you do change the traffic split (e.g., using a multi-armed bandit as described above), calculate those pieces separately. For instance, if we go from 50/50 traffic to 90/10, calculate conversion rates separately for 50/50 and 90/10.
 2. Beware of ad and landing page platforms that automatically change traffic splits. When a page appears to be performing better, the platform may decide to give it more traffic to maximize conversions, since you are measuring conversion rate to decide the winner. But this can mess up aggregate results due to Simpson's paradox.

Section Summary: Statistics Overview & A/B Testing

This appendix is labeled optional, but the concepts it covers determine whether you can trust the numbers from your holdout tests, geo experiments, and A/B tests. You don't need to become a statistician, but understanding hypothesis testing, confidence intervals, and the traps that invalidate experiments will help you ask the right questions and avoid expensive mistakes.

The chapter grounds everything in the framework of hypothesis testing, borrowed from medicine. You start with a null hypothesis (the ad doesn't work, the new page doesn't convert better), gather data, and ask whether the results you observed would be unlikely if the null hypothesis were true. If unlikely enough, you reject the null and conclude the treatment worked. The key insight is that "statistically significant" doesn't mean "definitely true." It means you've crossed a threshold where you're willing to act, knowing you'll be wrong some percentage of the time.



The section introduces Type I and Type II errors in practical terms. Type I is declaring a winner that isn't really better (false positive). Type II is failing to declare a winner who is actually better (false negative). For landing page tests where the cost of being wrong is low, you might reasonably accept 90% confidence (rather than the traditional 95%). The right threshold depends on the asymmetry of consequences in your specific situation.

Power analysis gets explained as the pre-test sanity check you should run. Before launching an experiment, estimate the lift you expect and calculate whether you have enough traffic to detect it. If the math says you need a year of data to reach significance, kill the test before it starts and try something with a bigger expected effect. This saves you from the common trap of running underpowered tests that can never conclusively tell you anything.

The multi-armed bandit section offers a way to optimize while you test. Instead of holding traffic at 50/50 even when one variant is clearly winning, you can shift traffic toward the likely winner proportionally to your confidence. If B has a 70% chance of being better, send it 70% of the traffic (rough rule of thumb). You can generally capture more conversions during the test. But the section warns that this only works if you analyze the phases separately.

Simpson's Paradox is illustrated with a vivid example: Page B beats Page A on weekdays and weekends separately, yet loses in aggregate. This happens when you change traffic allocation mid-test, and the new allocation correlates with a confounding variable (like day of week). The lesson is: if you do change traffic allocation mid-test, analyze each period independently.



Why should you care?

Every measurement technique in this book, from holdout tests to geo experiments to MMM validation, ultimately rests on statistical inference. If you understand what a confidence interval means, you'll know when you encounter noisy results and be able to tell real signals. If you don't run power analysis, you might waste weeks on tests that were doomed from the start. If you understand Simpson's Paradox, you won't fall prey to a seeming contradiction. You don't need to do the math yourself, but you need to know enough to catch when something's wrong. This appendix gives you that foundation, turning statistics from a black box into a tool you can actually interrogate.